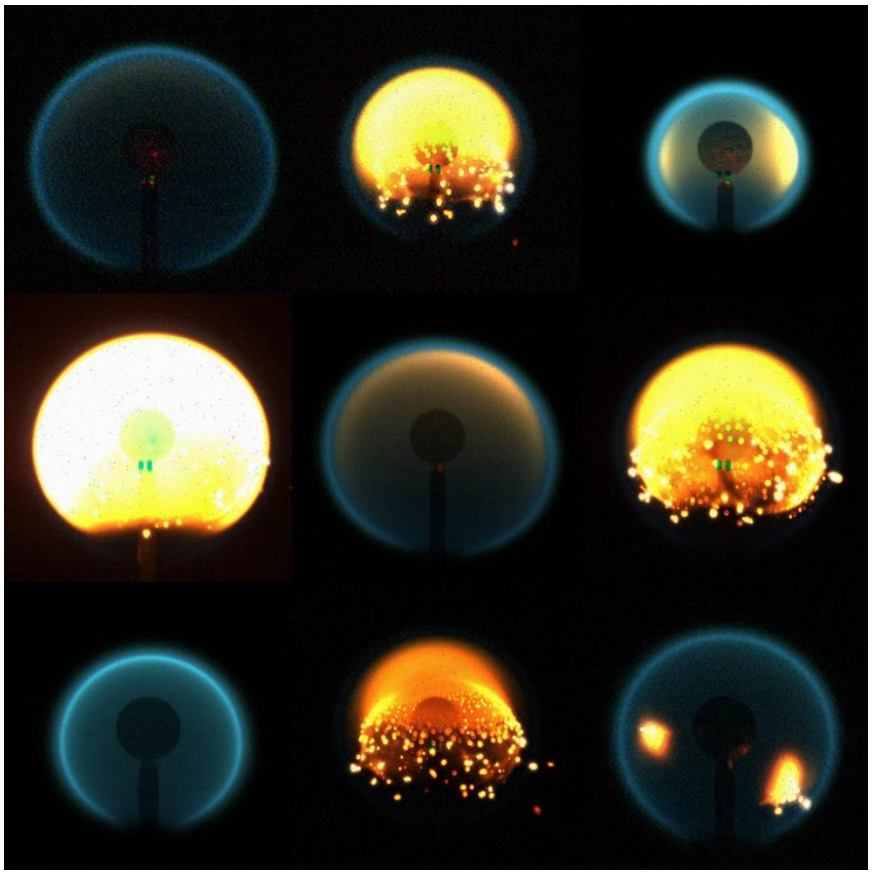
Advanced Combustion via Microgravity Experiments (ACME)
Flame Design

ACME/Flame Design Final Report (2023*)

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Example flame images, where this composite appeared in the 2021 NASA Science calendar and the Oct. 2022 issue of the National Geographic Magazine.

**Data analysis and the publication of results is still underway, so it is expected that this report will be updated in the future with at least a revision of the list of papers and presentations.*

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1. Overview

The Flame Design experiment is one of six independent experiments that make up the Advanced Combustion via Microgravity Experiments (ACME) project for which the tests were conducted within the Combustion Integrated Rack (CIR) on the International Space Station (ISS).

The experiment's Principal Investigator (PI) is Prof. Richard L. Axelbaum of the Washington University of St. Louis's Dept. of Energy, Environmental & Chemical Engineering. The Co-Investigator (Co-I) is Prof. Peter B. Sunderland of the University of Maryland's Dept. of Fire Protection Engineering. The ISS research also benefitted greatly from the work of Phillip H. Irace who was a doctoral student at the Washington University of St. Louis throughout the on-orbit operations.

Meanwhile, Dennis P. Stocker of the NASA Glenn Research Center is the Project Scientist (PS) for ACME including Flame Design and the rest of the ACME investigations. In ACME's preparatory drop testing, etc., Dr. Fumiaki Takahashi of the Case Western Reserve University (CWRU) served as the Deputy Project Scientist. For ACME's spaceflight operations, the NASA science team included Drs. Uday G. Hegde and Vedha Nayagam who are also with CWRU.

The ISS operations for Flame Design were conducted in two rounds of testing which occurred in April-July 2019 and Oct. 2021 to Feb. 2022. The two rounds occurred during Increments 59/60 and 66 and respectively featured normal and inverse flames. The inverse flame tests were conducted with the chamber filled with a mixture of nitrogen and ethylene (C_2H_4) while an oxygen/nitrogen mixture flowed from the burner. As a result, the inverse flames were effectively inside out, e.g., where the soot formed outside rather than inside of the flames. As a notable exception, the first day of testing in the second round was carried out in the normal flame configuration, with propane (with or without nitrogen) flowing into an oxygen/nitrogen atmosphere. The tests on that day were conducted with the long (near) thermocouple rake to acquire data to calibrate optical measurements of the flame temperature through Thin Filament Pyrometry (TFP). Meanwhile, the short (far) thermocouple rake was used on all other Flame Design test days.

Except for the single propane (C_3H_8) test day previously mentioned, all ISS tests were conducted with the fuel ethylene (C_2H_4), the inert nitrogen, and the spherical flame configuration. While other conditions had been identified as desired, NASA's intent was focused rather comprehensive research.

2. Science Overview

A. Background

This background is the Executive Summary from the experiment's Science Requirements Document (SRD) of April 2010.

The majority of practical combustion processes involve nonpremixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken flames, leading to performance penalties and possible extinction. The proposed study considers a novel method of approaching combustion to address

these concerns. The approach is termed *flame design*, as it takes a fundamental understanding of diffusion flames and, within the inherent constraints imposed, *designs* flames that best optimize efficiency and minimize pollutants.

Flame design involves affecting the basic structure of the flame in that the relationship between the temperature and concentration fields is varied to accomplish a stated objective. The method involves exchanging nitrogen (or any inert, including carbon dioxide) from the oxidizer stream to the fuel stream such that the stoichiometric mixture fraction, *Zst*, can be varied with or without affecting adiabatic flame temperature, *Tad*. The structure of a high *Zst* flame is substantially different from that of the standard hydrocarbon/air flame, where *Zst* is relatively low. By varying *Zst* at constant *Tad*, flames have been produced that are soot-free at any strain rate (permanently blue) and have extinction scalar dissipation rates up to 40 times those of their fuel-air counterparts.

The nitrogen exchange approach (or some variant of it) has been the subject of many studies by the PI and Co-Is (Du and Axelbaum, 1995, 1996, Chao *et al.*, 1998, Sunderland *et al.*, 2003, 2004, Liu *et al.*, 2005, Chen and Axelbaum, 2005, Kumfer *et al.*, 2006, 2008, Santa *et al.*, 2007a, 2007b, Skeen *et al.*, 2009) and other investigators (Sung *et al.*, 1995; Lin and Faeth, 1996; Kang *et al.*, 1997; Hwang and Chung, 2001). Although these studies have been informative, they typically involved normal-gravity flames, where buoyancy and inherent strain complicated the interpretation of results. For example, these ambiguities have led to different theories to explain the phenomenon of permanently-blue diffusion flames. It is not known whether this phenomenon is due more to a change in the direction of convection across the flame or to a change in the detailed structure of the flame. This issue can be addressed by performing microgravity experiments because the unique character of nonbuoyant microgravity flames allows the effects of convection direction and flame structure to be independently varied. By ejecting fuel into an oxidizer in a buoyancy free system, the convective flow across the flame is unambiguously from fuel to oxidizer. Alternatively, by ejecting oxidizer into fuel, the flow is from oxidizer to fuel. Furthermore, the long residence times that can be obtained in the spherical flame geometry will allow for validation of a theory that predicts the sooting limits at the limit of infinite Damköhler number, *Da*.

Studies of the effect of *Zst* on scalar dissipation rate at extinction have indicated that there is a unique *Zst* where the flame is most resistant to extinction. These findings also suggest that by employing inert exchange to vary *Zst*, the peak temperature can be shifted to coincide with the location of radical production, thus allowing the flame to burn at much lower temperatures. The optimum minimum temperature at extinction has led us to propose the concept of a pseudo-flammability limit for nonpremixed flames. Again, this fundamental information is most unambiguously demonstrated in quasi-steady, microgravity flames wherein the effects of flame structure can be systematically studied.

The proposed work will also allow, for the first time, a strain-free experimental study of the effects of *Zst* on soot inception and flame extinction processes. Microgravity experiments will involve gaseous hydrocarbon diffusion flames supported on porous spherical burners and coflow burners.[1] Soot inception limits and flame extinction limits will be obtained as functions of *Zst*, amount of inert, flow direction, and pressure. Extensive analytical and computational modeling

will support the experimental investigation. Models will include gas-phase and burner-surface radiation.

Specifically, soot-inception limits in spherical flames will be measured by establishing a flame at a given *Zst* and allowing peak temperature to decrease with time until the yellow luminosity is eliminated. The effect of flow direction across the flame will be studied by performing coflow experiments (desired) where the fuel issues into the oxidizer and oxidizer issues into fuel. Gas temperatures will be measured by thin film pyrometry and thermocouples to identify the soot-inception temperature.

In addition, soot volume fraction measurements will be made to evaluate the effects of *Zst* and flow direction on soot formation and growth, and to better understand the phenomena of permanently-blue flames. Flame extinction limits will also be studied in a manner similar to that of soot inception limits, and temperatures near extinction will be measured.

The experiments and modeling will yield unique data that will allow for an understanding of how *Zst* affects soot-inception and flame extinction and how *Zst* can be used to design optimum flames. In addition, the effects of flow direction will also be studied and the importance of flow direction and flame structure to permanently-blue flames will be assessed. Finally, the proposed concept of a pseudo-flammability limit for nonpremixed flames will be evaluated and these limits will be ascertained for the test fuels. These results will be useful for the development of future combustion devices that will take advantage of oxygen enrichment technology to *design* more stable and less polluting flames.

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[1] The coflow flame experiments are *desired*. This means that many critical goals of the experiments can be obtained without these experiments but they would add great value to the results.

B. Rationale for Microgravity

This is the 'Justification for Extended Duration Microgravity' section (2.5) of the experiment's SRD from April 2010. References and the appendices are not included in this document but can be found in the SRD. The included equation is formatted differently in this copy.

Previous ground-based microgravity experiments have demonstrated the value of the spherical, burner-supported nonpremixed flames for flame studies. The one-dimensional flame offers a simple, fundamental configuration to improve our understanding of nonpremixed flame behavior and to explore such effects as radiation and thermophoresis, which are more pronounced under microgravity conditions (Atreya *et al.*, 1992, Law *et al.*, 2001). The burner-supported spherical flame is in some respects similar to droplet burning but it holds the advantages that larger flames are possible and, with sufficiently long time, can approach quasi-steady state. Furthermore, the proposed microgravity coflow flames allow the direction of convection across the flame to be varied.

Although the spherically-symmetric flame geometry has many advantages, it is not practical to produce such a flame at normal gravity. The reason for this can be understood by considering the Richardson number, *Ri*, which is the ratio of the buoyancy force to the inertial force and is given by

$$\mathrm{Ri} = g\,(\Delta T\,L) / (T\,u^2)\,. \qquad (2.1)$$

Here g is the acceleration of gravity, T is temperature, L is the characteristic length scale, and u burner surface, and $\Delta T = (Tad - Tb)$, we find that Ri is on the order of 1000 in normal gravity. In fact, it is much higher than this because the more appropriate length and velocity scales are the flame diameter and the velocity at the flame, respectively. The value of Ri can be reduced by judicious choice of fuels and inert; Law *et al.* (2001) have shown that nearly spherical flames can be created in normal gravity. However, Ri for these flames is still on the order of unity, implying that while the flames appear spherical, natural convection is still present. Furthermore, hydrogen must be used as a fuel and be supplied in the ambient so such flames are not relevant to this study. A reduction in Ri can be achieved by reducing the flame size, but substantial reduction in L is not practical because our goal is to produce a flame that is sufficiently large that its structure can be characterized. Furthermore, heat loss to the burner would be excessive for small flames. Consequently, for general fuels and inerts the only practical method of obtaining $\mathrm{Ri} << 1$ is to reduce g by four orders of magnitude. This can be accomplished in a number of microgravity environments, including ground-based facilities, e.g., drop towers for short times, as well as flight experiments, e.g., the International Space Station for longer times. A quantitative evaluation of the acceptable g levels for these experiments is given in Appendix A and is discussed below.

The drop-tower results detailed in Section 1.3 have demonstrated the value of the spherically-symmetric burner stabilized flame. The ability to reverse flow direction has helped to elucidate the dominant mechanism responsible for permanently-blue flames, yet the results are not conclusive because transient effects could dominate these observations. Transient effects arise from two sources: the ignition process and the long times required for spherical microgravity flames to reach quasi-steady state. The ignition process is not spherically symmetric and thus there is an asymmetry imposed by ignition that must be damped out. More importantly, soot particles that are created during ignition can be trapped in the interior of the flame due to thermophoresis. The particles are trapped for the same reasons that a soot shell is produced in microgravity droplet combustion (Choi *et al.*, 1991, Avedisian and Callahan, 2000). Thus, it is possible for soot to exist in flames after the ignition event even though the flame has ceased producing soot.

The reason the spherically-symmetric burner-stabilized flame takes so long to reach quasi-steady state is that the radial velocity scales with $1/r^2$. Thus, at only a few burner diameters away from the sphere, diffusion is the dominant mechanism of transport. Since the characteristic diffusion time τ_d is given by r^2/D, τ_d is on the order of tens of seconds a few centimeters from the burner. Decreasing the flame size will decrease the size of the thermal field and, consequently, the time to reach quasi-steady state, but there are limits to the minimum flame size that is possible. These limits are imposed by the need to avoid heat loss to the burner and ensure flame sphericity. Experiments performed in the 5 s drop facility indicate that for many of the run conditions needed for the proposed experiments the flame is still growing significantly after 5 s. Analytical (see Appendix A) and numerical calculations predict that up to 30 s of jitter-free (less than 10^{-5} g) microgravity is needed to approach quasi-steady state for these run conditions. No ground-based facilities are available to accomplish this.

In some ways, the spherical flame experiments are the nonpremixed analog to premixed flame balls because in both systems diffusion dominates the development of the far field temperature distribution. Therefore, this work is subject to the same needs as SOFBALL (Ronney *et al.*, 1998)

in that long duration microgravity is needed and g-jitter must be negligible because the thermal field is very large and convection is small, i.e., Ri is large even for small *g*.

To provide a quantitative assessment of the acceptable gravity level, an analysis was performed, as detailed in Appendix A. In this analysis, the acceleration induced on a fluid packet by natural convection is formulated by considering a gravitational acceleration given by a sinusoidal function. The resulting acceleration function for the fluid packet is next integrated to obtain the gravity induced flow velocity. This velocity can be converted to an additional mass flux that transports the flame front and causes distortion of the flame. By specifying the maximum percentage of distortion allowed in the experiment (10% here), the acceptable gravity level is determined, and this result is given in Fig. 2.1 for the four flames of Table 1.1. As shown, the required *g* levels are on the order of 10 μg. This is too for microgravity aircraft, but is consistent with the anticipated environment of the Combustion Integrated Rack (CIR). Finally, it is important to emphasize that the proposed experiments illustrate an important and novel use of microgravity for combustion research. In microgravity we are able to hold *Zst* fixed and in coflow tests vary the direction of convection through the flame. There are no analogous experiments at 1g that can accomplish this.

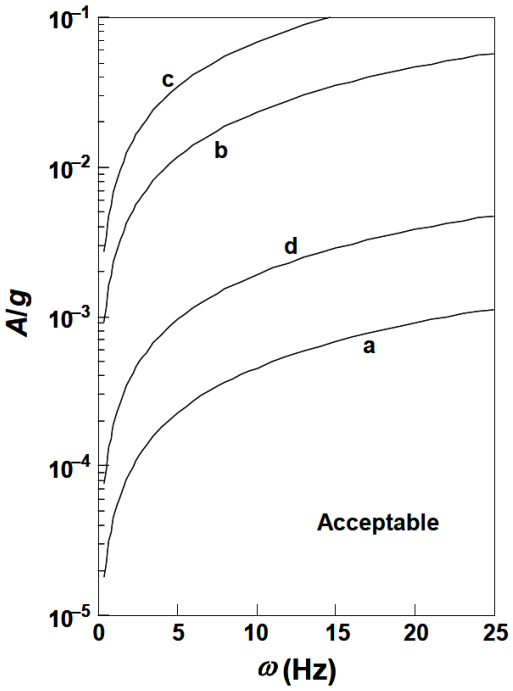


Figure 2.1. Allowable g-levels as a function of frequency for the four flames (a) – (d) described in Table 1.1. A/g is the acceleration nondimensionalized with respect to normal gravity.

Experiments have been performed in normal gravity coflow flames to understand the effects of flame structure and flow direction on soot formation (Kumfer *et al*., 2006, 2008, Skeen *et al.*, 2009). As mentioned previously, owing to the effects of buoyancy on the flow field, the flow direction cannot be prescribed in normal gravity coflow flames as it can in microgravity spherical flames or microgravity coflow flames. For example, it is well known that for fuel issuing into an ambient of air, buoyancy accelerates the flow field and causes the streamlines to be from oxidizer to fuel in the lower regions of the flame and from fuel to oxidizer in the upper regions (Santoro *et al.*, 1987). The importance of buoyancy to the flow field will vary with *Zst* because the velocity and thus Ri are strongly affected by dilution. In other words, the relative amount of entrainment will vary with *Zst* in buoyant flames. Thus, normal gravity flames have ambiguities with respect to flow direction as *Zst* is varied. Nonetheless, coflow flames have been valuable in confirming the role of *Zst* on flame structure and the effects of residence time on soot inception (Kumfer *et al.*, 2006, 2008, Skeen *et al.*, 2009).

Microgravity gas jet flames in quiescent surroundings have been observed in drop towers and the Space Shuttle and the results have revealed significant differences between the two in terms of

sooting behavior, suggesting that the drop tower results were far from steady state (Urban *et al.*, 1998). Velocities in coflow flames are similar to those in gas jet flames in quiescent environments. Therefore the impact of buoyancy on coflow flames is expected to have a similar impact to that reported in Urban *et al.* (1998). Thus, drop tower coflow experiments are not a viable option for studying quasi-steady microgravity coflow flames. The characteristic times in these flames are long and thus the scalar fields develop much slower that in normal gravity, indicating that long duration microgravity is needed.

C. Science Objectives

The proposed objectives text presented here is the 'Experimental Objectives' section (2.3) of the experiment's SRD from April 2010. Desired ISS testing with the fuel methane (CH4), inert carbon dioxide (CO_2), and coflow flame configuration was not conducted.

Thus motivated, the primary objectives of this flight experiment are listed here in priority order:

A. Obtain soot-inception limits of quasi-steady normal and inverse spherical diffusion flames as functions of flow rate, *Zst*, amount of inert (N_2 or CO_2**), and pressure** for C_2H_4 and CH_4 flames. Identify the corresponding temperatures in hot regions. Obtain similar limits and temperatures for inverse and normal coflow flames**.

B. Obtain detailed measurements in quasi-steady normal and inverse spherical diffusion flames for C_2H_4 and CH_4*. Determine the effects of flow rate, *Zst*, amount of inert (N_2 or CO_2**), and pressure** on flame temperature, size, color, and soot volume fraction. Evaluate the possible existence of steady flames. Obtain similar measurements for inverse and normal coflow flames**.

C. Obtain extinction limits as functions of *Zst*, amount of inert (N_2 or CO_2**), and pressure** for normal and inverse spherical C_2H_4 and CH4 flames. Identify the corresponding temperatures in hot regions. Identify the presence of radiative or kinetic extinction. Evaluate whether pseudo-flammability limits can be obtained for these flames. Measure blowoff limits for inverse and normal coflow flames**.

*desired; **highly desired.

D. Science Success Criteria

This text is from the Flame Design portion of the 'Success Criteria' section of the ACME Integrated Science Requirements Document (ISRD, ACME-SRD-001.B) from Sept. 2013, which is revision B of the version prepared for the Requirements Definition Review (RDR). Desired ISS testing with the fuel methane (CH4), inert carbon dioxide (CO_2), and coflow flame configuration was not conducted. And while the chamber pressure was varied in ISS testing, tests were not conducted at pressures as low as 0.2 atm (specified below under 6.3.3).

6.3.1. Minimum Success

- Obtain color video of one normal and one inverse spherical flame that passes its sooting limit, passes its radiative extinction limit, and has a total burn time that exceeds 20 s.
- Obtain color video of three spherical flames of C_2H_4 flowing into air as a functions of time and *m*.

6.3.2. Complete Success (i.e., completion of required test matrix)

All of the above plus:

- Perform tests NS1 – 32 and IS1 – 32 or a similar set of spherical flame tests.

- Obtain soot inception limits and extinction limits for normal and inverse spherical flames as functions of *Zst* and *Tad*.
- Measure peak temperatures as a function of time and *m* for normal and inverse spherical flames.
- Measure temperature distributions at (or near) the sooting and extinction limits of normal and inverse spherical flames.
- Measure soot volume fraction profiles for normal and inverse spherical diffusion flames at low and high *Zst*.

6.3.3. Superior Success (desired)

All of the above plus:

- Obtain color video, soot inception limits, and extinction limits of normal and inverse coflow flames at quasi-steady conditions as a function of *m*, *Zst* and *Tad*.
- Obtain soot inception limits and extinction limits with CO_2 diluent as functions of *Zst* and *Tad* for normal and inverse spherical and coflow flames.
- Obtain soot inception limits and extinction limits for C_2H_4 as functions of *Zst* and *Tad* for normal and inverse spherical and coflow flames at pressures of 0.2 and 0.5 atm.
- Obtain soot inception limits and extinction limits for CH_4 as functions of *Zst* and *Tad* for normal and inverse spherical and coflow flames.

3. Hardware Overview

A. General Description

Except for limited updates and clarifications in the boxed notes, this section is copied from section 2.A of the April 2021 version of 'ACME Engineering Implementation' which should be consulted for additional details.

This section provides a general description of the ACME engineering from the '3.0 Experimental' section of the following 2014 conference paper:

> *Dennis P. Stocker, Fumiaki Takahashi, J. Mark Hickman, and Andrew C. Suttles, "Gaseous Non-Premixed Flame Research Planned for the International Space Station," Spring Technical Meeting of the Central States Section of the Combustion Institute, Tulsa, OK, 16–18 March 2014.*

IMPORTANT: *The 2104 paper, from which this section was extracted, proceeded completion of the spaceflight hardware which was mostly launched in 2017. As a result, it does not accurately describe the final engineering in some regards. Therefore updates, clarifications, additional details, shortcomings, references, etc. have been included in boxed notes like this following the original description. As a minor note, the sizes of figures were adjusted in this report, but none were cropped.*

As previously discussed, the ACME experiments will be conducted within the Combustion Integrated Rack (CIR) using a chamber insert, depicted in Figure 2, which was designed to accommodate gas delivery to a variety of exchangeable burners. Besides the chamber insert, other ACME-unique hardware includes a high-definition color camera and an avionics package, which is for data acquisition and control.

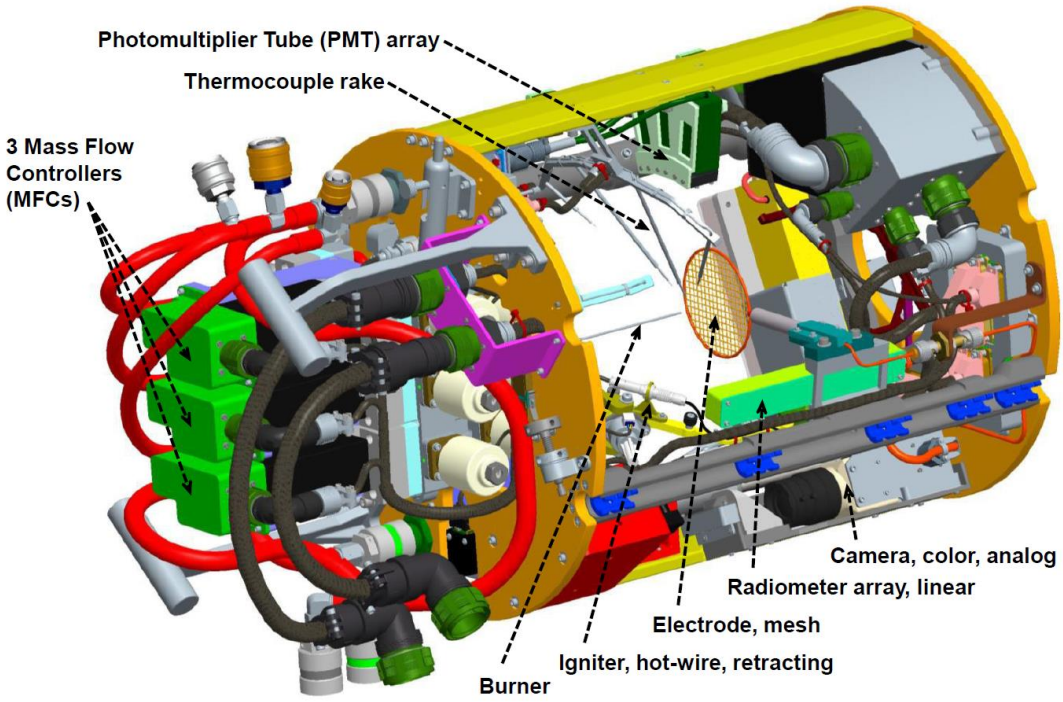


Figure 2: Design of the Advanced Combustion via Microgravity Experiments (ACME) chamber insert that will be used in the Combustion Integrated Rack (CIR) on ISS.

ACME's chamber insert, which is ~0.4 m in diameter, provides common mounting for a variety of burners on the chamber axis (but only one at a time) and includes a retracting resistive igniter. As depicted in Figure 3, there will initially be gas-jet burners (with inner diameters ranging from 0.4 to 3.2 mm), a small coflow burner (where the two concentric tubes have inner diameters of 2.1 and 25 mm), porous spherical burners (with diameters ranging from 6.35 to 12.7 mm), and two flat porous burners for the BRE experiment (with inner diameters of 25 and 50 mm). Most burners will be equipped with a surface thermocouple, and each BRE burner will have two heat flux transducers. The internal pressure of the porous burners will be measured to allow monitoring of the pressure differential across the outlet surface. The igniter will have a crew-exchangeable tip and arm (to accommodate the various burners) and its stepper-motor drive will allow for precise positioning.



Figure 3: ACME burner types, including from left to right, gas-jet burners, a co-flow burner, porous spherical burners, and flat porous burners for the BRE experiment.

Ultimately, the spherical burners were limited to a single diameter of 6.35 mm, and the fabricated gas-jet burners were limited to two sizes with nominal inner diameters of 1.3 and 2.1 mm.

However, only the larger gas-jet burner was used during the spaceflight tests. The burners are addressed in detail in section 3.C of the ACME Engineering Implementation report.

Unfortunately, the spherical burners did not always yield spherically symmetric flames, presumably because of (1) greater burner flow through the joint between the porous sphere and supply tube, and (2) heat loss to the support tube (encompassing the supply tube) causing local flame quenching.

Two types of igniter tips have been used during ACME testing. Most tests have been conducted using a loop igniter tip like those which had been successfully used in the droplet combustion experiments which preceded ACME research in the CIR. However, some tests were conducted with a coil igniter tip, where that included the CFI-G investigation and the second-round testing for Flame Design, Burning Rate Emulator (BRE), and Structure and Response of Spherical Diffusion Flames (s-Flame), with the exception of the latter's first day testing. For more information about the igniter tips, see section 3.D of the ACME Engineering Implementation report.

The chamber insert includes three mass flow controllers (visible on the left of Fig. 2) for the delivery of fuel, oxidizer or inert (e.g., helium or carbon dioxide), and nitrogen to the burner. The maximum fuel flow is 2 slpm (on a nitrogen basis), whereas much higher flows, e.g., 10 slpm (on a nitrogen basis), are possible for the oxidizer and nitrogen lines. The crew can exchange the mass flow controllers to provide appropriate ranges that maximize the flow accuracy. The gas delivery system enables on-orbit dilution of the fuel or oxidizer (but not both simultaneously), coflow, inverse flames (e.g., oxidizer delivery to the center of a coflow burner), and premixed flames, although the latter are not planned for any of the current ACME experiments. Gases can also be delivered premixed to the space station where that is necessary in some circumstances, such as the blending of different fuel gases. CIR's standard bottle sizes are 1, 2.25, and 3.8 liters but ACME will sometimes use smaller bottles as a hazard control to limit the deficient reactant (e.g., fuel) that can be released into the combustion chamber.

ACME tests were not conducted with carbon dioxide as a diluent. Furthermore, the fuel diluted with helium (for s-Flame) was premixed prior to launch rather than on orbit. Ultimately, oxidizer flows of up to 20 slpm (on a nitrogen basis) were possible, while the nitrogen flow was limited to a maximum of 5 slpm. ACME did not use gas bottles smaller than 1 liter as indicated above; the bottle fill pressure was instead limited as an additional hazard control.

ACME's primary flame diagnostics are a suite of imaging systems including five video cameras and two illumination systems, of which three cameras are part of the CIR facility and are already in use on orbit. All but one of the cameras is mounted outside of the chamber. Collectively, the imaging systems will enable color, CH*, and OH* imaging; pyrometry; measurement of the soot volume fraction; and operational support. However, it is unlikely that all functions would be exercised at any one time as will become apparent in the subsequent discussion.

Color imaging is provided by two cameras: a high-definition digital camera mounted outside of the chamber and a standard analog camera mounted on the chamber insert (with a turning mirror). Both cameras are part of the ACME hardware that is now in development. The analog camera signal will be downlinked in near real time to support the experiment operations. Light-emitting diodes within the chamber can be used to illuminate the igniter and burner to both

support the operations and provide reference images. The 12-bit digital camera records at up to 30 full frames (1360x1024) per second, and is equipped with a motorized lens with independent control of the zoom, iris, and focus. The digital camera will be equipped with a filter assembly with four options, where selection of a clear "filter" allows for color imaging. Color imaging can also be carried out with a blue-green BG7 Schott glass filter, which balances the color planes to avoid saturation in the red while maintaining sensitivity in the blue and green. The natural flame color can be approximately reconstructed, but the primary purpose of the filter is pyrometry and measurement of the soot volume fraction, again via the techniques of [23-24]. Alternately, CH* imaging can be conducted with this camera because the filter assembly includes a 430-nm bandpass filter with a 10-nm full width at half-maximum transmission (FWHM). A 450-nm bandpass filter (again with a 10-nm FWHM) can be used with the CH* filter to effectively subtract interfering broadband soot emission. However, the filter assembly rotates slowly so such subtraction effectively requires a separate test with each filter and of course reproducible flame conditions.

The video from the analog operations (i.e., ops) camera, also referred to as the internal camera, was not archived in space. The video was instead recorded on the ground in the operations center. Its gain was sometimes adjusted for improved imaging of dim flames, e.g., approaching extinction.

The sensitivity of the color digital camera, often referenced as the ACME DataCam (i.e., Data Camera), to the dim microgravity flames was such that it was often used with long exposure times and thus low framing rates, for example on the order of 1 frame per second. But as an alternate, it was sometimes instead used with high gain values.

The clear "filter" is of fused silica. Unfortunately, the thickness of the blue-green BG7 Schott glass filter and the added antireflective (AR) coating caused too great a reduction in the red, so it does not balance the intensity of the three color planes as intended and thereby can limit the accuracy of temperature determination. Meanwhile, rather than the 450-nm bandpass filter described above, the high-definition color camera was equipped with a bandpass filter for imaging formaldehyde emissions. For more information, see the Data Camera Characterization Report (ACME-RPT-010 Rev. A).

Pyrometry can alternately be accomplished with CIR's monochrome High-Bit Depth Multispectral (HiBMs) camera, equipped with a Liquid-Crystal Tunable Filter (LCTF), which allows for imaging at multiple spectra, but requires a quasi-steady flame for measurement at each desired band. This camera has a telecentric optical system and is recorded at 512x512 and 30 frames per second (fps). Although it has a 1024x1024 pixel array, binning has been necessary because of signal loss through the LCTF. Both soot pyrometry and Thin Filament Pyrometry (TFP) are possible, where ACME is equipped with a crew-replaceable array of 5 silicon carbide (SiC) fibers. The fibers are orthogonal to the flame axis and the array can be translated along the flame axis to the desired position downstream of the burner, e.g., based on the downlink of the analog video. During set up, the crew can manually orient the array so that it is orthogonal to either ACME's color digital camera or the HiBMs camera with the LCTF. An alternate fiber array including a platinum wire will be used for inflight characterization, which in combination with preflight calibration will enable determination of the soot volume fraction [24].

In contrast to the preceding paragraph, ACME did not use a HiBMs camera for pyrometry, but instead relied on use of ACME's color DataCam for such temperature measurements. For an example of how this can be done, see the techniques described in the following papers:

P.B. Kuhn, B. Ma, B.C. Connelly, M.D. Smooke, and M.B. Long, "Soot and Thin-filament Pyrometry Using a Color Digital Camera," Proceedings of the Combustion Institute, 33 (2011) 743-750.

B. Ma, M. B. Long, "Absolute light calibration using S-type thermocouples," Proceedings of the Combustion Institute, 34 (2) (2013) 3531-3539.

TFP tests were conducted with 14-micron fibers spaced apart by 5 mm for a total span of 20 mm (in the case of 5 intact ceramic fibers), neglecting optional translation of the array. Tests were conducted using arrays with and without the platinum wire in the furthest downstream position. Tests were also sometimes conducted with some of the fibers broken and subsequently removed, leaving only three fibers in the final testing for the BRE and Coflow Laminar Diffusion Flame (CLD Flame) experiments and for all CFI-G testing.

CIR has a second HiBMs camera, without a filter, that will be used in combination with CIR's illumination package for alternate measurements of the soot volume fraction. This will be accomplished via an extinction technique previously employed in microgravity [3,6,26], where the camera and the collimated light source (with its two fiber-coupled laser diodes) are positioned at opposite windows. The images are recorded at 1024x1024 and 30 fps. Thus far, the primary purpose of this combined system has been the measurement of the droplet diameter for determination of the droplet burning rate.

A HiBMs camera without the Liquid Crystal Tunable Filter (LCTF) was used throughout ACME's operations. In the first round of the CLD Flame experiment, it was used without a filter for flame imaging, but this proved to be largely unsuccessful because the camera is very insensitive to blue. Through the remainder of ACME testing, it was equipped with a red filter corresponding to the CIR illumination package for the rest of the ACME testing. Unfortunately, that filter is sufficiently broad that significant incandescence can pass through the filter limiting its utility for determinations of soot volume fraction via light extinction, e.g., through the technique described in the following papers:

D.L. Urban, Z.G. Yuan, P.B. Sunderland, K.C., Lin, Z. Dai, and G.M. Faeth, "Smoke-Point Properties of Nonbuoyant Round Laminar Jet Diffusion Flames," Proceedings of the Twenty-Eighth Symposium (International) on Combustion 28 (2000) 1965-1972.

F.J. Diez, C. Aalburg, P.B. Sunderland, D.L. Urban, Z.-G. Yuan, G.M. Faeth, "Soot properties of laminar jet diffusion flames in microgravity," Combustion and Flame, 156 (2009) 1514-1524.

P.S. Greenberg and J.C. Ku, "Soot Volume Fraction Imaging," Applied Optics, 36 (22) (1997) 5514-5522.

Nonetheless, soot volume fraction can also be determined from the color DataCam images.

During ACME operations, the HiBMs camera was typically archived at 15 fps because of data limits for transfer and downlink operations. And because operations were constrained by data limits, image data from this camera were sometimes deleted without downlinking to enable additional testing. And in some testing, HiBMs images weren't even acquired given their lesser value.

Unfortunately, the HiBMs backlight illumination is significantly obstructed by the bracket for the electrode mesh further limiting its utility for the E-FIELD Flames experiment. This was a design

trade-off necessitated primarily by ACME's large number of cameras and other optical instruments.

CIR's Low-Light-Level Ultraviolet (LLL-UV) package is a monochrome camera with an intensifier (for 280-700 nm) and a 310-nm bandpass filter (with a 10-nm FWHM) for OH* imaging. Again as a result of the signal loss through the filter, binning is employed such that the OH* images are recorded at 512x512 and 30 fps.

During ACME operations, the LLL-UV camera was only used for the Structure and Response of Spherical Diffusion Flames (s-Flame) experiment.

While not discussed in the preceding section, another intensified camera was used throughout ACME testing, although its image data was sometimes deleted or not even acquired to allow the performance of additional tests. This was the Cool Flames Investigation (CFI) camera which was generally configured with its CH filter, where exceptions included CFI-G and s-Flame testing with particularly dim flames. Due to design trade-offs, its Field of View (FOV) was partially obstructed by the arms of the Thin Filament Pyrometry (TFP) fiber array when it was moved inwards to the flame. More information about the CFI intensified camera can be found in its characterization report, CIR-RPT-4600.*

The ACME chamber insert will also be equipped with non-imaging optical diagnostics including both thermopile radiometers and Photomultiplier Tubes (PMTs). There is a crew-exchangeable linear array of radiometers of which there are two versions supporting the initial set of experiments. The array has a fixed position which is parallel to the chamber axis and centered (along the chamber length) relative to the chamber windows and thus most of the other instrumentation. One array is equipped with a single wide-view radiometer with two narrow-view radiometers that are downstream and two more that are upstream, where all views are orthogonal to the chamber axis, as depicted in Figure 4. The narrow-view radiometers are intended to support studies of gas-jet flames, where ACME can accommodate shorter burners, making the "lower" detectors useful. The other array, also seen in Fig. 4, provides for three wide-view measurements of the flame, where one view is masked to nominally block radiant emission from the burner. It was specifically designed for the BRE experiment but will be a resource available to other ACME experiments. Given that the radiometer array can be exchanged by the crew, alternate arrays could be developed.

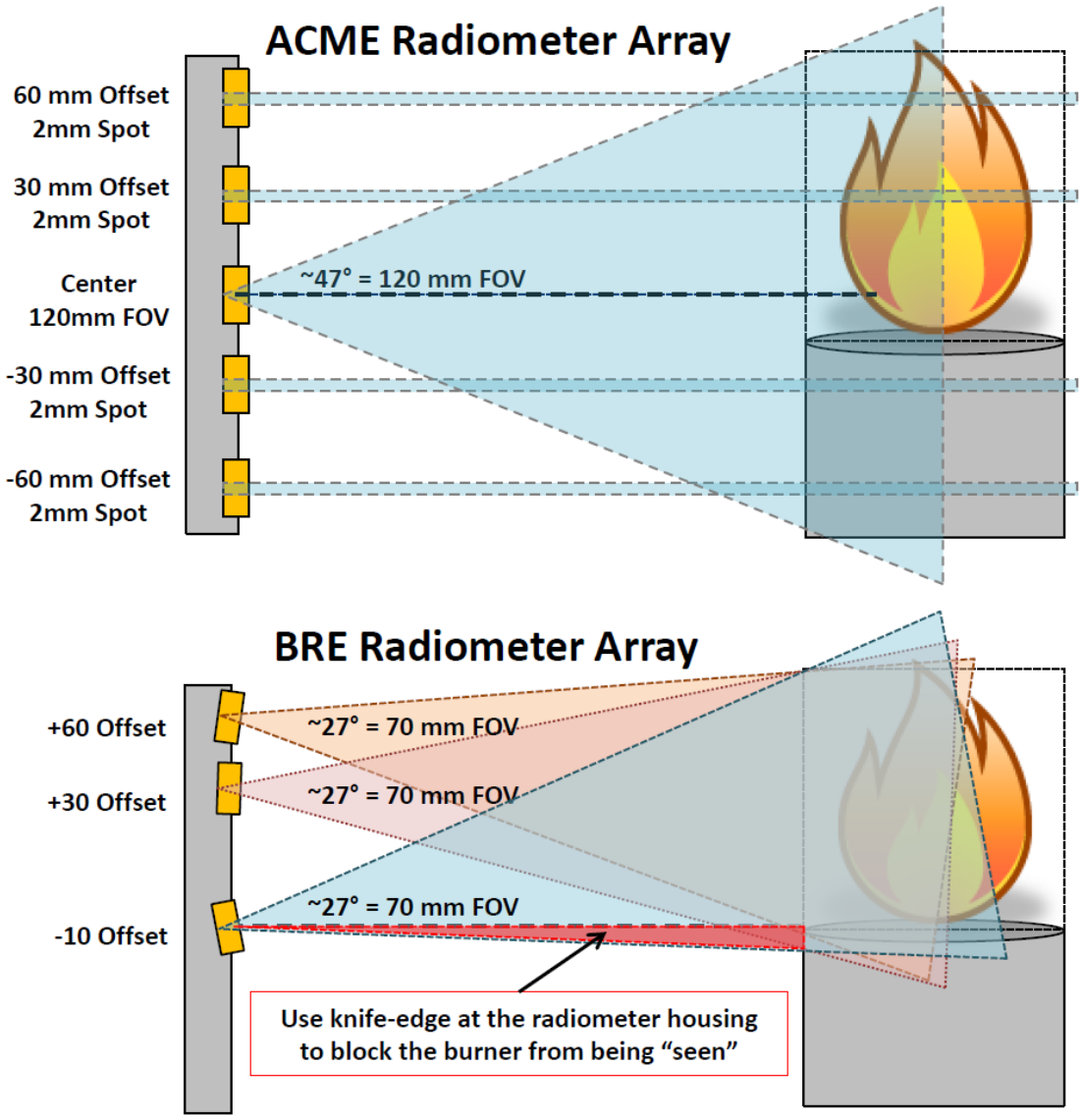


Figure 4: Crew-exchangeable radiometer arrays on the ACME chamber insert: (top) standard (ACME) array of 5 radiometers, and (bottom) array with three radiometers, designed for the BRE experiment, where the lowest detector is masked to avoid view of the burner.

A third radiometer array was used in the CFI-G experiment and the second round of the Flame Design experiment, where it is important to understand that only a single radiometer array can be used at a time. Unlike ACME's other radiometer arrays, each sensor in the CFI-G array has two fixed gains, i.e., a high gain and low gain. It is described in the Cool Flames Radiometer Characterization Report (CIR-RPT-4600).

The narrow-view radiometers in the standard array are only fully sensitive in small spots, but they are each sensitive in a much greater Field of View (FOV) than represented in the preceding figure. They do not have lenses; their FOV is a result of an aperture where most of their FOV is within the penumbra which is only visible to a portion of the sensor. Figures representing the measured FOVs

of the as-built standard and BRE radiometer arrays are found in section 3.F of the ACME Engineering Implementation report.

Unfortunately, the radiometers in the BRE array typically saturated in the first round of BRE testing, so the standard array was used in its second round. However, those radiometers similarly saturated in many of round 2 tests. In this regard, each of ACME's BRE and standard radiometers have a fixed gain.

Radiometers in the standard ACME array regrettably saturated in most of Flame Design's first round of testing so it was replaced with the CFI-G array in the second round.

For more information about the standard radiometer array used for most ACME testing, see the Radiometer Characterization Report (ACME-RPT-011). Meanwhile, you can learn about the array used in the first round of BRE testing from the BRE Radiometer Characterization Report (ACME-RPT-012).

An additional wide-view radiometer, that is fixed and not crew exchangeable, is positioned downstream of the burner and flame at 30° from the chamber axis. This detector is primarily intended to support the spherical flame experiments and will have a quicker response than the more sensitive detectors that will used in the linear arrays. Like the radiometers on the two linear arrays, it will have a spectral range of nominally 0.2-11 micron and its field of view will be established by an appropriately selected aperture.

The FOV of the off-set radiometer is also shown in section 3.F of the ACME Engineering Implementation report. Learn more about this sensor from the Radiometer Characterization Report (ACME-RPT-011).

To detect the quenching of dim flames, the chamber insert will be equipped with a set of three PMTs. Each PMT will have a wide-angle view, a sight line that is orthogonal to the chamber axis, and will be centered (along the chamber length) relative to the chamber windows so that it is aligned with most of the other instrumentation. The PMTs' spectral range will be 230-700 nm, i.e., visible and near ultraviolet, but one will have a 310-nm bandpass filter and another will have a 430-nm bandpass filter (where both are with a 10-nm FWHM). The pair of detectors will enable measurement of OH* and CH* emission, respectively. The third PMT will be equipped with a neutral density filter to prevent saturation. Like many other components, the PMT array can be removed from the chamber insert by the crew.

Unlike the radiometers, each PMT had a readily adjustable gain. However, the PMT signals sometimes saturated when the gains were set too high. See the PMT Characterization Report (ACME-RPT-013) to learn more about the three Photomultiplier Tube (PMTs).

The chamber insert is also equipped with six far-field thermocouples to measure the spread of the thermal field in the spherical flame experiments. They are of limited utility for other experiments because they are positioned upstream of the burner tip. Therefore, they will generally not be installed by the crew to reduce the chance of accidental damage. There will be two versions of a four-thermocouple rake to allow a choice of near and far measurements. Other

arrays could be designed to support future experiments, but they cannot extend downstream of the burner tip without interfering with other (e.g., optical) measurements.

The two lone thermocouples, which are both at nominally 200 mm from the center of the spherical burner, were installed throughout ACME testing. The tip of one was seen to be slightly bent early in testing, as discussed and depicted in section 3.G of the ACME Engineering Implementation report.

The short (or far) version of the thermocouple rake was used throughout ACME testing, except for the final day of CFI-G testing and first round-two day of Flame Design testing, which both featured normal flames, even though second rounds of each study were otherwise with inverse flames. Factors which led to the limited use of the short thermocouple rake included the need for crew time to switch between rakes and the potential adverse effects, such as heat loss and quenching, of the long (close) rake. The short (far) rake's thermocouple tips should nominally be at 45, 60, 100, and 100 mm from the spherical burner center. Meanwhile, the thermocouples of the long (near) rake should nominally be at 13, 25, 50, and 100 mm from the spherical burner center.

For thermocouples at the same distance from the burner center, the angle between the rays from the burner center to the two thermocouples should be 90 degrees. The bead size of the thermocouples was not measured, but it is undoubtedly at least twice the size of the 0.2 mm diameter wire.

The two thermocouples closest to the burner outlets on each four-thermocouple rake are type B, while the others are type K.

Experiment-unique hardware is also mounted on the chamber insert. For example, to accommodate the E-FIELD Flames experiment, the insert is equipped with a high-voltage supply and a crew-removable electrode mesh which can be set between +10 and -10 kV. The circular mesh is orthogonal to the chamber axis and is positioned (during set up) at either ~35 or ~50 mm downstream of the insulated burner. The ion current between the burner and electrode mesh will be determined via measurement of the voltage drop across a known resistor and application of Ohm's Law.

Additional images of the chamber insert can be seen in the ACME Engineering Implementation report, where the hardware specific to E-FIELD Flames is discussed further in section 2.B of that report.

Additional supporting measurements will be made, where the chamber pressure is a simple example. The ACME experiments will use the CIR's gas chromatograph for pre- and post-test analysis of the chamber atmosphere, e.g., for oxygen, nitrogen, carbon dioxide, carbon monoxide, and fuel gases. Acceleration measurements will be provided by the Space Acceleration Measurement System (SAMS) [27], which is the primary instrument for such measurements on ISS and previously served that role for the space shuttle.

The CIR's Gas Chromatograph Instrumentation Package (GCIP) was not used for the CFI-G, CLD Flame, or E-FIELD Flames investigations but was sometimes used during BRE, Flame Design, and s-Flame testing. Its use required significant time and as such effectively reduced the number of

tests that could be conducted. However, estimates of the chamber contents were computed assuming that the fuel in a normal flame test (for example) was either fully or not at all oxidized, e.g., for unsustained ignition in the latter case.

The CIR facility was equipped with a 3-axis SAMS sensor head and the acceleration measurements, along with relevant documentation, are accessible via the home page for the Principal Investigator Microgravity Services (PIMS), https://gipoc.grc.nasa.gov/wp/pims/home/. To be clear, the acceleration measurements are not archived with the ACME experiment data in the Physical Sciences Informatics (PSI) website, www.nasa.gov/PSI.

All ACME experiments will be constant volume (i.e., rather than constant pressure) studies, where the chamber's free volume will be about 90 liters. Burner flows will generally be constrained to limit the pressure rise, and tests will typically be 0.5-2 minutes in duration.

The chamber's free volume is approximately 83 liters depending on the installed hardware.

The ACME experiments will be configured, but not conducted, by the ISS crew. Instead, the experiments will be commanded from the ground, specifically from the Telescience Support Center (TSC) at the NASA Glenn Research Center in Cleveland, Ohio. They can be conducted in a nominally automated mode using pre-programmed scripts. However, it will also be possible to effectively operate the experiments in a "manual" mode, where changes can be made in response to the analog video downlink and the numerical data available in the telemetry stream. For example, a flow rate could be adjusted to approach a stability or soot limit. Step and ramp changes can be made in the flow rate(s) or electric field potential. The TFP fiber array can be translated uniformly through a soot-free flame to get a continuum measurement of the temperature field. In summary, ACME's experiment control will allow for broad flexibility.

The TSC is now also referred to as the Glenn ISS Payload Operations Center (GIPOC). ACME tests were always conducted in a nominally automated mode.

B. Hardware Shortcomings

From the Flame Design PI, Rich Axelbaum:

> "The only significant disappointment with the project was with the diagnostics, particularly with respect to gas temperature and soot volume fraction. They were not at the level that the team would have hoped. Much of this was due to the fact that the diagnostics were outdated by the time they were used in our experiments on the ISS, and thus the diagnostics and calibrations were not at the level that we expect in a laboratory experiment. Of course, the ISS has constraints that limit the ability to incorporate state-of-the-art diagnostics, and this likely made this weakness unavoidable, but it would have been valuable to have had more flexibility in terms of being able to upgrade the diagnostics with more modern cameras and instrumentation."

ACME's sintered spherical burners did not always yield spherically symmetric flames which are centered on the burner's center, where that combination is important for true spherical symmetry. While nominally spherical at moderate flow rates, the flame's center became increasing offset from the burner center as the flow rate was increased. It is very likely that the sintered attachment of the sphere to the supply tube is more porous than the sintered sphere

itself because the flame is seen to be shifted toward the supply tube. There was significant improvement when the starting burner (serial number 1002) was replaced mid-way through round 1 with an alternate burner (serial number 1003), which continued to be used for spherical flame testing throughout the remainder of ACME operations. Nonetheless, the symmetry was not perfect, which is a shortcoming of the Flame Design data.

On 27 Dec. 2021 (GMT 361), it was determined that the Thin Filament Pyrometry (TFP) fiber array was not aligned to the ACME Data Camera (DataCam) as intended, i.e., so that the fibers would be orthogonal relative to the camera. Unfortunately, this was the fourth day of testing in which the fiber array wasn't aligned, where it regrettably wasn't recognized earlier. Given the importance of the TFP-derived measurements of the flame temperature, testing was halted until the array was aligned (with other crew tasks) by astronaut Raja Chari on 17 Jan. 2022 (GMT 017).

The signal from the standard ACME radiometer array used in round 1 of Flame Design testing was saturated for nearly every test, yielding non-useful data. For round 2, that sensor array was replaced with the one designed for Cool Flames Investigation with Gases (CFI-G) which yielded satisfactory measurements.

ACME's Mass Flow Controllers (MFCs) used to control the flow of fuel, nitrogen, and oxidizer did not always perform adequately, where the flow rates were not always appropriately steady or at the intended value. However, the resulting effects on the flame behavior are tentatively assumed to be minor.

In addition to sections 3.A-B, shortcomings are discussed in Appendices B and C of this report and in the *ACME Engineering Implementation* document. The latter document includes discussion of deviations and waivers (in section 4.B), etc.

4. Preliminary Results

A. Changed Test Plans

A change was made to the CIR combustion chamber's pressure limit on 17 Jan. 2022 (by astronaut Raja Chari) to enable Flame Design testing at chamber pressures of up to 3 atm in the final round-2 test days, whereas the limit had been 1.5 atm in preceding testing. In his request for this change, Flame Design PI Rich Axelbaum explained:

> "One of the fundamental goals of Flame Design is to achieve sooting limits. We are unsure if we will be able to attain a range of sooting flames with the current ambient fuel constraints. One way to increase soot formation, and thus the range of the test matrix, is via higher pressures, so this capability can help to ensure we meet our science goals. Another fundamental goal of Flame Design is to understand flame extinction. Our recent analysis suggests that pressure is a critical variable when predicting flame extinction, and in can influence flame chemistry in ways that we had not anticipated. Obtaining extinction data over a wide range of pressures could also be valuable for spacecraft fire safety."

An additional normal-flames test day was added to the start of the second round of testing (even though that was focused on inverse flames) to acquire data useful for calibrating the Thin Filament Pyrometry (TFP) measurements. With that goal, the added test day, conducted on 28 Oct. 2021

(GMT 301), was carried out with the long (near) thermocouple rake, where all other Flame Design tests were conducted with the short (far) rake. In his request for this additional test day, the PI stated in part:

> "The ability to validate the TFP is very important for most of the ACME experiments. Thus, measuring the near-field temperatures under conditions where we have TFP data would be highly desirable, almost required. … These results would be the only measurements ACME has with a direct comparison with TFP, and since we already know exactly which flames to run to get good data, we know that we will be successful in obtaining this data and validating the TFP data. … Reiterating, this extra day would directly benefit all of the temperature data of Flame Design, and all of the spherical flame ACME experiments, by providing at least one direct validation of the TFP method. As you know, we have uncertainties about the accuracy of the TFP technique we are using in the ACME rig, and having these T/C measurements could have a profound impact on the ACME results."

As discussed in the previous section, the decision was made to use the CFI-G radiometer array in Flame Design's second round of testing rather than the standard ACME array that often saturated in the earlier testing.

Coiled igniter tips were also used in the second round of testing instead of the loop igniter tips used in the first round, where the coiled were believed to be more robust.

Other changes to the test plans were minor in comparison.

B. Overview of Experiment Test Operations

While not yet prepared, additional details will be made available in the ACME/Flame Design Experimental Procedure document.

The narrative description of the experimental procedure is an excerpt from the 'Methods / Experimental' section of the following conference paper.

> *Y.-C. Chien, J. Tinajero, D. Stocker, U. Hegde, and D. Dunn-Rankin, "Electric Field Effects on Flames in Microgravity on the International Space Station," 2018 Spring Technical Meeting of the Central States Section of The Combustion Institute, Minneapolis, MN, 20-22 May 2018.*

While the paper was written for the Electric-Field Effects on Laminar Diffusion Flames (E-FIELD Flames) experiment, much of the discussion applies to ACME in general, including the Flame Design investigation. The figure referenced in this excerpt is not included herein. Clarifications and additional information, not in the conference paper, are included in boxed notes like this.

The CIR experiments are configured, but not conducted, by the ISS crew. Thus far, ACME's activity has included the majority of the crew members, regardless of their nationalities. Most often the astronauts and cosmonauts exchange gas bottles, mass flow controllers, and igniter tips. In the latter case, the igniter tip exchange can occur after failure or proactively before failure when the crew already has changes to make changes in the interior of the insert. While these activities may seem mundane, the crew play a critical role in setting up the hardware for each experiment. As an extreme example, there were crew activities on 8 separate days in Sept.-Oct. 2017 to configure the CIR and ACME hardware for the first ACME experiment, namely the Coflow Laminar Diffusion Flame (CLD Flame) experiment. In contrast, only two crew days were required to reconfigure the

hardware for the first E-FIELD Flames tests because of the initial use of the same coflow burner. Planning is critical because crew activity is scheduled three weeks in advance, for which detailed procedures must be prepared and uplinked to the ISS.

As a correction, including two maintenance activities, there were 11 crew days in Sept.-Oct. 2017 to prepare for the initial CLD Flame testing on the space station.

All CIR tests are remotely commanded from the ground from NASA's Glenn ISS Payload Operations Center (GIPOC) in Cleveland, Ohio. Typically, four to six people staff four consoles during the testing. The CIR hardware is controlled from the Rack Officer (RO) and Data Management Officer (DMO) consoles, and the ACME hardware is controlled from the Payload Developer (PD) console. Two shifts of personnel are scheduled to staff these three consoles on test days because they typically last 14 hours with the many pre- and post-testing activities. Given that the tests are remotely commanded, the operations are carried out in the day and evening in Cleveland, e.g., from 7 AM to 9 PM. A day's tests, i.e., with flames, are normally conducted in a period of 10 hours of less. Tests are normally conducted on three weekdays per two weeks. On intervening weekdays, the image data is downlinked and the crew often swap one or both gas bottles. Transferring and downlinking data from a day's tests also requires a 2-shift day for the RO and DMO consoles, where it is always necessary to verify that data was downlinked without errors and that data is not missing. Those two positions are also staffed to guide and oversee the crew activity. As a result, the CIR consoles are normally actively staffed on four weekdays per week. A team of four carry out the detailed planning for the CIR and ACME console staff, and the daily operational plans typically exceed 50 pages on days including testing.

The science console allows for review of the analog video and non-imaging telemetry (at 1 Hz) in near real time but doesn't allow for commanding. In the case of E-FIELD Flames, arrangements were made so that the analog video is streamed to a secure computer at UC Irvine to allow the research team to participate in test operations and to recommend adjustments to the day's planned parameters. That team does not see the telemetry during the testing, but is in communication via phone and e-mail with their NASA research partners at the GRC ISS Payload Operations Center (GIPOC). While there are many preparatory steps, the tests are ultimately conducted using an uploaded software script and parameter files. In advance of the test day, the UC Irvine science team plans the nearly 60 parameters in the voltage sweep script and specifies other parameters that are configured outside of the script, such as camera settings. It is also possible to effectively conduct tests in a 'manual' mode, where parameters are changed in response to the analog video downlink or telemetry. For example, the electric field potential can be adjusted to approach a stability or soot limit.

Provisions were made to enable the investigators from all ACME experiments to remotely view the analog video while they communicated verbally with NASA science personnel in the operations center. All ACME flame tests, including those for Flame Design, were conducted in a nominally automated manner using software scripts and extensive parameter files.

Typically, ten E-FIELD Flames tests are conducted per test day. Despite the short test durations, the low rate results from several unavoidable factors. After a test is conducted, the 125-Hz non-imaging data is downlinked for immediate review by the science team. A small sample of images are often downlinked to ensure that the camera exposures are appropriate, where the times for

those images are typically selected based on non-imaging data to provide extreme examples. The review helps to verify parameters or plan for adjustments in subsequent tests, e.g., to select the voltage range after determining where the ion current saturates for the given flame conditions. After a test is conducted, a fan circulates the chamber atmosphere after which the chamber is vented (to space) back to atmospheric pressure, if needed. Midway through a day's E-FIELD Flames tests, the atmosphere is circulated for 15 minutes through a filter cartridge filled with molecular sieves to capture any soot and adsorb at least a portion of the combustion products, e.g., carbon dioxide, water vapor, and incomplete combustion products. The tests are only conducted when both Ku and S band communication is available through NASA's Tracking and Data Relay Satellite (TDRS) System. Depending on the various orbits, there are periodic communication outages with a loss of signal (LOS) that can be a short handover (from one satellite to another) or can last as long as 30 minutes.

An average of 16 tests were conducted per Flame Design test day, where ignition was typically unsuccessful (e.g., un-sustained) in a couple of tests.

In contrast to E-FIELD Flames, recirculation during Flame Design inverse-flame test operations was only conducted prior to venting at the end of a test day. During normal-flame testing, it was usually conducted once in the midst of testing, as done with E-FIELD Flames. Exceptions were test days 19115, 19123, and 19137 (the first three days with flames); 19161 (when no flames were ignited); 19206 (the last test day in the first round of testing), and 21301 (the first day of the second round of testing). Another exception was on 19150 when it was carried out during the midst of testing and then again when the combustion chamber was vented and refilled for additional tests on the same day.

A variety of constraints limit the number of tests that can be performed, both overall and on a daily basis. The next combustion experiments in the CIR will feature solid fuels and focus on flame spread and material flammability. That follow-on plan limits the duration of ACME operations, where each ACME experiment is allotted four months for testing. Although it requires additional crew time, these allotments are split into two rounds so that the investigators have a break from testing for careful review of the early results so that they can maximize the scientific outcome of their second round. This paper reports on early results from the first round of E-FIELD Flames.

While nitrogen is available from the space station, the combustion testing is limited by the available gas bottles that were and are launched to support the tests. For E-FIELD Flames and most of ACME's experiments, the test matrices used to plan those gas resources were established in 2010. At that time, NASA limited the scope of the ACME tests matrices with the intent of focused experiments with limited objectives rather than comprehensive experiments. As such, the general target was 50 tests per experiment. For most ACME experiments, the limiting gas constraint is the oxygen/nitrogen bottles used to fill the chamber. This resource constraint is exacerbated by the CIR because it is equipped with a pair of redundant safety timers to limit the fuel that can flow into the combustion chamber (where the timers can be set for the oxidizer flow in the case of inverse flame testing). The timers can only be reset by venting the combustion chamber. Therefore, the timer limit combined with other constraints effectively corresponds to the day's maximum flow time for the fuel (or more generally, the limiting reactant). For E-FIELD Flames, the safety timers are set to the maximum value of 999 seconds, which corresponds to the 10 tests typically completed per day.

The maximum limit of 999 seconds for the valve safety timers was used for all ACME testing.

Another constraint is the available time in a day, where the console personnel and ISS resources, such as the video downlink, are planned for a limited number of hours. When necessary for the research, NASA does have continuous 24-hour/day operations for some non-CIR experiments. But as discussed, the ACME tests are typically two minutes or less in duration, so the testing is almost always limited to the planned durations.

Data limits are yet another constraint, where the on-orbit hardware can store a limited amount of data. More importantly, 127,000 MB is the most that can be reasonably managed in a two-shift transfer and downlink day. Therefore, tests days are limited to this amount of data, where image data from select cameras is often archived at reduced rates to mitigate this challenge. Nonetheless, the limit sometimes forces decisions to abandon recorded images.

This detailed description of operations and hardware provides the context under which the ACME experiments are conducted. …

Table 1 is a summary log of the flames ignited for Flame Design during the two rounds of testing, including a limited overview of the test conditions. They were copied from this report's Appendix C. 30-Day Operational Accomplishments Report for ISS Increment 66 which comprehensively reflects all Flame Design testing on the ISS.

Table 1.

Flame Design log of flames ignited in Increments 59/60 and 66							
Test day[1]	**Year**	**Month**	**Date**	**Spherical burner[2]**	**Burner reactant bottle**	**Atmosphere concentration[3]**	**Flames ignited[4]**
First round of tests, featuring normal flames, in April to July 2019							
19115	2019	4	25	serial number 1002	100% C2H4	21% O2	11
19123		5	3			21% O2	19
19137			17			21% O2	20
19142			22			40% O2	8
19150			30			30%, 40% O2	15
19156		6	5			40% O2	13
19161			10			40% O2	0
19171			20	serial number 1003		40% O2	12
19175			24			40% O2	14
19182		7	1			40% O2	15
19189			8			40% O2	15
19196			15			40% O2	13
19200			19			30%, 40% O2	12
19206			25			21% O2	21
Second round of tests, mostly featuring inverse flames, in Oct. 2021 to Feb. 2022							
21301	2021	10	28	serial number 1003	100% C3H8	40% O2	18
21328		11	24		85% O2	27%, 8% C2H4	9
21340		12	6			17%, 12% C2H4	13
21349			15			65%, 27.5% C2H4	15
21361			27			20% C2H4	20
22018	2022	1	18			17%, 10% C2H4	12
22024			24			19% C2H4	20
22027			27			10% C2H4	16
22035		2	4			17%, 10% C2H4	15

[1]**Test day** includes the year as the first two digits followed by the 3-digit day of the year.
[2]**Spherical burner** refers to the specific 0.25-inch diameter spherical burner used in the tests. The porous sintered burners varied in quality, where this was especially evident at high burner flow rates.
[3]**Atmosphere concentration** (as diluted with nitrogen on a volume basis) was varied between tests, where approximate concentrations - neglecting the vitiation from preceding tests - are listed for the tests that ignited. *The chamber pressure was also sometime varied but that secondary parameter is not included in this summary table.*
[4]**Flames ignited** is the number ignited on that day (neglecting the day change on ISS).

C. Summary of Preliminary Results

Abstract for *Critical temperature and reactant mass flux for radiative extinction of ethylene microgravity spherical diffusion flames at 1 bar* (2023):

In microgravity combustion, where buoyancy is not present to accelerate the flow field and strain the flame, radiative extinction is of fundamental importance, and has implications for spacecraft fire safety. In this work, the critical point for radiative extinction is identified for normal and inverse ethylene spherical diffusion flames via atmospheric pressure experiments conducted aboard the International Space Station, as well as with a transient numerical model. The fuel is ethylene with nitrogen diluent, and the oxidizer is an oxygen/nitrogen mixture. The burner is a porous stainless-steel sphere. All experiments are conducted at constant reactant flow rate. For normal flames, the ambient oxygen mole fraction was varied from 0.2 to 0.38, burner supply fuel mole fraction from 0.13 to 1, total mass flow rate, *mtotal* , from 0.6 to 12.2 mg/s, and adiabatic flame temperature, *Tad* , from 2000 to 2800 K. For inverse flames, the ambient fuel mole fraction was varied from 0.08 to 0.12, burner supply oxygen mole fraction from 0.4 to 0.85, *mtotal* from 2.3 to 11.3 mg/s, and *Tad* from 2080 to 2590 K. Despite this broad range of conditions, all flames extinguish at a critical extinction temperature of 1130 K, and a fuel-based mass flux of 0.2 g/m^2 -s for normal flames, and an oxygen- based mass flux of 0.68 g/m^2-s for inverse flames. With this information, a simple equation is developed to estimate the flame size (i.e., location of peak temperature) at extinction for any atmospheric-pressure ethylene spherical diffusion flame given only the reactant mass flow rate. Flame growth, which ultimately leads to radiative extinction if the critical extinction point is reached, is attributed to the natural development of the diffusion-limited system as it approaches steady state and the reduction in the transport properties as the flame temperature drops due to increasing flame radiation with time (radiation-induced growth.)

Abstract for *On the existence of steady state gaseous microgravity spherical diffusion flames in the presence of radiation heat loss* (2023):

Whether steady-state gaseous microgravity spherical diffusion exist in the presence of radiation heat loss is an important fundamental question and has important implications for spacecraft fire safety. In this work, experiments aboard the International Space Station and a transient numerical model are used to investigate the existence of steady-state microgravity spherical diffusion flames. Gaseous spherical diffusion flames stabilized on a porous spherical burner are employed in normal (i.e., fuel flowing into an ambient oxidizer) and inverse (i.e., oxidizer flowing into an ambient fuel) flame configurations. The fuel is ethylene and the oxidizer oxygen, both diluted with nitrogen. The flow rate of the reactant gas from the burner is held constant. It is found that steady-state gaseous microgravity spherical diffusion flames can exist in the presence of radiation heat loss, provided that the steady-state flame size is less than the flame size for radiative extinction, and the flame develops fast enough that radiation heat loss does not drop the flame temperature below the critical temperature for radiative extinction (1130 K). A simple model is provided that allows for the identification of initial conditions that can lead to steady-state spherical diffusion flames. In the spherical, infinite domain configuration, the characteristic time for the diffusion-controlled system to effectively reach steady-state is found to be on the order of 100,000 s. Despite a narrow range of attainable conditions, flames that exhibit steady-state behavior are observed aboard the ISS for up to 870 s, even with the constraint of a finite boundary. Steady-state flames are simulated using the numerical model for over 100,000 s.

Abstract for *Observations of long duration microgravity spherical diffusion flames aboard the international space station* (2021):
Spherical diffusion flames of gaseous fuels supported on a porous spherical burner are investigated experimentally in microgravity for long durations aboard the International Space Station (ISS). Experimental results are compared with a transient numerical model. The experiments involve normal spherical diffusion flames (i.e., fuel flowing into oxidizer) burning ethylene in an oxygen/nitrogen mixture at atmospheric pressure. The stoichiometric mixture fraction and adiabatic flame temperature are varied by diluting the fuel with nitrogen and varying the oxygen concentration in the ambient. The diagnostics consist of color video, thin-filament pyrometry for gas temperature, thermocouple measurements of burner temperature, and photomultiplier tube measurements of broad and narrow-band flame emission. Burn times range from 28 –336 s. All flames grow monotonically after ignition. Peak gas temperature generally decreases with time. Radiative extinction is observed when, at constant reactant flow rate, the flame expands until radiation heat loss approaches the chemical heat release. Before a flame experiences complete extinction, it may enter an oscillatory mode in which part of the flame partially extinguishes and then reforms. The oscillations grow in magnitude until complete extinction. Sufficiently small flames with low radiation losses exhibit quasi-steady behavior, but burner heating is significant, and the experimental duration is dictated by the temperature limit of the porous sphere.

Publications:

- P.H. Irace, K.A. Waddell, Denis Constales., P.B. Sunderland, R.L. Axelbaum, Critical temperature and reactant mass flux for radiative extinction of ethylene microgravity spherical diffusion flames at 1 bar, Proceedings of the Combustion Institute, in press (2023).
- P.H. Irace, K Waddell, D. Constales, G. Yablonsky, M. Kim, P.B. Sunderland, R.L. Axelbaum, On the existence of steady state gaseous microgravity spherical diffusion flames in the presence of radiation heat loss, Proc. Combust. Inst., in press (2023).
- P.H. Irace, H. Lee, K. Waddell, L. Tan, D.P. Stocker, P.B. Sunderland, R.L. Axelbaum, Observations of Long Duration Microgravity Spherical Diffusion Flames Aboard the International Space Station, Combustion and Flame, 229 (2021) 111373, 1-9.

D. Surprises and New Discoveries

- All the ISS flames observed by this project extinguish at a critical extinction temperature of 1130 K, and at a fuel mass flux of 0.2 g/m^2-s for normal flames, and at an oxygen mass flux of 0.68 g/m^2-s for inverse flames.
- Microgravity diffusion flame growth, which ultimately leads to radiative extinction if the critical extinction point is reached, is attributed to the natural development of the diffusion-limited system as it approaches steady state and the reduction in the transport properties as the flame temperature drops due to increasing flame radiation with time (radiation-induced growth).
- Steady-state gaseous microgravity spherical diffusion flames can exist in the presence of radiation heat loss, provided that the steady-state flame size is less than the flame size for radiative extinction, and the flame develops fast enough that radiation heat loss does not drop the flame temperature below the critical temperature for radiative extinction (1130 K).

- In the spherical, infinite domain configuration, the characteristic time for the diffusion-controlled system to effectively reach steady-state is found to be on the order of 10^5 s.
- Radiative extinction is observed when, at constant reactant flow rate, a flame expands until radiation heat loss approaches the chemical heat release. Before a flame experiences complete extinction, it may enter an oscillatory mode in which part of the flame partially extinguishes and then reforms. The oscillations grow in magnitude until complete extinction.

E. Accomplishment of Approved Objectives

All the approved objectives, i.e., not designated as desired or highly desired in Section 2.C, were accomplished.

F. Additional Objectives Achieved

See 4.D, Surprises and New Discoveries, above.

G. Impacts, Benefits, and Products

The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants.

H. Patents

No patents have been filed from this research.

5. Follow-On Experiments

The range of available experimental conditions was limited, including the fuel and oxygen concentrations and the ambient pressure. In particular, for safety reasons, the O_2 mole fraction limit for normal tests was 40% and the C_2H_4 mole fraction limit for inverse flames was 65%. Tests with CH_4 as fuel and CO_2 as diluent, were slated for the test matrix by the investigators, but not allowed. Finally, co-flow burner tests at the conditions considered by this project could be especially valuable.

Appendix A. Abbreviations and Acronyms

#####	ACME days are identified by year (2 digits) and the day of the year (3 digits, 001-366)
#####x#	ACME tests are identified by day (5 digits; see above), a letter, and a 1-digit number
0-g	microgravity
1-g	normal gravity (on the surface of the Earth)
ACME	Advanced Combustion via Microgravity Experiments (*CIR project*)
ADP	Acceptance Data Package
ASGSR	American Society for Gravitational and Space Research
AVP	avionics package
BRE	Burning Rate Emulator (*an ACME experiment*)
C_2H_4	ethylene
C_3H_8	propane
CH_2O	formaldehyde
CASIS	Center for the Advancement of Science in Space
CCD	Charge-Coupled Device (*such as a camera sensor*)
CFI	Cool Flames Investigation (*an MDCA experiment preceding ACME in the CIR*)
CFI-G	Cool Flames Investigation with Gases (*an ACME experiment*)
CH_4	methane
CIA	Chamber Insert Assembly (*inserted into the CIR combustion chamber*)
CIR	Combustion Integrated Rack (*ISS facility in which ACME experiments are conducted*)
CLD Flame	Coflow Laminar Diffusion Flame (*an ACME experiments*)
CM	Combustion Module (*space shuttle precursor to the CIR*)
cm	centimeter
CO_2	carn dioxide
Co-I	Co-Investigator
CRS	Commercial Resupply Service (*a designation of ISS cargo missions)*
CSA	Canadian Space Agency
CSV	comma-separated values
CWRU	Case Western Reserve University
Da	Damköhler number
DataCam	Data Camera (*ACME's primary camera which was characterized for pyrometry, etc.)*
E-FIELD Flames	Electric-Field Effects on Laminar Diffusion Flames (*an ACME experiment*)
EMI	electromagnetic interference
EMS	Experiment Mounting Structure
ERD	Engineering Requirements Document
ESA	European Space Agency
Fiji	Fiji Is Just ImageJ (*a version of ImageJ preloaded with many plug-in features*)
Flame Design	Flame Design (*an ACME experiment*)
FOMA	Fuel and Oxidizer Management Assembly (*CIR's gas management subsystem*)
FOV	field of view
fps	frames per second
GC	Gas Chromatograph (*see GCIP*)
GCIP	Gas Chromatograph Instrumentation Package (*for measuring CIR chamber contents*)
GigE	Gigabit Ethernet (*an interface standard for high performance cameras*)
GIPOC	Glenn ISS Payload Operations Center (*also known as the TSC*)
GMT	Greenwich Mean Time (*where most references in this report are the day*)
H_2	hydrogen

HiBMs	High Bit-depth Multi-spectral camera (*developed for the CIR facility*)
HOBJ	Halcon object (*binary uncompressed format; used by the ACME Data Camera*)
HV	high voltage
HVPS	High-Voltage Power Supply
ICDERS	International Colloquium on the Dynamics of Explosions and Reactive Systems
ID	inner diameter (*e.g., of burner tube*)
ImageJ	Image Processing and Analysis in Java (*open-source image processing software*)
IPSU	Image Processing and Storage Unit (*CIR device for recording images on ISS)*
IPSU-G	IPSU for GigE cameras (*including the DataCam and CFI intensified camera*)
ISRD	Integrated Science Requirements Document (*for a set of experiments*)
ISS	International Space Station
JAXA	Japan Aerospace Exploration Agency
kPa	kilopascal
kV	kilovolt
LLL-UV	Low Light Level Ultraviolet camera ((*developed for the CIR facility*)
LSP	Laminar Soot Processes (*a shuttle study conducted in the Combustion Module*)
MDCA	Multi-user Droplet Combustion Apparatus (*a CIR chamber insert*)
MFC	Mass Flow Controller
Mist	Water-Mist Fire Suppression Experiment (*a Combustion Module experiment*)
MIUL	Materials Identification/Usage List
mm	millimeter
MSG	Microgravity Science Glovebox (*ISS facility in which SLICE was conducted*)
N_2	nitrogen
NSF	National Science Foundation
NTSC	National Television System Committee
O_2	oxygen
OMA2	Optical Multi-channel Analyzer 2 (*open-source software for image processing)*
ops	operations
PARIS	Passive Rack Isolation System (*with which the CIR is equipped, but not used for ACME*)
PI	Principal Investigator
PIMS	Principal Investigator Microgravity Services
PMT	Photomultiplier Tube (*where ACME is equipped with three PMT modules*)
PS	Project Scientist
PSI	Physical Science Informatics System
psia	pounds per square inch
PT	pressure transducer
QD	quick disconnect
qty	quantity
RSA	Russian Space Agency, i.e., Roscosmos
s	second
SAMS	Space Acceleration Measurement System
SDP	Safety Data Package
s-Flame	Structure and Response of Spherical Diffusion Flames (*an ACME experiment*)
SiC	Silicon Carbide (*material of the TFP fibers*)
SLICE	Structure & Liftoff In Combustion Experiment (*CLD Flame precursor*)
slpm	standard liters per minute (*a measure of the mass or molar flow rate of a gas*)
SOFBALL	Structure Of Flame Balls At Low Lewis-number (*a Combustion Module study*)

SoFIE	Solid Fuel Ignition and Extinction (*project to follow ACME in the CIR facility*)
SORD	Science Operations Requirements Document (*an ACME supplement to the ISRD*)
SRD	Science Requirements Document
T/C (or t/c)	Thermocouple
TDRS	Tracking and Data Relay Satellite (System)
TFP	Thin Filament Pyrometry (*a method for gas-phase temperature measurements*)
TIFF or TIF	Tagged Image File Format (e.g., *for the CFI intensified, HiBMs, and LLL-UV cameras*)
TM	thermistor
TS	Transport Stream (*a video format*)
TSC	Telescience Support Center (*also known as the GIPOC*)
UV	ultraviolet
Ze	Zeldovich number
zero-g	zero gravity, i.e., microgravity
ZIN	ZIN Technologies, Inc. (*ACME engineering and operations provider*)
µA	microamp

While this list of abbreviations and acronyms is long, it is not comprehensive. Contact ACME Project Scientist Dennis Stocker (216-433-2166, dennis.p.stocker@nasa.gov) for additional clarification.

Appendix B. 30-Day Operational Accomplishments Report for ISS Increments 59 and 60

17 December 2019

30-day Operational Accomplishments Report

… for the International Space Station (ISS) Increments 59 and 60

ACME Advanced Combustion via Microgravity Experiments Flame Design

Flame Design is one of several independent experiments that are part of the ACME project, which uses a single modular set of hardware to carry out studies of flames burning gaseous fuels in the ISS' Combustion Integrated Rack (CIR). The investigation will be carried out in two rounds of tests, where the first occurred in Increments 59 and 60. It is currently expected that the second round of tests will be conducted during Increment 63.

Prepared by the PI team and Dennis Stocker (dennis.p.stocker@nasa.gov), the ACME Project Scientist, who is the primary contact for E-FIELD Flames and ACME information.

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I. Introduction

The Flame Design experiment is conducted in the Combustion Integrated Rack (CIR) on the International Space Station, as part of the Advanced Combustion via Microgravity Experiments (ACME) project. The primary goal of the Flame Design experiment is to improve the understanding of soot inception and control, in order to enable the optimization of oxygen enriched combustion and the 'design' of non-premixed flames that are both robust and soot free. An outside review panel declared that Flame Design "… could lead to greatly improved burner designs that are efficient and less polluting than current designs."

Flame Design investigates the soot inception and extinction limits of spherical microgravity flames created with porous spherical burners. Tests are conducted with various concentrations of both the fuel (e.g., ethylene) and oxygen in order to determine the role of the flame structure on the soot inception.

More specifically, Flame Design has the following experimental objectives which are listed here in priority order. It must be emphasized that NASA does not currently plan to carry out the desired or highly desired testing with carbon dioxide, pressure as a variable (except to a limited degree), coflow flames, or methane.

A. Obtain soot-inception limits of quasi-steady normal and inverse spherical diffusion flames as functions of flow rate, *Zst*, amount of inert (N2 or CO2**), and pressure** for C2H4 and CH4* flames. Identify the corresponding temperatures in hot regions. Obtain similar limits and temperatures for inverse and normal coflow flames**.
B. Obtain detailed measurements in quasi-steady normal and inverse spherical diffusion flames for C2H4 and CH4*. Determine the effects of flow rate, *Zst*, amount of inert (N2 or CO2**), and pressure** on flame temperature, size, color, and soot volume fraction. Evaluate the possible existence of steady flames. Obtain similar measurements for inverse and normal coflow flames**.
C. Obtain extinction limits as functions of *Zst*, amount of inert (N2 or CO2**), and pressure** for normal and inverse spherical C2H4 and CH4 * flames. Identify the corresponding temperatures in hot regions. Identify the presence of radiative or kinetic extinction. Evaluate whether pseudo-flammability limits can be obtained for these flames. Measure blowoff limits for inverse and normal coflow flames**.

*desired; **highly desired

The majority of practical combustion processes involve the non-premixed combustion of hydrocarbons. These processes must be efficient, and the resulting emissions must be low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing inert concentrations and reducing pressures, but these techniques generally weaken flames, leading to performance penalties and possible extinction.

The Flame Design study considers an innovative method of approaching combustion to address these concerns. The approach is termed *flame design*, as it takes a fundamental understanding of diffusion flames and, within the inherent constraints imposed, *designs* flames that best optimize efficiency and minimize pollutants.

Flame design involves affecting the basic structure of the flame, in that the relationship between the temperature and concentration fields is varied to accomplish the desired objective. This method involves shifting the nitrogen from the air (or oxygen/nitrogen mixture) to the fuel stream, such that the mixing that occurs at the flame can be varied with or without affecting the flame temperature. [This can also be done with inerts other than nitrogen and oxygen/inert

mixtures other than air.] Such nitrogen exchange has enabled the production of the flames that are soot-free at all flow conditions, i.e., which are 'permanently blue.'

The Flame Design experiment involves gaseous hydrocarbon diffusion flames supported on a porous spherical burner. Soot inception limits and flame extinction limits are obtained as functions of the nitrogen exchange, flow direction, and pressure. Extensive analytical and computational modeling support the experimental investigation, where they are facilitated by the spherical flame shape enabled by the microgravity environment. Models include gas-phase and burner-surface radiation.

Buoyant gravity-driven flow complicates the interpretation of the existing results and has led to different theories to explain the permanently-blue diffusion flames. It is not known whether this phenomenon is due more to a change in the direction of convection across the flame, or to a change in the detailed structure of the flame.

The challenges of normal-gravity studies are avoided by performing the tests on the International Space Station (ISS), where the non-buoyant microgravity flames allow the effects of convection direction and flame structure to be independently varied. By ejecting fuel from the burner into an oxidizer (e.g., air) in a buoyancy free system, the convective flow across the flame is unambiguously from fuel to oxidizer. Alternatively, by ejecting oxidizer from the burner into an atmosphere of fuel, the flow is from oxidizer to fuel.

Furthermore, microgravity studies of flames allow for a range of simplifications and provide various advantages over normal-gravity studies. For example, the flicker of flame, which is a buoyancy-driven (i.e., gravity-driven) instability, is eliminated yielding quasi-steady flames in microgravity. Length scales are increased in microgravity flames facilitating analysis of the flame structure. Momentum-dominated flames, which are important for most practical combustion, can be studied at low velocities to simplify analysis. Microgravity flames tend to have a much stronger sensitivity to their atmosphere and exhibit a much broader range of characteristics than normal-gravity flames because of the near absence of buoyant entrainment. The long residence times in microgravity flames can lead to strong soot production, but many microgravity flames are soot free. The absence of buoyancy makes microgravity flames advantageous for studies of limit and stability behavior where chemical kinetics are important. With these advantages, microgravity testing can lead to new understanding of the complex nature of combustion – regardless of the gravitational environment of the combustion application.

Flame Design should yield unique data that allows for a better understanding of how nitrogen exchange affects soot-inception and flame extinction, and how it can be used to design optimum flames. In addition, the effects of flow direction are also studied, and the importance of the flow direction and flame structure to permanently-blue flames is assessed.

II. Methods/Research Operations

A. Discussion of Method/Protocol

Flame Design and other ACME experiments are configured, but not conducted, by the ISS crew. The crew's role is to set up the hardware, gas bottles, etc. for each set of tests. The most common changes include the gas bottles on the Combustion Integrated Rack (CIR) and the mass flow controllers on the modular ACME chamber insert.

All testing is commanded from the ground, specifically from the Glenn ISS Payload Operations Center (GIPOC). The science portion of the testing is typically conducted in a nominally automated mode using pre-programmed scripts. However, it may alternately be conducted in a 'manual' mode, where changes are made to a control parameter in response to the analog video downlink and the numerical data available in the telemetry stream. For example, the flow rate can be adjusted to approach a flame's soot limit. This was not done during Increments 59 and 60, where the console commanding during a test was typically limited to halting the test when it was observed in the analog ops video that the flame had extinguished.

Testing is not conducted when thrusters fire or spacecraft dock or undock from ISS to ensure an acceptable acceleration environment.

Data is downlinked between each set of tests, i.e., day of testing, to allow for preliminary analysis and requisite planning for subsequent testing.

III. Results

A. Baselined Activities Completed – Pre-flight, In-flight, Post-Flight

The first round of Flame Design tests, during which a total of 188 flames were ignited, was conducted during Increments 59-60, more specifically from April-July. These tests featured normal flames, where gaseous fuel was ejected from a porous spherical burner into an oxygen/nitrogen atmosphere. In contrast, the upcoming second round will feature tests with inverse flames, where the oxidizer is ejected from the burner into a fuel/nitrogen atmosphere. As a result, the second round flames will be 'inside out.' It is currently expected that the second round of tests will be conducted during Increment 63. The 'interruption' in the Flame Design testing was intentionally planned years in advance to enable the PI team to review the initial results in detail and with that knowledge plan the second round of tests to maximize its scientific outcome.

Flame Design does not have living subjects like a life science experiment, but the tests are independently conducted with the gaseous fuel ethylene (C2H4). It is a simple hydrocarbon gas for which the chemistry is well understood and it is typically highly sooting because of its carbon double bond. The tests were conducted with both pure fuel

and fuel mixed with nitrogen. In the latter case, the ACME hardware allows for simultaneous flow of fuel (from a launched bottle) and ISS nitrogen to the burner, allowing for on-orbit fuel dilution and testing at a continuum of fuel concentrations.

No samples (e.g., of soot) were collected for Flame Design; all of the ISS results were electronically transmitted to the ground. That is furthermore the case for all currently planned ACME experiments.

A summary log of the flames ignited during Increments 59-60, including a limited overview of the test conditions, is found in Table 1. The table includes a limited overview of the test conditions. For completeness, a row was included for testing on June 10 (GMT 161) even though it was quickly discovered that the hot-wire igniter tip had failed at the end of preceding test day. As a result, flames were not ignited until ten days later after astronaut Christina Koch had replaced the igniter tip.

Table 1.

Flame Design log of flames ignited in Increments 59-60							
Test day[1]	**Year**	**Month**	**Date**	**Spherical burner[2]**	**Burner reactant bottle**	**Atmosphere concentration[3]**	**Flames ignited[4]**
First round of tests, featuring normal flames, in April to July 2019							
19115	2019	4	25	serial number 1002	100% C2H4	21% O2	11
19123		5	3			21% O2	19
19137			17			21% O2	20
19142			22			40% O2	8
19150			30			30%, 40% O2	15
19156		6	5			40% O2	13
19161			10			40% O2	0
19171			20	serial number 1003		40% O2	12
19175			24			40% O2	14
19182		7	1			40% O2	15
19189			8			40% O2	15
19196			15			40% O2	13
19200			19			30%, 40% O2	12
19206			25			21% O2	21

[1]**Test day** includes the year as the first two digits followed by the 3-digit day of the year.
[2]**Spherical burner** refers to the specific 0.25-inch diameter spherical burner used in the tests. The porous sintered burners varied in quality, where this was especially evident at high burner flow rates.
[3]**Atmosphere concentration** (as diluted with nitrogen on a volume basis) was varied between tests, where approximate concentrations - neglecting the vitiation from preceding tests - are listed for the tests that ignited. *The chamber pressure was also sometime varied but that secondary parameter is not included in this summary table.*
[4]**Flames ignited** is the number ignited on that day (neglecting the day change on ISS).

Koch also swapped the porous spherical burner (serial number 1002) along with the igniter tip on June 10 (GMT 161), where it was determined that the initial burner's performance was not satisfactory with high burner flow rates. The flame ceased being spherical and its center didn't coincide with the burner center. The results indicated that the sintered attachment of the sphere to the supply tube was more porous than the sintered sphere itself, where the flame was shifted toward the supply tube. The alternate burner (serial number 1003) showed similar behavior, but to a significantly reduced degree. As a result, some of the early Flame Design testing was repeated after the burner swap.

Pre-flight and post-flight activities are generally not addressed in this report.

B. List of Preflight, In-flight, Postflight Anomalies

1. Mission anomalies

There were no significant anomalies hindering Flame Design operations during Increments 59-60, but there was a pertinent anomaly that had previously occurred. It was discovered on Feb. 4 (GMT 035) that the pressure was low in a check gas bottle for the CIR's Gas Chromatograph Instrumentation Package (GCIP). The bottle had been on the space station for 9 years, and the low pressure was attributed to a nominal leak rate. The check gas was for calibration of the GCIP measurements and not needed for regular operations which proceeded unhindered. The bottle was replaced on June 13 to enable the necessary calibration.

2. Any other difficulties in obtaining scientific data

Flame Design tests sometimes automatically aborted (due to the controlling software script), where that was typically prior to gas flow and the activation of the igniter. As a result, there was little lost other than the time in the failed test attempt. Such aborts normally occur because of communication issues with an Image Processing and Storage Unit (IPSU). These are infrequent, but there are sometimes days when ACME tests have aborted repeatedly.

ACME tests are of course also unsuccessful when the flame fails to ignite. The first test of the day has sometimes failed to ignite, presumably because the fuel line has not been sufficiently primed. Ignition was sometimes unsuccessful because the settings for the automatic flame detection are such that detection threshold is reached because of the igniter glow rather than a flame. As a result of the detection, the hot-wire igniter is deactivated and retracted, which is of course an issue if the flame has yet to ignite. This is typically avoided by detecting ignition with measurement of the flame's OH* chemiluminescence with a Photomultiplier Tube (PMT) module. The OH* chemiluminescence, in the ultraviolet at 310 nm, is a good indicator of an ignited flame rather than the glowing igniter, which emits primarily in the infrared and visible spectrum. However, the combination of igniter settings and PMT gains can lead to false ignition detection.

C. Completeness/Quality of Data

The number of flames ignited in the round 1 testing significantly exceeds the normal-flame tests which were specified as required in the ACME Integrated Science Requirements Document (ISRD). A total of 41 normal-flame tests were identified as required for the Flame Design experiment. In great contrast, 188 flames of this type were ignited in the experiment, so more than 4.5 times as many flames were ignited than required.

However, as discussed in Section III.A, the performance of the first spherical burner was not always satisfactory – where the flame was not always sufficiently spherical or centered on the burner, suggesting extreme asymmetry in the flow field and thus

questionable data. Yet, 102 flames were ignited with the second burner, which is nearly 2.5 times the number of required normal-flame tests.

ACME's sintered spherical burners do not yield spherically symmetric flames which are centered on the burner's center, where that combination is important for true spherical symmetry. This lack of symmetry is especially evident at high burner flows. It is very likely that the sintered attachment of the sphere to the supply tube is more porous than the sintered sphere itself because the flame is seen to be shifted toward the supply tube. There was significant improvement when the starting burner (serial number 1002) was replaced mid-way through round 1 with an alternate burner (serial number 1003). But the symmetry is not perfect and this is a shortcoming of the Flame Design data.

Portions of ACME's recorded image data is sometimes deleted without downlinking. This is done because of operational constraints which limit the amount of data that can be managed per test day. Given the constraint, the decision is sometimes made to sacrifice some image data (e.g., from the HiBMs camera) to allow additional tests to be conducted. This is intentionally done in order to maximize the scientific return of each day of testing within the project constraints.

Data quality is of course dependent on the experimental hardware. But neglecting that limitation on data quality, the cameras are sensitive to radiation where pixels gradually become 'hot' (i.e., dead) over time. While these bright spots are addressed in image post-processing, they are a degradation of the image quality.

ACME's Mass Flow Controllers (MFCs) used to control the flow of fuel and nitrogen have unfortunately not always performed adequately, where the flow rates have not always been appropriately steady. The resulting effects on the flame behavior are tentatively assumed to be minor, but this still needs to be assessed.

The ACME camera is not of sufficient quality to demonstrate the novel aspects of the experiments to the public, but it appears to be satisfactory from a science perspective.

It is possible that the tests are subtly affected by the acceleration environment, but there is no obvious indication of this and this has thus far not been correlated with available measurements from the Space Acceleration Measurement System (SAMS). In this regard, note that the Combustion Integrated Rack (CIR), in which the experiment is conducted, is equipped with a Passive Rack Isolation System (PaRIS).

Regardless of the shortcomings discussed above, a great wealth of experimental data has been secured and Flame Design's first round is considered to be at least minimally successful despite the imperfect symmetry. Depending on the results of the data processing, the round may rate as highly successful.

D. Status of Data Analysis

The raw non-imaging data acquired from round 1 of Flame Design testing (e.g., gas flow rates, burner surface temperature, photomultiplier tube measurements of flame emissions, radiometer measurements of flame thermal emission, far-field gas temperatures, and pressures) has been converted to engineering units . All of the raw (hobj) images obtained from the ACME camera have been successfully converted to tiff files for further analysis. Thin filament pyrometry is being used to obtain fiber temperature measurements in ACME camera images that have SiC fibers inserted in the flame. Soot temperature measurements using two-color ratio pyrometry are being acquired from ACME camera images that have soot and no fibers. Burn time and extinction diameter have been acquired for radiatively extinguishing flames using the photomultiplier tubes and analog camera, respectively. HiBMs camera images are being examined for use in calculating the soot volume fraction. Intensified camera images have yet to be analyzed.

A one-dimensional computational model is being implemented to assist in the investigation of spherical diffusion flames in microgravity. Also, collaborations are underway with scientists at the Russian Academy of Sciences for further simulations. The Russian team is focusing on detailed chemistry and more realistic boundary conditions, while the US team is focusing on a general understanding of the flame dynamics.

E. Preliminary Research Findings

Flame Design results have shown that spherical diffusion flames in microgravity have unique characteristics that are valuable for fundamental studies of diffusion flames. Flame size increases with reactant flow rate and decreases with ambient oxygen concentration. Radiative heat loss scales with flame size such that very small flames - just above their kinetic extinction limits - have low radiative loss, and asymptotically approach steady-state behavior. Larger flames increase in size over time until the rate of radiative heat loss exceeds the rate of chemical heat release, and the flame extinguishes via radiative extinction. Before total extinction occurs in these flames, they experience an unstable, oscillatory mode, in which they partially extinguish and reform. The oscillations grow in magnitude until total extinction. For constant ambient oxygen concentration, larger flames (with higher fuel flow rates) radiatively extinguish faster than smaller flames (with lower flow rates) due to higher radiative losses and decreasing chemical heat release. Small, asymptotically quasi-steady flames are found to have low radiative losses and nearly constant chemical heat release. A thorough analysis of low flow rate flames was completed in order to investigate the existence of steady-state spherical flames, which has been a subject of debate in literature. This analysis is described in the submission to the 38th International Symposium on Combustion cited below.

1. P. Irace, H. Lee, K. Waddell, L. Tan, D. Stocker, D. Urban, P. Sunderland, and R. Axelbaum, “Steady-state burner stabilized spherical diffusion flames in microgravity,” 38th International Symposium on Combustion, Adelaide, Australia, 2020.

The abstract from this submission states:

> This study is an examination of steady-state behavior in spherical diffusion flames in microgravity on the International Space Station (ISS). Gaseous spherical diffusion flames stabilized on a porous sphere were employed in a normal configuration (i.e., fuel issuing into an ambient oxidizer). The use of a porous sphere allowed for a spherical flame with a constant mass flow rate, a condition that is not attainable with free-droplet combustion. Unlike previous microgravity spherical flame studies, time is not a limitation for these experiments, and the potential to achieve steady-state flames can be explored for the first time. Stoichiometric mixture fraction, Z_{st}, and flame temperature were varied by diluting the fuel with nitrogen, or changing the ambient oxygen concentration, either 21 or 40 vol% oxygen. Flame size was found to increase with increasing fuel flow rate and decrease with increasing ambient oxygen concentration. The large spherical diffusion flames were observed to grow until extinction, while small flames were found to exhibit an asymptotically steady behavior due to their low radiative losses and faster transport. Experimental measurements of flame size were in quantitative agreement with numerical results with one-step chemistry, provided the diffusivity was appropriately adjusted. For the small flame sizes that are necessary to achieve steady-state flames, the burner acts as a heat sink, and as the burner temperature increases, its cooling effect on the flame diminishes, and the flame temperature increases. Accordingly, fiber temperatures measured using two-color ratio pyrometry showed increasing temperature with time for the flames examined. Flame emissions of OH* (310 nm) measured with a photomultiplier tube and a bandpass filter showed a steady intensity after the initial ignition transient. The flames investigated appear to be asymptotically approaching a steady-state size and temperature, however, a decisive conclusion cannot be made because of burner heating.

Early Flame Design results from the ISS testing were presented in the following oral presentations:

1. Richard Axelbaum, Peter Sunderland, and David Urban, “ACME - Flame Design,” ISS Increments 61 & 62 Science Symposium, 17-19 Sept. 2019.
2. P. Irace, R.L. Axelbaum, and P.B. Sunderland, “Radiative and Kinetic Extinction of Gaseous Spherical Diffusion Flames in Microgravity,” 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, 20-23 Nov. 2019.

F. Communication Imagery

1. Still Imagery

Photos are available online at www.flickr.com/photos/space-flames. Although the Facebook page at www.facebook.com/space.flames is meant to provide updates on ACME and related combustion research, it is good source for recent photos, e.g., of flames, that may not be available on the Flickr site. Additional

photographs may be requested from the Project Scientist, Dennis Stocker (dennis.p.stocker@nasa.gov).

2. **Video**
   An example video from the Flame Design experiment can be found with the web-based article at:
   https://www.nasa.gov/mission_pages/station/research/news/combustion-research-microgravity-clean-burning-fuel-space-station/.

   'The Power of Studying Combustion on the ISS,' which was developed with input from the ACME Project Scientist, is available at https://science.nasa.gov/science-news/news-articles/the-power-of-studying-combustion-on-the-iss.

   Video(s) describing the ACME project can be found at:
   https://www1.grc.nasa.gov/space/iss-research/iss-fcf/cir/acme/.

   Videos related to Flame Design and other microgravity combustion research can also be seen on Facebook at www.facebook.com/space.flames.

   Contact Dennis Stocker (dennis.p.stocker@nasa.gov) for access to those videos or for additional videos.

IV. Additional Comments from science team

Constantly sooting flames were only observed in 40% oxygen and high fuel concentrations. In 21% oxygen, small flames did not soot because of heat loss to the burner and large flames did not soot because high radiative losses caused the flame temperature to quickly drop below the sooting limit. Preliminary analysis of the HiBMs images suggests that significant emission of light from soot particles and dissymmetry of the soot cloud will make acquiring accurate soot volume fraction measurements difficult, if possible. The cloud of soot particles also shows relatively extreme dissymmetry, further complicating our ability to obtain soot volume fraction measurements.

Appendix C. 30-Day Operational Accomplishments Report for ISS Increment 66

Table I was corrected where the report submitted in July 2022 incorrectly identified the fuel used on 28 Oct. 2021 as 100% ethylene (C2H4) when it was 100% C3H8.

6 July 2022

30-day Operational Accomplishments Report

… for the International Space Station (ISS) Increments 66

ACME Advanced Combustion via Microgravity Experiments Flame Design

Flame Design is one of six independent experiments that are part of the ACME project, which used a single modular set of hardware to carry out studies of flames burning gaseous fuels in the ISS' Combustion Integrated Rack (CIR). The investigation was carried out in two rounds of tests which occurred in Increments 59/60 and 66. This report comprehensively addresses both rounds of Flame Design testing.

Prepared by the PI team and Dennis Stocker (dennis.p.stocker@nasa.gov), the ACME Project Scientist, who is the primary contact for Flame Design and ACME information.

Principal Investigator (PI):	Prof. Richard L. Axelbaum, Washington University in St. Louis, MO
Co-Investigator (Co-I):	Prof. Peter B. Sunderland, University of Maryland, College Park, MD
Co-Investigator (Co-I):	Dr. David L. Urban, NASA Glenn Research Center, Cleveland, OH
Lead collaborator:	Dr. Sergei Frolov, N.N. Semenov Institute of Chemical Physics, Russian Academy of Sciences, Moscow, Russia
NASA Project Scientist (PS):	Dennis P. Stocker, NASA Glenn Research Center, Cleveland, OH
PS team members:	Dr. Uday G. Hegde, Case Western Reserve University, Cleveland, OH
	Dr. Vedha Nayagam, Case Western Reserve University, Cleveland, OH

I. Introduction

The Flame Design experiment is conducted in the Combustion Integrated Rack (CIR) on the International Space Station, as part of the Advanced Combustion via Microgravity Experiments (ACME) project. The primary goal of the Flame Design experiment is to improve the understanding of soot inception and control, in order to enable the optimization of oxygen enriched combustion and the 'design' of non-premixed flames that are both robust and soot free.

An outside review panel declared that Flame Design "… could lead to greatly improved burner designs that are efficient and less polluting than current designs."

Flame Design investigates the soot inception and extinction limits of spherical microgravity flames created with porous spherical burners. Tests are conducted with various concentrations of both the fuel (e.g., ethylene) and oxygen in order to determine the role of the flame structure on the soot inception.

More specifically, Flame Design has the following experimental objectives which are listed here in priority order. It must be emphasized that NASA does not currently plan to carry out the desired or highly desired testing with carbon dioxide, pressure as a variable (except to a limited degree), coflow flames, or methane.

D. Obtain soot-inception limits of quasi-steady normal and inverse spherical diffusion flames as functions of flow rate, *Zst*, amount of inert (N2 or CO2**), and pressure** for C2H4 and CH4* flames. Identify the corresponding temperatures in hot regions. Obtain similar limits and temperatures for inverse and normal coflow flames**.
E. Obtain detailed measurements in quasi-steady normal and inverse spherical diffusion flames for C2H4 and CH4*. Determine the effects of flow rate, *Zst*, amount of inert (N2 or CO2**), and pressure** on flame temperature, size, color, and soot volume fraction. Evaluate the possible existence of steady flames. Obtain similar measurements for inverse and normal coflow flames**.
F. Obtain extinction limits as functions of *Zst*, amount of inert (N2 or CO2**), and pressure** for normal and inverse spherical C2H4 and CH4 * flames. Identify the corresponding temperatures in hot regions. Identify the presence of radiative or kinetic extinction. Evaluate whether pseudo-flammability limits can be obtained for these flames. Measure blowoff limits for inverse and normal coflow flames**.

*desired; **highly desired

The majority of practical combustion processes involve the non-premixed combustion of hydrocarbons. These processes must be efficient, and the resulting emissions must be low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing inert concentrations and reducing pressures, but these techniques generally weaken flames, leading to performance penalties and possible extinction.

The Flame Design study considers an innovative method of approaching combustion to address these concerns. The approach is termed *flame design*, as it takes a fundamental understanding of diffusion flames and, within the inherent constraints imposed, *designs* flames that best optimize efficiency and minimize pollutants.

Flame design involves affecting the basic structure of the flame, in that the relationship between the temperature and concentration fields is varied to accomplish the desired objective. This

method involves shifting the nitrogen from the air (or oxygen/nitrogen mixture) to the fuel stream, such that the mixing that occurs at the flame can be varied with or without affecting the flame temperature. [This can also be done with inerts other than nitrogen and oxygen/inert mixtures other than air.] Such nitrogen exchange has enabled the production of the flames that are soot-free at all flow conditions, i.e., which are 'permanently blue.'

The Flame Design experiment involves gaseous hydrocarbon diffusion flames supported on a porous spherical burner. Soot inception limits and flame extinction limits are obtained as functions of the nitrogen exchange, flow direction, and pressure. Extensive analytical and computational modeling support the experimental investigation, where they are facilitated by the spherical flame shape enabled by the microgravity environment. Models include gas-phase and burner-surface radiation.

Buoyant gravity-driven flow complicates the interpretation of the existing results and has led to different theories to explain the permanently-blue diffusion flames. It is not known whether this phenomenon is due more to a change in the direction of convection across the flame, or to a change in the detailed structure of the flame.

The challenges of normal-gravity studies are avoided by performing the tests on the International Space Station (ISS), where the non-buoyant microgravity flames allow the effects of convection direction and flame structure to be independently varied. By ejecting fuel from the burner into an oxidizer (e.g., air) in a buoyancy free system, the convective flow across the flame is unambiguously from fuel to oxidizer. Alternatively, by ejecting oxidizer from the burner into an atmosphere of fuel, the flow is from oxidizer to fuel.

Furthermore, microgravity studies of flames allow for a range of simplifications and provide various advantages over normal-gravity studies. For example, the flicker of flame, which is a buoyancy-driven (i.e., gravity-driven) instability, is eliminated yielding quasi-steady flames in microgravity. Length scales are increased in microgravity flames facilitating analysis of the flame structure. Momentum-dominated flames, which are important for most practical combustion, can be studied at low velocities to simplify analysis. Microgravity flames tend to have a much stronger sensitivity to their atmosphere and exhibit a much broader range of characteristics than normal-gravity flames because of the near absence of buoyant entrainment. The long residence times in microgravity flames can lead to strong soot production, but many microgravity flames are soot free. The absence of buoyancy makes microgravity flames advantageous for studies of limit and stability behavior where chemical kinetics are important. With these advantages, microgravity testing can lead to new understanding of the complex nature of combustion – regardless of the gravitational environment of the combustion application.

Flame Design should yield unique data that allows for a better understanding of how nitrogen exchange affects soot-inception and flame extinction, and how it can be used to design optimum flames. In addition, the effects of flow direction are also studied, and the importance of the flow direction and flame structure to permanently-blue flames is assessed.

II. Methods/Research Operations

B. Discussion of Method/Protocol

Flame Design and other ACME experiments were configured, but not conducted, by the ISS crew. The crew's role was to set up the hardware, gas bottles, etc. for each set of tests. The most common changes included the gas bottles on the Combustion Integrated Rack (CIR) and the mass flow controllers on the modular ACME chamber insert.

All ACME testing was commanded from the ground, specifically from the Glenn ISS Payload Operations Center (GIPOC). The science portion of ACME testing was conducted in a nominally automated mode using pre-programmed scripts. While it was possible to make some 'manual' changes outside of the script, e.g., to approach a flame's soot limit, during a test that was not done in any ACME testing, neglecting minor exceptions, e.g., to terminate tests after the flames had extinguished.

Testing was not conducted when thrusters fired or spacecraft docked or undocked from ISS to ensure an acceptable acceleration environment.

Data was downlinked between each set of tests, i.e., day of testing, to allow for preliminary analysis and requisite planning for subsequent testing.

III. Results

G. Baselined Activities Completed – Pre-flight, In-flight, Post-Flight

A total of 326 flames were ignited for the Flame Design experiment during Increments 59/50 and 66, where 188 and 138 flames were ignited in the two increments, respectively.

The first-round tests featured normal flames, where gaseous fuel was ejected from a porous spherical burner into an oxygen/nitrogen atmosphere. In contrast, the second round - except for its first test day - was carried out with inverse flames, in which the oxidizer is ejected from the burner into a fuel/nitrogen atmosphere. As a result, the second-round flames were 'inside out.' The 'interruption' in the Flame Design testing was intentionally planned years in advance to enable the PI team to review the initial results in detail and with that knowledge plan the second round of tests to maximize its scientific outcome.

Flame Design did not have living subjects like a life science experiment, where the tests were conducted with the gaseous fuel ethylene (C_2H_4). It is a simple hydrocarbon gas for which the chemistry is well understood and it is typically highly sooting because of its carbon double bond. The tests were conducted with both pure fuel and fuel mixed with nitrogen. In the latter case, the ACME hardware allows for simultaneous flow of fuel

(from a launched bottle) and ISS nitrogen to the burner, allowing for on-orbit fuel dilution and testing at a continuum of fuel concentrations.

No samples (e.g., of soot) were collected for Flame Design; all of the ISS results were electronically transmitted to the ground. That is furthermore the case for all ACME experiments.

A summary log of the flames ignited during Increments 59/60 and 66, including a limited overview of the test conditions, is found in Table 1. The table includes a limited overview of the test conditions. For completeness, a row was included for testing on June 10 (GMT 161) even though it was quickly discovered that the hot-wire igniter tip had failed at the end of preceding test day. As a result, flames were not ignited until ten days later after astronaut Christina Koch had replaced the igniter tip.

Koch also swapped the porous spherical burner (serial number 1002) along with the igniter tip on June 10 (GMT 161), where it was determined that the initial burner's performance was not satisfactory with high burner flow rates. The flame ceased being spherical and its center didn't coincide with the burner center. The results indicated that the sintered attachment of the sphere to the supply tube was more porous than the sintered sphere itself, where the flame was shifted toward the supply tube. The alternate burner (serial number 1003) showed similar behavior, but to a significantly reduced degree. As a result, some of the early Flame Design testing was repeated after the burner swap.

Pre-flight and post-flight activities are generally not addressed in this report.

Table 1.

Flame Design log of flames ignited in Increments 59/60 and 66							
Test day[1]	**Year**	**Month**	**Date**	**Spherical burner[2]**	**Burner reactant bottle**	**Atmosphere concentration[3]**	**Flames ignited[4]**
First round of tests, featuring normal flames, in April to July 2019							
19115	2019	4	25	serial number 1002	100% C2H4	21% O2	11
19123		5	3			21% O2	19
19137			17			21% O2	20
19142			22			40% O2	8
19150			30			30%, 40% O2	15
19156		6	5			40% O2	13
19161			10			40% O2	0
19171			20	serial number 1003		40% O2	12
19175			24			40% O2	14
19182		7	1			40% O2	15
19189			8			40% O2	15
19196			15			40% O2	13
19200			19			30%, 40% O2	12
19206			25			21% O2	21
Second round of tests, mostly featuring inverse flames, in Oct. 2021 to Feb. 2022							
21301	2021	10	28	serial number 1003	100% C3H8	40% O2	18
21328		11	24		85% O2	27%, 8% C2H4	9
21340		12	6			17%, 12% C2H4	13
21349			15			65%, 27.5% C2H4	15
21361			27			20% C2H4	20
22018	2022	1	18			17%, 10% C2H4	12
22024			24			19% C2H4	20
22027			27			10% C2H4	16
22035		2	4			17%, 10% C2H4	15

[1]**Test day** includes the year as the first two digits followed by the 3-digit day of the year.
[2]**Spherical burner** refers to the specific 0.25-inch diameter spherical burner used in the tests. The porous sintered burners varied in quality, where this was especially evident at high burner flow rates.
[3]**Atmosphere concentration** (as diluted with nitrogen on a volume basis) was varied between tests, where approximate concentrations - neglecting the vitiation from preceding tests - are listed for the tests that ignited. *The chamber pressure was also sometime varied but that secondary parameter is not included in this summary table.*
[4]**Flames ignited** is the number ignited on that day (neglecting the day change on ISS).

H. List of Preflight, In-flight, Postflight Anomalies

3. Mission anomalies

There were no significant anomalies hindering Flame Design operations during Increments 59-60, but there was a pertinent anomaly that had previously occurred. It was discovered on 4 Feb. 2019 (GMT 035) that the pressure was low in a check gas bottle for the CIR's Gas Chromatograph Instrumentation Package (GCIP). The bottle had been on the space station for 9 years, and the low pressure was attributed to a nominal leak rate. The check gas was for calibration of the GCIP measurements and not needed for regular operations which proceeded unhindered. The bottle was replaced on June 13 to enable the necessary calibration.

On 27 Dec. 2021 (GMT 361), during Increment 66, it was determined that the Thin Filament Pyrometry (TFP) fiber array was not aligned to the ACME Data Camera (DataCam) as intended, i.e., so that the fibers would be orthogonal relative to the camera. Unfortunately, this was the fourth day of testing in which the fiber array wasn't aligned, where it regrettably wasn't recognized earlier. Given the importance of the TFP-derived measurements of the flame temperature, testing was halted until the array was aligned (with other crew tasks) by Raja Chari on 17 Jan. 2022 (GMT 017).

4. Any other difficulties in obtaining scientific data

Flame Design tests sometimes automatically aborted (due to the controlling software script), where that was typically prior to gas flow and the activation of the igniter. As a result, there was little lost other than the time in the failed test attempt. Such aborts normally occured because of communication issues with an Image Processing and Storage Unit (IPSU). These were infrequent, but there were some days when ACME tests aborted repeatedly.

ACME tests were of course also unsuccessful when the flame fails to ignite. In the first round of Flame Design testing, the igniter tip failed twice requiring crew time to be scheduled to replace it before the subsequent test day could be conducted. Regrettably, it was sometimes discovered during ACME testing that the igniter had failed during the final test of the preceding test day, where an example was on 10 June 2019 (19161) when no flames were ignited as a result (as discussed in section III.A). An igniter check was subsequently added to the ACME end-of-test-day procedure to avoid such recurrences. However, on another test day, it was believed that the igniter tip had failed but ignition was instead prevented because the chamber pressure exceeded the pressure switch limit. Ignition was achieved after the chamber pressure was appropriately reduced.

The first test of the day has also sometimes failed to ignite, presumably because the fuel line has not been sufficiently primed. Ignition was also sometimes unsuccessful because the settings for the automatic flame detection were such that the detection threshold was reached because of the igniter glow rather than a flame. As a result of

the detection, the hot-wire igniter was deactivated and retracted, which was of course an issue when the flame has yet to ignite. This was typically avoided by detecting ignition with measurement of the flame's OH* chemiluminescence with a Photomultiplier Tube (PMT) module. The OH* chemiluminescence, in the ultraviolet at 310 nm, is a good indicator of an ignited flame rather than the glowing igniter, which emits primarily in the infrared and visible spectrum. However, the combination of igniter settings and PMT gains can lead to false ignition detection.

I. Completeness/Quality of Data

The 326 flames ignited in the two rounds of greatly exceeds the 76 tests which were specified as required in the ACME Integrated Science Requirements Document (ISRD). As a result, more than 4.2 times as many flames were ignited than required.

However, as discussed in Section III.A, the performance of the first spherical burner was not always satisfactory – where the flame was not always sufficiently spherical or centered on the burner, suggesting extreme asymmetry in the flow field and thus questionable data. Yet, 240 flames were ignited with the second burner, which is over 3.1 times the number of required tests.

ACME's sintered spherical burners do not yield spherically symmetric flames which are centered on the burner's center, where that combination is important for true spherical symmetry. This lack of symmetry was especially evident at high burner flows. It is very likely that the sintered attachment of the sphere to the supply tube is more porous than the sintered sphere itself because the flame is seen to be shifted toward the supply tube. There was significant improvement when the starting burner (serial number 1002) was replaced mid-way through round 1 with an alternate burner (serial number 1003). But the symmetry is not perfect and this is a shortcoming of the Flame Design data.

Portions of ACME's recorded image data were sometimes deleted without downlinking. This was done because of operational constraints which limit the amount of data that can be managed per test day. Given the constraint, the decision was sometimes made to sacrifice some image data (e.g., from the HiBMs camera) to allow additional tests to be conducted. This was intentionally done in order to maximize the scientific return of each day of testing within the project constraints.

The signal from the ACME radiometer array used in round 1 of Flame Design testing was saturated for nearly every test, yielding non-useful data. For round 2 of FLame Design, the radiometer array was replaced with the radiometer array designed for Cool Flames Investigation with Gases (CFI-G) which yielded satisfactory measurements.

Data quality is of course dependent on the experimental hardware. But neglecting that limitation on data quality, the cameras are sensitive to radiation where pixels gradually

become 'hot' (i.e., dead) over time. While these bright spots are addressed in image post-processing, they are a degradation of the image quality.

ACME's Mass Flow Controllers (MFCs) used to control the flow of fuel, nitrogen, and oxidizer did not always performed adequately, where the flow rates were not always appropriately steady or at the intended value. The resulting effects on the flame behavior are tentatively assumed to be minor, but this still needs to be assessed.

The ACME camera is not of sufficient quality to demonstrate the novel aspects of the experiments to the public, but it appears to have been satisfactory from a science perspective.

It is possible that the tests were subtly affected by the acceleration environment, but there was no obvious indication of such effects. However, the Flame Design data has not been correlated with available measurements from the Space Acceleration Measurement System (SAMS). Although the Combustion Integrated Rack (CIR) is equipped with a Passive Rack Isolation System (PaRIS), it was not used for ACME because of the significant need for crew time that would have resulted.

Regardless of the shortcomings discussed above, a great wealth of experimental data has been secured and Flame Design's first round is considered to be at least minimally successful despite the imperfect symmetry. Depending on the results of the data processing, the round may rate as highly successful.

Similarly, a great wealth of experimental data was secured during Flame Design's second round of testing. Flame Design's second round is considered to be at least minimally successful, and depending on the results of data processing, may be rated as highly successful.

J. Status of Data Analysis

The raw non-imaging data acquired from round 1 of Flame Design testing (e.g., gas flow rates, burner surface temperature, photomultiplier tube measurements of flame emissions, far-field gas temperatures, and pressures) has been converted to engineering units. All the raw (hobj) images obtained from the ACME camera have been successfully converted to tiff files for further analysis. Thin filament pyrometry is being used to obtain fiber temperature measurements in ACME camera images that have SiC fibers inserted in the flame. Soot temperature measurements using two-color ratio pyrometry are being acquired from ACME camera images that have soot and no fibers. Burn time and extinction diameter have been acquired for radiatively extinguishing flames using the photomultiplier tubes and analog camera, respectively. HiBMs camera images are being

examined for use in calculating the soot volume fraction. Intensified camera images have yet to be analyzed.

The raw non-imaging data acquired from round 2 of Flame Design testing (e.g., gas flow rates, burner surface temperature, photomultiplier tube measurements of flame emissions, radiometer measurements of flame thermal emission, far-field gas temperatures, and pressures) has been converted to engineering units. All the raw (hobj) images obtained from the ACME camera have been successfully converted to tiff files for further analysis. Thin filament pyrometry is being used to obtain fiber temperature measurements in ACME camera images that have SiC fibers inserted in the flame. Burn time and extinction diameter have been acquired for radiatively extinguishing flames using the photomultiplier tubes and analog camera, respectively. Intensified camera images have yet to be analyzed.

A one-dimensional computational model is being implemented to assist in the investigation of spherical diffusion flames in microgravity. Also, collaborations are underway with scientists at the Russian Academy of Sciences for further simulations. The Russian team is focusing on the porous spherical burner and more realistic boundary conditions, while the US team is focusing on a general understanding of the flame dynamics.

K. Preliminary Research Findings

Flame Design results have shown that spherical diffusion flames in microgravity have unique characteristics that are valuable for fundamental studies of diffusion flames. Flame size increases with reactant flow rate and decreases with ambient oxygen concentration. Radiative heat loss scales with flame size such that very small flames - just above their kinetic extinction limits - have low radiative loss, and asymptotically approach steady-state behavior. Larger flames increase in size over time until the rate of radiative heat loss becomes sufficiently large, and the flame extinguishes via radiative extinction. Before total extinction occurs in these flames, they experience an unstable, oscillatory mode, in which they partially extinguish and reform. The oscillations grow in magnitude until total extinction. For constant ambient oxygen concentration, larger flames (with higher fuel flow rates) radiatively extinguish faster than smaller flames (with lower flow rates) due to higher radiative losses and decreasing chemical heat release. A thorough analysis of low flow rate flames was completed in order to investigate the existence of steady-state spherical flames, which has been a subject of debate in literature. It was found that steady state microgravity spherical diffusion flames can exist. Furthermore, it was found that ethylene microgravity spherical diffusion flames at 1 bar extinguish at a critical extinction point defined by an extinction flame temperature and an extinction reactant mass flux at the flame. These analyses are described in the manuscripts accepted for the 39th International Symposium on Combustion cited below.

2. P.H. Irace, K.A. Waddell, D. Constales, G. Yablonsky, M. Kim, P.B. Sunderland, R.L. Axelbaum, On the existence of steady state gaseous microgravity spherical diffusion flames in the presence of radiation heat loss, Proceedings of the Combustion Institute 2022 (accepted).

The abstract from this submission states:

> Whether steady-state gaseous microgravity spherical diffusion exist in the presence of radiation heat loss is an important fundamental question and has important implications for spacecraft fire safety. In this work, experiments aboard the International Space Station and a transient numerical model are used to investigate the existence of steady-state microgravity spherical diffusion flames. Gaseous spherical diffusion flames stabilized on a porous spherical burner are employed in normal (i.e., fuel flowing into an ambient oxidizer) and inverse (i.e., oxidizer flowing into an ambient fuel) flame configurations. The fuel is ethylene and the oxidizer oxygen, both diluted with nitrogen. The flow rate of the reactant gas from the burner is held constant. It is found that steady-state gaseous microgravity spherical diffusion flames do exist in the presence of radiation heat loss, provided that the steady-state flame size is less than the flame size for radiative extinction, and the flame develops fast enough that radiation heat loss does not drop the flame temperature below the critical temperature for radiative extinction (1130 K). A simple model is provided that allows for the identification of initial conditions that can lead to steady-state spherical diffusion flames. In the spherical, infinite domain configuration, the characteristic time for the diffusion - controlled system to effectively reach steady-state is found to be on the order of 100,000 s. Despite a narrow range of attainable conditions, flames that exhibit steady-state behavior are observed aboard the ISS for up to 870 s, even with the constraint of a finite boundary. Steady-state flames are simulated using the numerical model for over 100,000 s.

2. P.H. Irace, K.A. Waddell, D. Constales, P.B. Sunderland, R.L. Axelbaum, Critical temperature and reactant mass flux for radiative extinction of ethylene microgravity spherical diffusion flames at 1 bar, Proceedings of the Combustion Institute 2022 (accepted).

The abstract from this submission states:

> In microgravity combustion, where buoyancy is not present to accelerate the flow field and strain the flame, radiative extinction is of fundamental importance, and has implications for spacecraft fire safety. In this work, the critical point for radiative extinction is identified for normal and inverse ethylene spherical diffusion flames via atmospheric pressure experiments conducted aboard the International Space Station, combined with a transient numerical model. The fuel is ethylene with nitrogen diluent, and the oxidizer is an oxygen/nitrogen mixture. The burner is a porous stainless-steel sphere. All experiments are conducted at constant reactant flow rate. Conditions for the normal flames: ambient oxygen mole fraction from 0.2 to 0.38, burner supply fuel mole fraction from 0.13 to 1,

total mass flow rate, $\dot{m}_{total}$, from 0.6 to 12.2 mg/s, and adiabatic flame temperature, T_{ad}, from 2000 to 2800 K. Conditions for inverse flames: ambient fuel mole fraction from 0.08 to 0.12, burner supply oxygen mole fraction from 0.4 to 0.85, $\dot{m}_{total}$ from 2.3 to 11.3 mg/s, and T_{ad} from 2080 to 2590 K. It is found that despite the broad range of conditions, all of the flames extinguish at a critical extinction temperature of 1130 K, and at a fuel-based mass flux of 0.2 g/m^2-s for normal flames, and an oxygen-based mass flux of 0.68 g/m^2-s for inverse flames. With this information, a simple equation is developed to estimate the flame size (i.e., location of peak temperature) at extinction for any atmospheric-pressure ethylene spherical diffusion flame given only the reactant mass flow rate. Flame growth, which ultimately leads to radiative extinction if the critical extinction point is reached, is attributed to the natural development of a diffusion limited system as it approaches steady state and the dependence of transport properties on temperature as temperature is reduced by radiation (i.e., radiation-induced growth).

Flame Design publications with ISS data include:

1. P.H. Irace, H. Lee, K. Waddell, L. Tan, D.P. Stocker, P.B. Sunderland, R.L. Axelbaum, Observations of long duration microgravity spherical diffusion flames aboard the International Space Station, Combustion and Flame, 229 (2021) 111373. https://doi.org/10.1016/j.combustflame.2021.02.019
2. P.H. Irace, K. Waddell, D. Constales, P.B. Sunderland, R.L. Axelbaum, Critical temperature and reactant mass flux for radiative extinction of ethylene microgravity spherical diffusion flames at 1 bar, Proc. Combust. Inst. (accepted).
3. P.H. Irace, K Waddell, D. Constales, G. Yablonsky, M. Kim, P.B. Sunderland, R.L. Axelbaum, On the existence of steady state gaseous microgravity spherical diffusion flames in the presence of radiation heat loss, Proc. Combust. Inst. (accepted).

Conference presentations, posters, etc. with ISS results have included:

1. P. Irace, L. Tan, K. Waddell, H.J. Lee, D.P. Stocker, D.L. Urban, P.B. Sunderland, R.L. Axelbaum, A Galaxy of Soot Particles, Art Competition Entry, ASGSR, Denver, 2019.
2. P. Irace, H.J. Lee, K. Waddell, L. Tan, D.P. Stocker, D.L. Urban, P.B. Sunderland, R.L. Axelbaum, Steady-State and Radiatively Extinguishing Spherical Diffusion Flames in Microgravity, Abstract, ASGSR, Denver, 2019.
3. P.H. Irace, H.J. Lee, K. Waddell, L. Tan, D.P. Stocker, P.B. Sunderland, R.L. Axelbaum, *Observations of Long Duration Microgravity Spherical Diffusion Flames Aboard the International Space Station*, Abstract, Annual ASGSR Meeting, 2020.
4. P. Irace, L. Tan, K. Waddell, H.J. Lee, D. Stocker, D.L. Urban, P.B. Sunderland, R.L. Axelbaum, *Flame Ignition in Space*, Art Competition Poster, ASGSR, 2020.
5. K. Waddell, P.B. Sunderland, S. Medvedev, S. Frolov, P.H. Irace, R.L. Axelbaum, *Flame and Burner Temperature Measurements and Predictions in Spherical Diffusion Flames*, U.S. National Combustion Meeting, 2021, 9 pp.

6. P.H. Irace, K. Waddell, Z. Xu, D. Constales, P.B. Sunderland, R.L. Axelbaum, *Microgravity Spherical Diffusion Flames: The Critical Point for Radiative Extinction and the Dynamics to Reach It*, U.S. National Combustion Meeting (2021).
7. K. Waddell, P.B. Sunderland, S. Medvedev, S. Frolov, P.H. Irace, R.L. Axelbaum, *Flame and Burner Temperature Measurements and Predictions in Spherical Diffusion Flames*, Abstract, ASGSR, Baltimore, 2021.

L. Communication Imagery

1. Still Imagery

Photos are available online at www.flickr.com/photos/space-flames. Although the Facebook page at www.facebook.com/space.flames is meant to provide updates on ACME and related combustion research, it is good source for recent photos, e.g., of flames, that may not be available on the Flickr site.

Additional photographs may be requested from the Project Scientist, Dennis Stocker (dennis.p.stocker@nasa.gov).

2. Video

An example video from the Flame Design experiment can be found with the web-based article at:
https://www.nasa.gov/mission_pages/station/research/news/combustion-research-microgravity-clean-burning-fuel-space-station/.

'The Power of Studying Combustion on the ISS,' which was developed with input from the ACME Project Scientist, is available at https://science.nasa.gov/science-news/news-articles/the-power-of-studying-combustion-on-the-iss.

Videos related to Flame Design and other microgravity combustion research can also be seen on Facebook at www.facebook.com/space.flames.

Contact Dennis Stocker (dennis.p.stocker@nasa.gov) for access to those videos or for additional videos.

IV. Additional Comments from science team

Constantly sooting flames were only observed in 40% oxygen and high fuel concentrations. In 21% oxygen, small flames did not soot because of heat loss to the burner and large flames did not soot because high radiative losses caused the flame temperature to quickly drop below the sooting limit. Preliminary analysis of the HiBMs images suggests that significant emission of light from soot particles and dissymmetry of the soot cloud will make acquiring accurate soot volume fraction measurements difficult, if possible. The cloud of soot particles also shows relatively extreme dissymmetry, further complicating our ability to obtain soot volume fraction measurements.

Appendix D. Operational Timeline

This summary timeline of key crew and remote operations was prepared by Dennis Stocker, where the red text indicates the key days when flames were ignited. The dates were based on plans, where it is possible that the dates for some simple tasks, e.g., hardware gather or stowage, may have shifted slightly.

The timeline also includes the launch and landing of crews who supported the Flame Design operations and more generally those of crews that occurred during Flame Design operations.

First round (limited to normal flames)

2018	**GMT**	**Activity**
Dec. 3	*337*	*Launch of Konenenko, McClain, and Saint-Jacques*
2019		
March 14	*073*	*Launch of Hague, Koch, and Ovchinin*
April 5	095	Hardware gathered for Flame Design by Koch
April 9	099	ACME chamber insert reconfigured for Flame Design by Koch
April 10	100	Hardware stowed by Koch
April 11	101	Bottle swaps by Kononenko
April 22	112	MFC testing & tuning, etc. for start of Flame Design
April 25	**115**	**Flame Design tests using 100% C_2H_4**
May 3	**123**	**Flame Design tests using 100% C_2H_4**
May 10	130	Igniter tip replacement by Saint-Jacques
May 17	**137**	**Flame Design tests using 100% C_2H_4**
May 20	140	Bottle swaps by Hague
May 22	**142**	**Flame Design tests using 100% C_2H_4**
May 30	**150**	**Flame Design tests using 100% C_2H_4**
May 31	151	Swap of nitrogen Mass Flow Controller and bottle by Hague
June 3	154	MFC testing & tuning
June 5	**156**	**Flame Design tests using 100% C_2H_4**
June 10	**161**	**Flame Design tests using 100% C_2H_4**
June 13	164	Bottle swaps by Hague
June 14	165	Hardware gather by Saint-Jacques; igniter tip replacement by Koch
June 20	**171**	**Flame Design tests using 100% C_2H_4**
June 24	**175**	**Flame Design tests using 100% C_2H_4**
June 25	*176*	*Landing of Konenenko, McClain, and Saint-Jacques*
June 28	179	Bottle swap by Koch
July 1	**182**	**Flame Design tests using 100% C_2H_4**
July 8	**189**	**Flame Design tests using 100% C_2H_4**
July 11	192	Swap of nitrogen Mass Flow Controller and bottle by Koch
July 15	**196**	**Flame Design tests using 100% C_2H_4**
July 17	198	Igniter tip replacement by Hague
July 19	**200**	**Flame Design tests using 100% C2H4**
July 20	*201*	*Launch of Morgan, Parmitano, and Skvortsov*
July 22	203	Bottle swap by Ovchinin (assisted by Koch)
July 25	**206**	**Flame Design tests using 100% C_2H_4**
2019	**GMT**	**Activity**
Oct. 3	*277*	*Landing of Hague, Ovchinin, and visiting astronaut Almansoori*

2020

Feb. 6	*037*	*Landing of Koch, Parmitano, and Skvortsov*

Second round (inverse flames except for the first day)

2021

April 9	*099*	*Launch of Dubrov, Novitskiy, and Vande Hei*
April 23	*113*	*Launch of Hoshide, Kimbrough, McArthur, and Pesquet*
Oct. 26	299	Bottle swap, etc. by Vande Hei
Oct. 28	**301**	**Flame Design tests in normal-flame mode using 100% C_3H_8**
Nov. 2	306	Chamber insert reconfiguration for inverse flame testing by Kimbrough
Nov. 3	307	Chamber insert, etc. reconfiguration by Vande Hei; bottle swaps by McArthur
Nov. 8	*312*	*Landing of Hoshide, Kimbrough, McArthur, and Pesquet*
Nov. 10	*314*	*Launch of Barron, Chari, Marshburn, and Maurer*
Nov. 22	326	Hardware checkout in preparation for inverse-flame testing
Nov. 24	**328**	**Flame Design tests in inverse-flame mode using 85% O_2**
Dec. 3	337	Bottle swap by Dubrov (with Vande Hei's assistance)
Dec. 6	**340**	**Flame Design tests in inverse-flame mode using 85% O_2**
Dec. 10	344	Bottle swap by Vande Hei
Dec. 15	**349**	**Flame Design tests in inverse-flame mode using 85% O_2**
Dec. 21	355	Bottle swaps by Chari
Dec. 27	**361**	**Flame Design tests in inverse-flame mode using 85% O_2**

2022

Jan. 5	005	Bottle swap by Vande Hei
Jan. 17	017	Thin Filament Pyrometry (TFP) fiber array aligned, etc. by Chari
Jan. 18	**018**	**Flame Design tests in inverse-flame mode using 85% O_2**
Jan. 24	**024**	**Bottle swap by Dubrov; Flame Design tests using 85% O_2**
Jan. 26	026	Bottle swaps by Dubrov
Jan. 27	**027**	**Flame Design tests in inverse-flame mode using 85% O_2**
Jan. 31	031	Bottle swap by Dubrov
Feb. 4	**035**	**Flame Design tests in inverse-flame mode using 85% O_2**
March 30	*089*	*Landing of Dubrov, Shkaplerov, and Vande Hei*
May 6	*126*	*Landing of Barron, Chari, Marshburn, and Maurer*

Appendix E. Astronauts who Conducted ISS Operations for Flame Design

This list was prepared by Dennis Stocker from his log of ACME operations.

The Flame Design experiment was supported by ten ISS crew members on 24 separate days, with 13 occurring during the first round of operations and 11 in the second round. The 8 men and 2 women are listed by the number of days in which they conducted supporting on-orbit activities and alphabetically.

Crew member	Year	Day(s) of year	# days
Koch, Christina (NASA)	2019	095, 099-100, 165, 179, 192, 203 (with Ovchinin)	7
Vande Hei, Mark (NASA)	2021	299, 307, 337 (with Dubrov), 344	5
	2022	005	
Dubrov, Petr (RSA)	2021	337	4
	2022	024, 026, 031	
Hague, T. Nick (NASA)	2019	140, 151, 164, 198	4
Chari, Raja (NASA)	2021	355	2
	2022	017	
Saint-Jacques, David (CSA)	2019	130, 165	2
Kimbrough, Shane (NASA)	2021	306	1
Kononenko, Oleg (RSA)	2019	101	1
McArthur, K. Megan (NASA)	2021	307	1
Ovchinin, Alexey (RSA)	2019	203	1

Together, the astronauts and cosmonauts represent three space agencies and three nations, namely Canada (CSA), Russia (RSA), and the United States (NASA).

Agency	# crew	Crew members	# days for agency
NASA	6	Koch, Vande Hei, Hague, Chari, Kimbrough, McArthur	19
RSA	3	Dubrov, Kononenko, Ovchinin	6
CSA	1	Saint-Jacques	2

In the first round of testing, which featured normal flames, most of the crew tasks were carried out by Christina Koch (NASA), Nick Hague (NASA), and David Saint-Jacques (CSA). In the second round, which featured inverse flames (except on the first test day), most crew tasks were conducted by Mark Vande Hei (NASA), Petr Dubrov (RSA), and Raja Chari (NASA).

Appendix F. ISS Operations per the Payload Weekly Science Summary

2019

First round (normal flames)

Increment 59/60

Week 5: April 5-11

This week, Christina Koch configured the ACME Chamber Insert to support the next ACME experiment, Flame Design. Also to support Flame Design, Oleg Kononenko replaced the bottles on CIR Manifold #2, CIR Manifold #4, and the CIR GC Helium Manifold. Flame Design is focused on understanding the soot processes of flames. Test points will start in a few weeks.

The April 11 reporting on the completion of BRE's first round was not included here.

Week 8: April 26 – May 2

On GMT 115, the first ACME Flame Design test points were performed. Pure gaseous ethylene fuel was ejected into air from a porous sphere, which, in microgravity, results in a one-dimensional, spherical flame. Thus, data analysis is greatly simplified because the flame shape doesn't need to be predicted, and computational modeling can focus on the detailed chemistry and transport of the molecules and energy.

The goal of Flame Design is to improve our understanding of soot processes so that flames can be "designed" to be soot free. This research could help lead to reduced soot and NOx—and thus reduced smog and acid rain—through nitrogen exchange, where the nitrogen is shifted from the air to the fuel. This oxygen-enriched combustion of diluted fuel could also facilitate carbon sequestration (i.e., capture and storage).

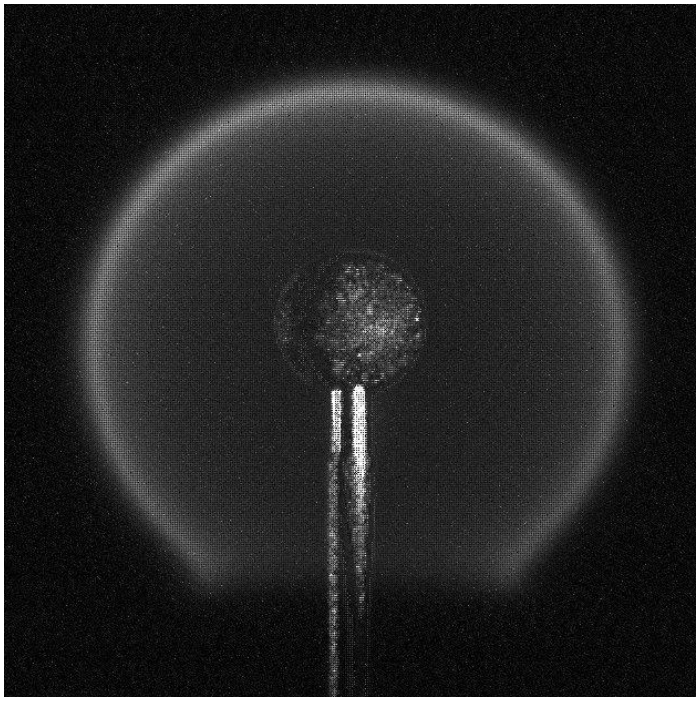


Figure 1. Flame Design Flame (Example Preliminary Monochrome Image)

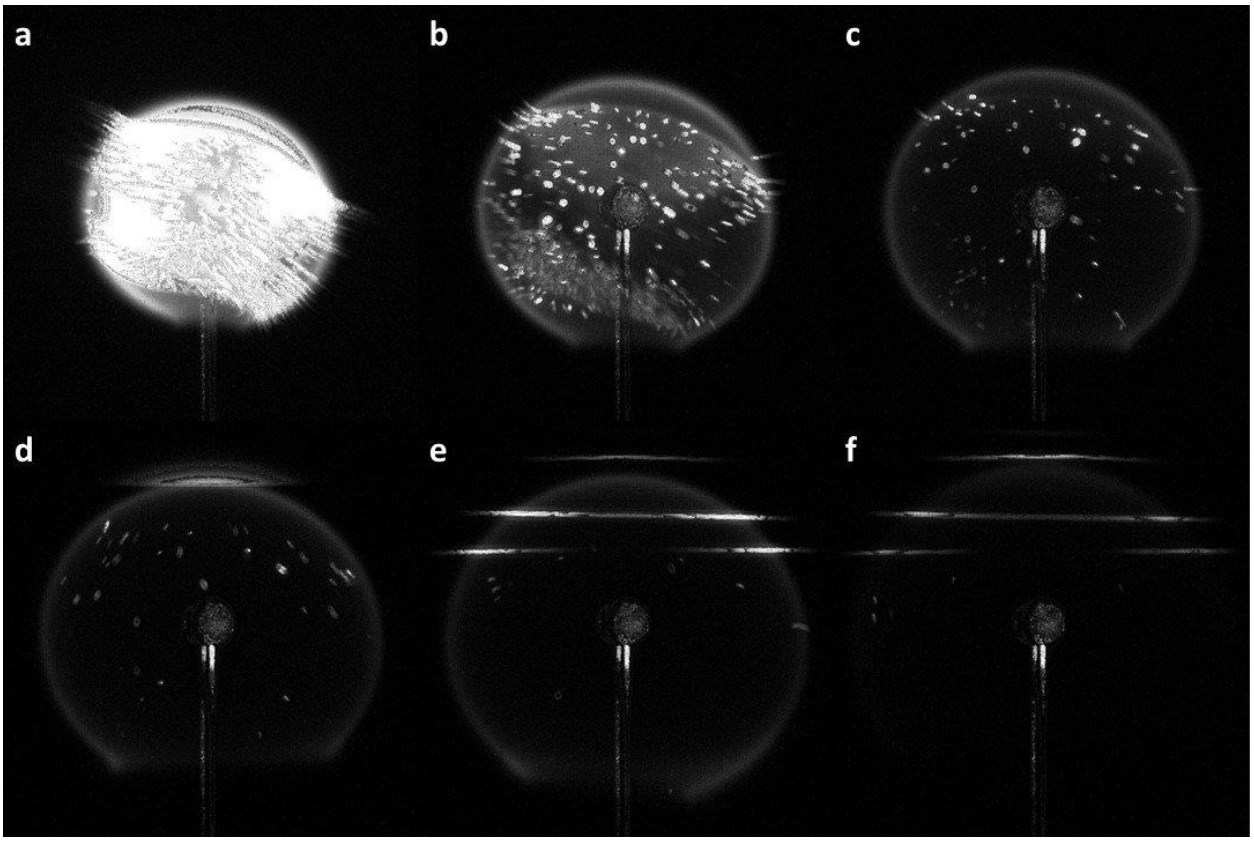
As can be seen in Figure 1, the flame is nominally spherical, but heat loss to the burner tube causes it to locally quench. Similarly, candle flames have an open gap close to the candle, again due to heat loss. These non-premixed flames are a bit like bubbles, where the flame is the shell where the fuel and oxygen combine and create heat, light, and chemical products like carbon dioxide and water. In premixed flames, the fuel is within the “bubble” and the oxygen is outside of it. Future Flame Design tests are planned where the flame will be inside out, i.e., inverted. To conduct those tests (planned for 2020), the CIR's Combustion Chamber will be filled with a fuel/nitrogen mixture while an oxygen/nitrogen mixture will flow from the burner.

Week 9: May 3-9

On GMT 123, 19 spherical flames were successfully ignited for ACME Flame Design. Gaseous ethylene fuel, typically diluted with nitrogen, was ejected into air from the porous sphere at the end of the burner. Compared to the test points on GMT 115, these tests were conducted with higher flow rates, which resulted in larger flames of shorter durations.

In most of the tests, the harp-like array of 14-micron ceramic fibers was translated into the flame to determine the radial temperature distribution. Where a fiber is visibly glowing, the temperature can be determined from its color, where yellow is hotter than orange, which in turn is hotter than red. The ACME color camera was carefully characterized prior to launch enabling a quantitative determination of the temperature by comparison of the red, green, and blue color planes in a measurement technique referred to as pyrometry.

Unfortunately, during the day’s final test point, the igniter tip burned out and will need to be replaced before performing more test points.



This sequence of preliminary monochrome flame images is a life history of an example Flame Design flame.

(a) Shortly after ignition, the ignition-created soot is being carried out of the flame by the gas flow from the porous spherical burner.

(b) The amount of soot has decreased dramatically and the blue flame (neglecting the bright spots of the soot particulate) has expanded and dimmed. Since the fuel is inside the flame and oxygen is outside, the flame expands outward until it reaches enough oxygen to consume the fuel. The gas flow is outward from the burner sphere, so the oxygen must diffuse inward due to its concentration gradient. That gradient, and thus the oxygen transport, diminishes (i.e., flattens) with time due to both the oxygen consumption and the release of combustion byproducts like carbon dioxide and water vapor. A flame becomes effectively blanketed by these byproducts, inhibiting the oxygen transport.

(c) The visible soot has mostly disappeared, and the flame has further grown and dimmed. The base of the flame has retreated from the burner tube because of heat loss.

(d) The fiber array has begun translating with the first ceramic fiber at the edge of flame, where its glow is visible at the top of the image. Again, the soot continued to disappear, and the flame continued to grow and retreat from the burner tube.

(e) Two fibers are in the flame and the third is close enough to visibly glow. The flame is a shell, where the fibers are cooler (i.e., dimmer in this gray-scale image) within the flame. A comparison of images (d) and (e) reveals that the flame base is oscillating, which is a sign of approaching extinction.

(f) The flame has nearly extinguished due to radiative heat loss and is very faint in comparison to the earlier images. The fibers are dimmer in image (f) than (e), indicating the decrease in flame temperature.

Week 10: May 10-16

On GMT 130, David Saint-Jacques replaced the burnt-out ACME Igniter Tip. He also installed the ACME Short Thermocouple Array, which will measure temperatures near the flame. ACME returned to performing Flame Design test points on GMT 137.

Week 11: May 17-23

On GMT 137 and 142, ACME Flame Design test points were performed in chamber environments of varying oxygen concentrations, including 21%, 30% and 40% oxygen (the remainder was nitrogen). The test points used gaseous ethylene fuel, which was sometimes diluted with nitrogen.

On GMT 140, Nick Hague replaced the bottles on CIR Manifold #2 and the CIR GC Argon Manifold to support Flame Design test points.

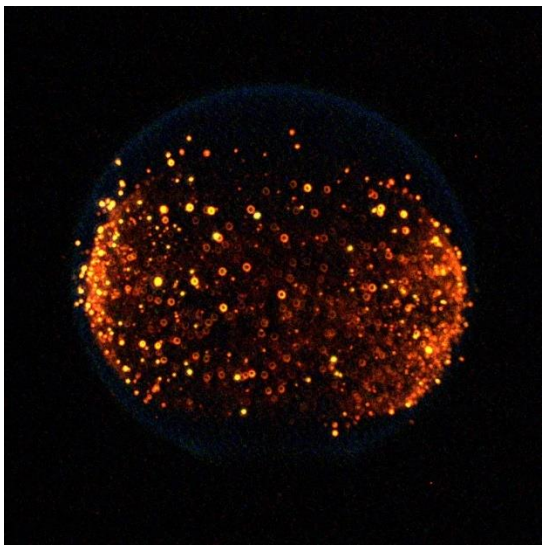
CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

This test point video first runs at normal speed, and then the end of it runs a second time at quarter speed to better show what is happening with the flame. This test point was performed in a chamber environment of air (21% oxygen/79% nitrogen). The flame almost completely extinguishes but then dramatically flashes back reestablishing the spherical flame before finally extinguishing. The asymmetry of this test's flashback makes the flame's death even more dramatic. The flame is brighter as it flashes back due to the added mixing of the oxygen and fuel that occurs where the flame has partially quenched.

Week 13: May 31 - June 6

On GMT 150, ACME Flame Design test points were performed in chamber environments ranging from 29-40% oxygen (the remainder was nitrogen). The test points used gaseous ethylene fuel, which was sometimes diluted with nitrogen. These flames were more robust and were often relatively stable in comparison to the earlier tests that had been conducted in ~21% oxygen.

On GMT 151, Nick Hague replaced the ACME Bypass Controller and the CIR Manifold #2 Bottle to support Flame Design test points with faster nitrogen flow rates and in oxygen concentrations up to 40%.



This figure shows a dim blue spherical flame with glowing soot that was created at ignition.

Week 14: June 7-13

On GMT 156, ACME Flame Design test conditions resulted in flames which continued to produce soot following ignition, yielding new information for this study of soot processes. While soot is both an environmental and health concern, it can be useful for some applications. As two examples, soot is helpful for radiant heat transfer and for illumination. The purpose of Flame Design is to determine the conditions controlling the presence or absence of soot in non-premixed flames (i.e., where the fuel and oxygen are separated by the flame's surface). With that understanding, engineers will be able to "design" flames that are either sooty or not depending on their needs.

CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

In this test point video, pure ethylene (C2H4), which is a sooty fuel due to its double bond, was burned in roughly 36% oxygen (with a balance of nitrogen). At ignition, the flame is very sooty. The broadband thermal radiation of the soot particles causes the bright yellow color of the flame. The fuel flow rate is then ramped up to more than twice the flow rate at ignition, where the flame grows larger as a result. The test conditions are such that the flame temperature is high enough to continue to produce soot particles, which are mostly carbon and form when the ethylene within the flame is hot enough to react with itself in the absence of oxygen (which is outside of the flame). But the flame later weakens as the oxygen transport drops and the radiative heat losses become important. The flame appears to no longer produce soot near the flame's end, but the glowing soot clusters are still apparent. With the long residence time resulting from the microgravity condition, the clusters become especially large.

Week 15: June 14-20

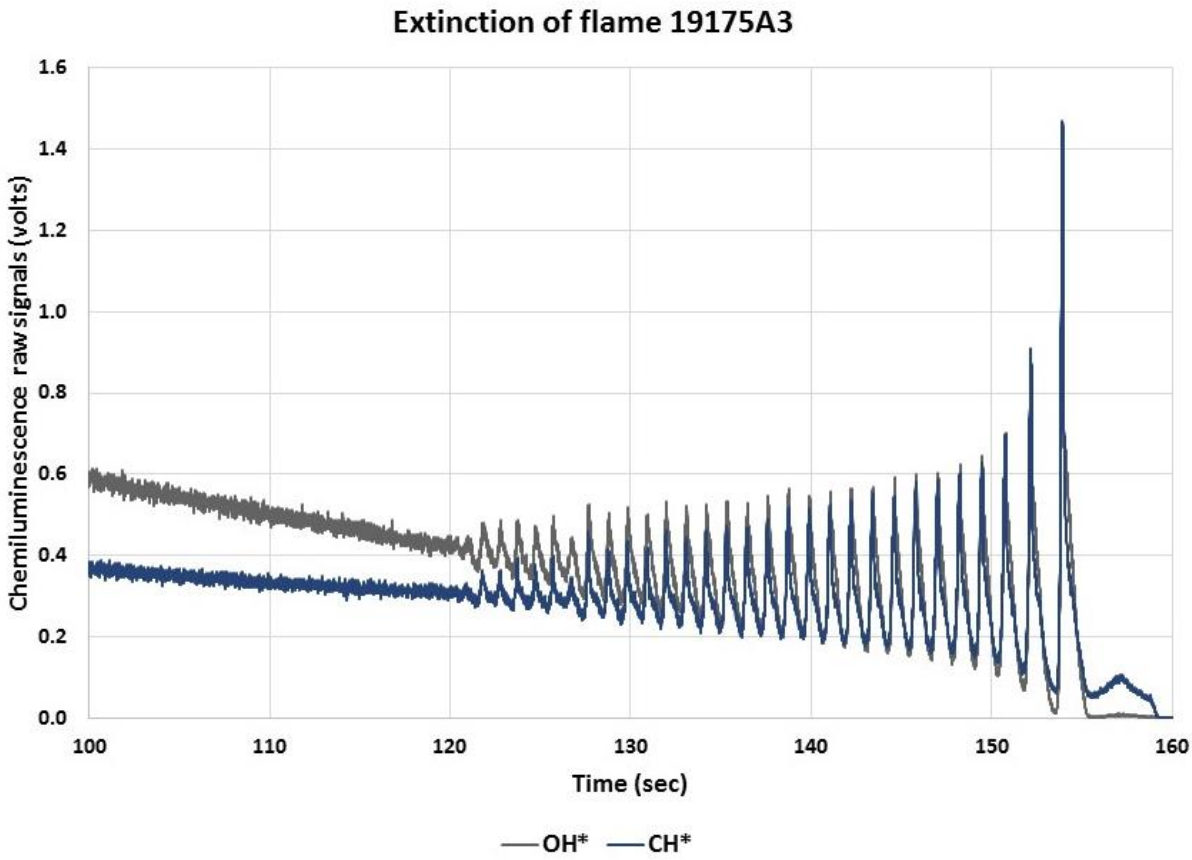
On GMT 164, Nick Hague replaced the bottles on the CIR Manifold #2 and the CIR Check Gas Manifold to continue supporting ACME Flame Design test points.

Also, in support of Flame Design test points, Christina Koch replaced the ACME Igniter Tip and the ACME Spherical Burner on GMT 165. ACME returned to performing test points on GMT 171.

Week 16: June 21-27

On GMT 171 and 175, ACME Flame Design test points were performed with the new spherical burner that was installed on GMT 165. This particular spherical burner is markedly better at achieving a spherical flame than the last spherical burner. With a spherical flame, data analysis is greatly simplified because the flame shape doesn't need to be predicted, and computational modeling can focus on the detailed chemistry and transport of the molecules and energy.

These tests were conducted in atmospheres of up to 40% oxygen and using gaseous ethylene fuel, which was sometimes diluted with nitrogen. The GMT 175 tests were conducted at slightly elevated pressures furthering this experiment's goal to understand the conditions that lead to sooty and soot-free flames.



While Flame Design's primary objective is to determine the conditions under which flames are soot-free or soot producing, a secondary objective is to establish flammability limits for these spherical flames. We have observed interesting phenomena where the flames sometimes self-extinguish after growing oscillations. These oscillations are apparent in the above graph, which shows the chemiluminescence from excited molecules that drop to a lower energy state by emitting a photon at a specific wavelength. CH* chemiluminescence is the source of most of the blue seen in hydrocarbon flames and OH* chemiluminescence is in the ultraviolet. These two measurements are made with Photomultipiler Tubes (PMTs), where each tube is equipped with a narrow-band filter corresponding to an excited molecule.

In the example shown in the graph, the flame's oscillations lasted about 30 seconds and occurred at roughly 1 Hz, but they slowed in frequency as they grew in strength. The oscillations are evident in the imaging, where the flame locally quenches—creating a hole in the flame. This hole allows for increased mixing of the fuel and oxygen, which in turn results in a recovery of the flame. The size of the opening in the flame (which normally separates the fuel and oxygen) grows with time until the entire flame extinguishes.

Interestingly, the graph shows that the CH* measurement was relatively steady before the start of the oscillations in comparison with the decaying OH* emission. This difference will be further examined to improve our understanding of flame extinction.

Week 17: June 28 – July 4

On GMT 179, Christina Koch replaced the CIR Manifold #2 Bottle to continue supporting ACME Flame Design test points.

On GMT 182, 15 spherical flames were successfully ignited for ACME Flame Design. The goal of Flame Design is to improve our understanding of soot processes so that flames can be "designed" to be soot free.

Week 18: July 5-11

On GMT 189, ACME Flame Design test points were performed using diluted ethylene fuel at a chamber pressure of 1.3 atm with varying oxygen concentrations of 34 to 40% oxygen (the remainder was nitrogen). The elevated oxygen and pressure can enhance the production of soot, where this research is focused on understanding the conditions that lead to sooty and soot-free flames.

On GMT 192, Christina Koch replaced the ACME Bypass Controller and the CIR Manifold #2 Bottle to support Flame Design test points with slower nitrogen flow rates and in oxygen concentrations of up to 40%.

CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

This test point video shows a sooty flame, and, in the absence of buoyancy, the soot stays within the flame and clusters. Some of the glowing soot clusters move out of the flame due to the outward flow of the diluted ethylene fuel through the burner's porous sphere. In contrast, some of the clusters get trapped within the flame due to a temperature gradient that moves them inward toward the cold burner and away from the hot flame.

After some time, the harp-like array of 14-micron ceramic fibers is translated into the flame to determine the radial temperature distribution. Using a carefully characterized camera, the temperature can be quantitatively determined from a glowing fiber's color, where yellow is hotter than orange, and orange is hotter than red. The motion of the fiber arm causes the soot clusters to move away from the fibers.

While the temperature measurements are being made, the flow of diluted ethylene is gradually reduced, which causes the flame to become smaller and unstable. Heat loss causes the flame to locally quench—creating a hole in the flame near the burner tube. This hole allows for increased mixing of the fuel and oxygen, which reinvigorates the flame and causes it to close again. The flame opens and closes, effectively consuming fuel in repeated gulps…a bit like a video-game character eating dots. But as the fuel flow is reduced further, the size of the open "mouth" gets smaller, as if the flame is becoming full. The flame becomes stable again until the flow becomes so small that the flame finally goes out.

This video illustrates the extreme complexities of even conceptually simple flames! Even though mankind has been using combustion for millennia, microgravity research can unlock new understanding, which would lead to improved efficiency and reduced pollutant emission in practical terrestrial combustion.

Week 19: July 12-18

On GMT 196, ACME Flame Design test points were performed using ethylene fuel (sometimes diluted) at a chamber pressure of 1.25 atm in about 40% oxygen (the remainder was nitrogen). Today's tests focused primarily on quasi-steady flames (where the flame radius remains essentially constant in time) and determining flame temperatures near the point where soot disappears. At the end of the day, the igniter tip burned out. On GMT 198, Nick Hague replaced the burnt-out igniter tip.

> **CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG***
> Upon ignition, the flame in this test point video exhibits a lot of soot, which is bright whitish-yellow in appearance. This extreme brightness masks the blue flame, which is the marker of the fuel oxidation region. With time, the soot diminishes, and the blue flame becomes more visible. The harp-like array of 14-micron ceramic fibers then slowly traverses in and out of the flame. The glow from the fibers allows for a good estimation of the local temperature.

Week 20: July 19-25

On GMT 200, ACME Flame Design test points were performed using ethylene fuel (sometimes diluted) at a chamber pressure of 1 atm in about 40% or 30% oxygen (the remainder was nitrogen). Today's tests studied quasi-steady flames and took flame temperature measurements using the ceramic fiber array. These tests were the last Flame Design tests to be performed in higher oxygen atmospheres.

On GMT 203, Alexey Ovchinin and Christina Koch replaced the CIR Manifold #2 Bottle to support Flame Design test points.

Week 21: July 26 – August 1

On GMT 206, we performed the final test points for the first round of testing for ACME Flame Design. Test points were performed using ethylene fuel (sometimes diluted) in a chamber environment of air (21% oxygen, 79% nitrogen at 1 atm). The flames for these tests were short-lived and non-sooty—unlike the flames from the tests at higher oxygen concentrations. The goal of Flame Design is to improve our understanding of soot processes so that flames can be "designed" to be soot free.

To prepare for the start of ACME s-Flame (Structure and Response of Spherical Diffusion Flames), Christina Koch replaced the bottles on CIR Manifolds #2 and #4 on GMT 211. While monitoring the bottle pressures, it was noted that the pressure of the bottle at Manifold #4 was slowly decreasing. The rack was powered down, and Nick Hague and Luca Parmitano both helped with troubleshooting operations. Currently, a forward plan is being developed.

2021

Second round (inverse flames expect for the first test day)

Increment 66

Week 2: Oct. 22-28

Last week, Shane Kimbrough replaced the gas bottles at CIR Manifold #2 and CIR Manifold #4, configured the gas bottle valves, and set the valve timer switches in CIR. These activities allowed us to perform the calibration of the CIR's FOMA (Fuel/Oxidizer Management Assembly) system. This calibration needs to be completed yearly to ensure that all the pressure transducers in the CIR are accurate. On GMT 299, Mark Vande Hei reset the valve timer switches, closed the bottle valve at CIR Manifold #1, and replaced the gas

bottle at CIR Manifold #4 to prepare for calibration test points for the next ACME experiment, Flame Design Part 2.

Week 3: Oct. 29 – Nov. 4

On GMT 301, we began the second round of testing for ACME Flame Design with 18 normal flames ignited using propane fuel. The goal of Flame Design is to improve our understanding of soot processes so that flames can be "designed" to be soot free. The remaining tests will look at inverse flames, which are formed when diluted oxygen flows from the burner into a Combustion Chamber filled with diluted fuel. Compared to normal flames, inverse flames are inside out so soot is formed outside rather than inside the flame. While soot processes are known to depend on flame temperature and the residence time of the fuel at high temperatures, a comparison of normal- and inverse-flame results will help us better understand the impact of flow direction on soot processes. On GMT 306 and 307, Shane Kimbrough, Megan McArthur and Mark Vande Hei configured the ACME Chamber Insert and replaced the gas bottles at CIR Manifold #2 and CIR Manifold #4 to support the inverse flame tests for Flame Design Part 2.

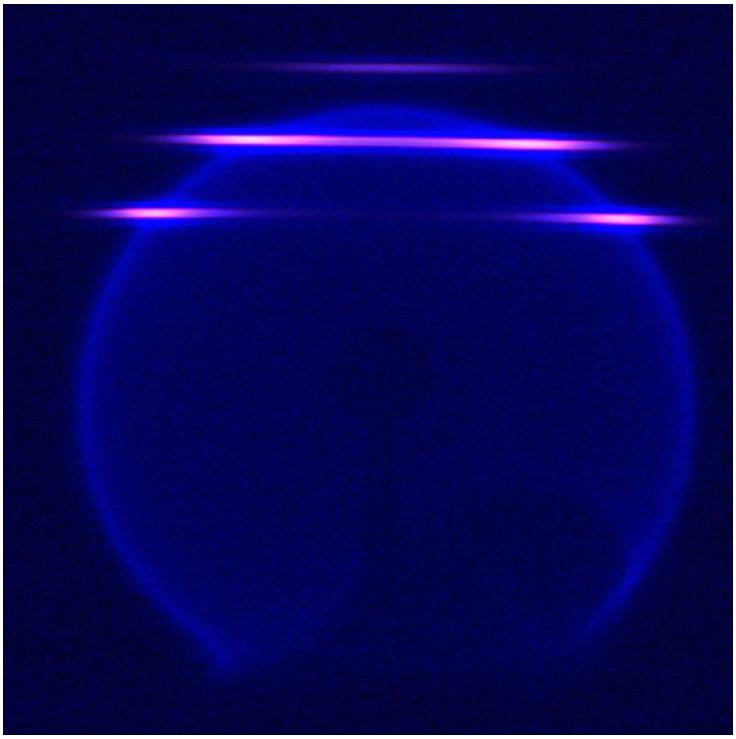


Figure 1. Flame Design Normal Propane Flame

This false-colored image of a soot-free spherical flame shows an array of 14-micron ceramic fibers making temperature measurements. The blue in the image has been enhanced so that the dim flame is visible in addition to the fibers. Normally, the hot fibers appear to be red, orange, and yellow—in order of increasing temperature—just like metal in a forge. The temperature profile along each fiber is optically determined from its varying hue through a technique known as Thin Filament Pyrometry (TFP).

The base of the flame has locally quenched due to heat loss to the burner's supply tube, which created a hole in the flame. This hole allowed for direct mixing of the fuel and oxygen, which enhanced flammability and causes the flame to momentarily reform. In this test, the spherical flame repeatedly opened and closed with oscillations that grew in strength until the entire flame extinguished.

Week 7: Nov. 26 – Dec. 2

On GMT 328, we ignited nine inverse flames for ACME Flame Design. For these tests, an oxygen/nitrogen mixture flowed from the burner into a combustion chamber, which was filled with an ethylene/nitrogen mixture. For one of these tests, we observed our longest sustained flame, which burned for over 14 minutes. At that time, operational constraints required its termination. For comparison, flame durations for all ACME experiments are typically shorter than two minutes with the longest—until now—under six minutes. This 14+ minute flame has led to new understanding of the stability of spherical flames.

While test measurements did show that this long-duration flame was slowly weakening with time, it might have burned significantly longer. We have found with higher burner flow rates that flames grow with time and extinguish due to radiative heat loss. If the flow rate is too low, the flame's small size leads to excessive burner heating. But under the right conditions, the flame can burn for minutes and perhaps even tens of minutes.

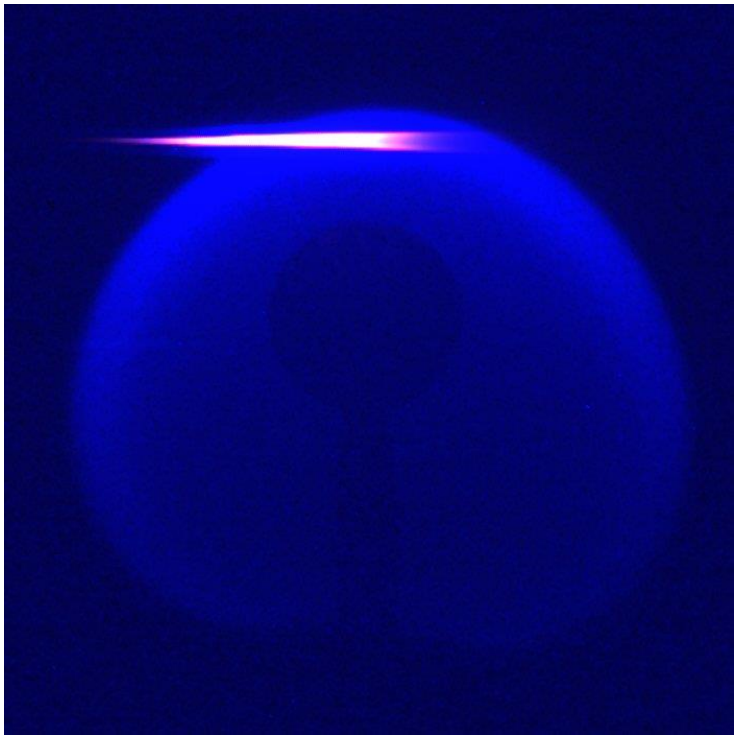


Figure 1. Flame Design Long-Duration Inverse Flame

This false-colored image was taken about halfway through the 14+ minute test. The blue in the image has been enhanced so that the dim flame is visible in addition to the ceramic fiber, which is glowing at the top of the flame. We use the fibers for optical determination of the flame's temperature.

Week 9: Dec. 10-16

On GMT 337, Petr Dubrov and Mark Vande Hei replaced the gas bottle at CIR Manifold #4 to support continuation of Flame Design test points. On GMT 340 and 349, we ignited 28 inverse flames for ACME Flame Design. For these tests, we filled a combustion chamber with a mixture of nitrogen and gaseous fuel (ethylene) and then flowed a mixture of nitrogen and oxygen from the burner. For both test days, the primary test parameters included varying the oxygen concentration and flow rate from the burner.

Another parameter on GMT 340 was varying the ethylene concentration; while on GMT 349, we varied the chamber pressure instead. On GMT 340, we used the ceramic fibers for measuring flame temperature; whereas on GMT 349, we focused on determining sooting limits to better understand the conditions which cause soot to form or not. On GMT 347, Mark Vande Hei replaced the gas bottle at CIR Manifold #4 to support continuation of Flame Design test points.

CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

In this test point video, we translated the 14-micron ceramic fibers into the flame to optically measure the temperature through a technique known as Thin Filament Pyrometry (TFP). The color of the fiber's glow is indicative of the temperature, where yellow is hotter than orange, which in turn is hotter than red. Due to excessive radiative heat loss, this large flame locally quenches and retreats from the burner supply tube (right-hand side). This hole allows the fuel and oxygen to directly mix, which enhances flammability and causes the hole to momentarily close. The cycle of partial quenching and recovery repeats with the oscillations growing in strength until the flame completely extinguishes.

Week 10: Dec. 17-23

On GMT 355, Raja Chari replaced the gas bottles at CIR Manifold #2 and CIR Manifold #4 to support continuation of Flame Design test points.

Week 11: Dec. 24-30

On GMT 361, we ignited 20 inverse flames for ACME Flame Design. We performed these tests with two different ethylene concentrations with a range of oxidizer concentrations and flow rates from the burner. The purpose of these tests was to determine at what conditions soot starts forming and to make detailed measurements of the flame at quasi-steady conditions (i.e., changes are occurring at such a slow rate that the flame can be considered constant).

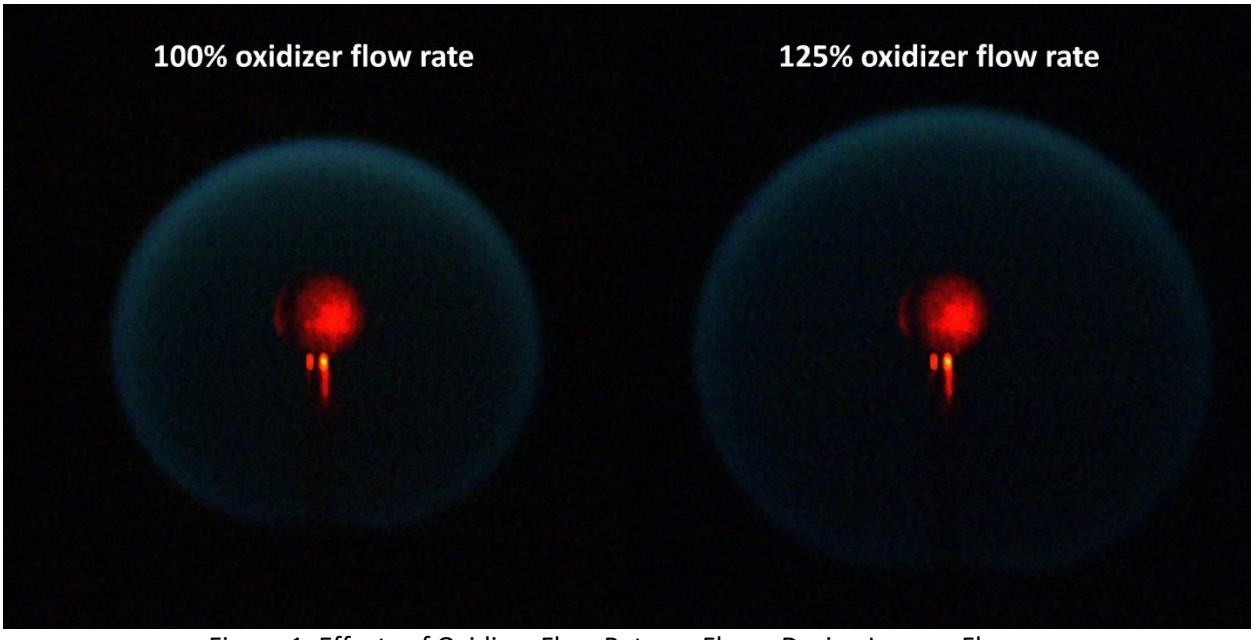


Figure 1. Effects of Oxidizer Flow Rate on Flame Design Inverse Flames

This composite image shows the effects of the oxidizer flow rate on inverse flames. In these tests, we varied the oxidizer flow rate from the burner while the concentrations of oxidizer and fuel in the chamber atmosphere were the same. These images were taken late in each test after the soot had dissipated. The higher flow rate resulted in a larger flame. However, a flame's radiative heat loss increases with its size, so the larger flame was dimmer. This image is false colored making the flame appear blue and the porous spherical burner appear red.

Week 12: Dec. 31 – Jan. 6

On GMT 005, Mark Vande Hei replaced the gas bottle at CIR Manifold #4 to support continuation of Flame Design test points.

2022

Week 14: Jan. 14-20

On GMT 017, Raja Chari configured the ACME Chamber Insert to support the remaining ACME Flame Design test points. On GMT 018, we ignited 12 inverse flames for Flame Design. We performed these tests with two different ethylene concentrations in a range of oxidizer concentrations and flow rates from the burner. The purpose of these tests was to use the ceramic fibers for measuring flame temperature and to study flames that oscillate prior to quenching. In one test, the oscillations continued for nearly 3 minutes before the flame quenched.

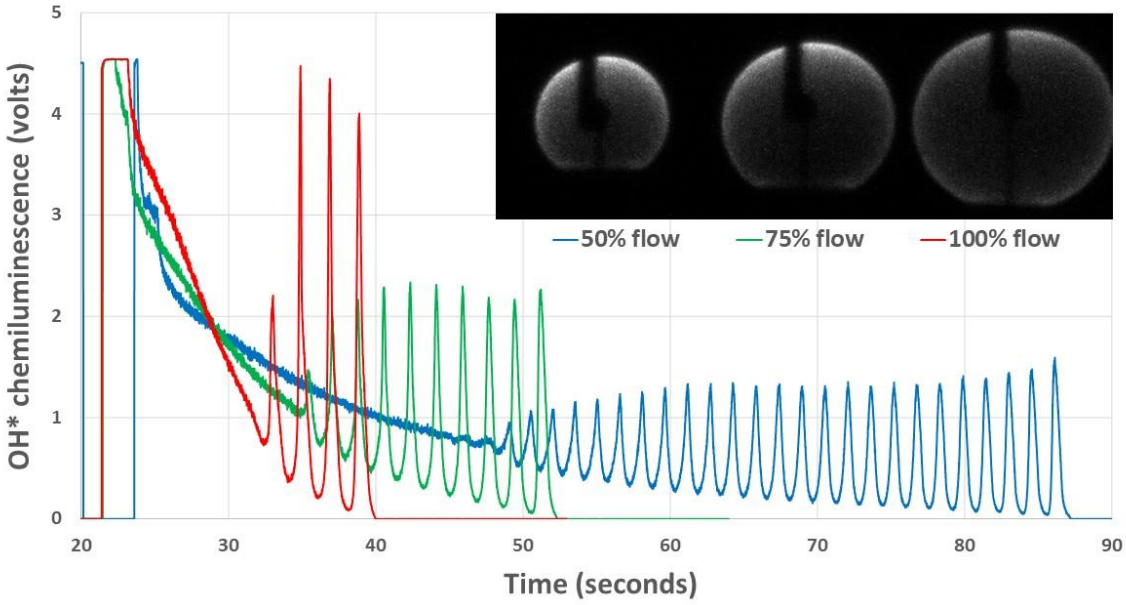


Figure 1. Effect of Oxidizer Flow Rate on Oscillatory Quenching of Flame Design Inverse Flames

Figure 1 is a preliminary graph of the oscillatory quenching of three inverse flames as a function of oxidizer flow rate. The measurement shown on the y-axis is of ultraviolet light from the molecule OH*, which transiently occurs in combustion. This light is a good indicator of flame stability.

The flame images show that higher oxidizer flow rates lead to larger flames, but larger flames experience greater radiative heat loss. So, these larger flames quench more quickly and with fewer oscillations. The

graph shows the largest flame with the highest flow rate (red) quenched after less than 10 seconds of oscillating; whereas, the smallest flame with half the flow rate (blue) oscillated for almost 40 seconds before quenching.

Week 15: Jan. 21-27

On GMT 024 and 026, Pyotr Dubrov replaced the gas bottles at CIR Manifold #2 and CIR Manifold #4 to support continuation of ACME Flame Design test points. On GMT 024, we ignited 20 inverse flames for Flame Design. We performed these tests with ethylene fuel in a range of oxidizer concentrations and flow rates from the burner. These tests were performed at two different chamber pressures. While pressure has a secondary effect on flames compared to the concentration of the reactants, its effect is still important and noticeable.

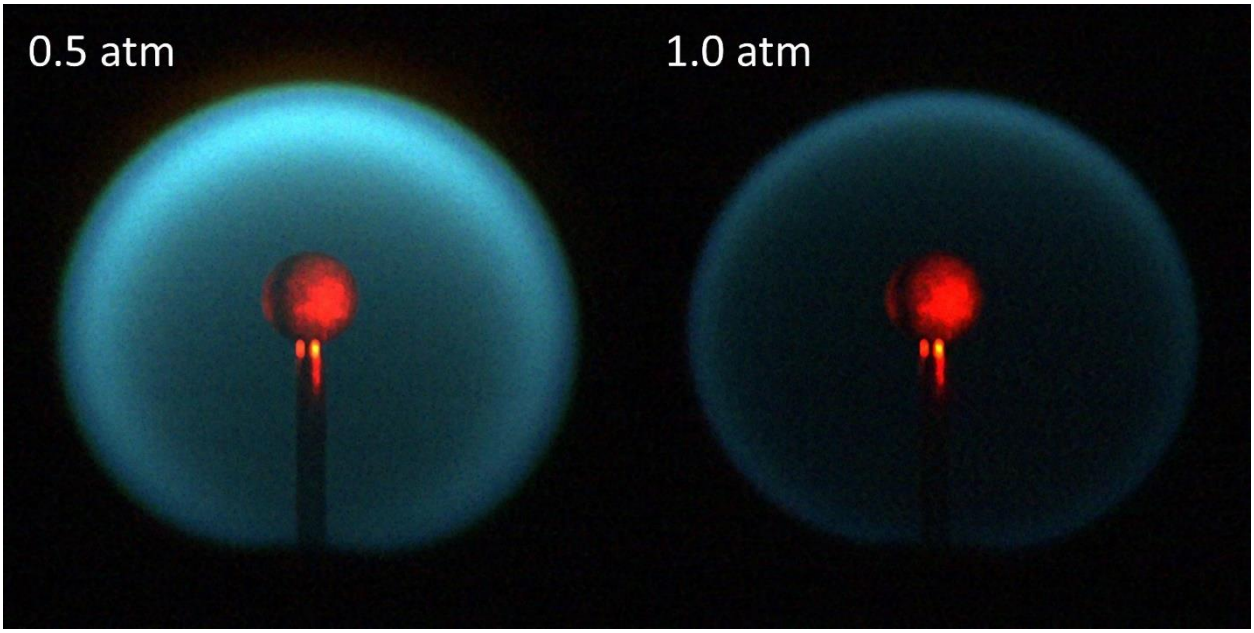


Figure 1. Effect of Pressure on Flame Design Inverse Flames

This composite image shows the effect of chamber pressure on inverse flames. Although the pressure didn't greatly affect the size of the flames, it still had a significant effect. The flame at lower pressure is much brighter because at higher pressures light-emitting molecules are more likely to be extinguished due to molecular collisions. Also, the low-pressure flame shows subtle signs of soot formation. At the top of the flame, the blue edge is tinted with yellow due to soot, which also creates a haze just above the flame. Inverse flames are inside out, where the soot is formed outside of the flame when the hot fuel combines with itself in the absence of oxygen (which is inside the flame). The porous spherical burner is illuminated in red from the illumination package we sometimes use during imaging.

Week 16: Jan. 28 – Feb. 3

On GMT 027, we ignited 16 inverse flames for ACME Flame Design. We performed these tests in a combustion chamber filled with an ethylene/nitrogen mixture to a pressure of 3 atm. We plan to compare these tests to earlier ones performed at lower pressures. During all tests, the 14-micron ceramic fibers were translated into the flame to optically measure the flame temperature through a technique known

as Thin Filament Pyrometry (TFP). On GMT 031, Pyotr Dubrov replaced the gas bottles at CIR Manifold #4 to support continuation of Flame Design test points.

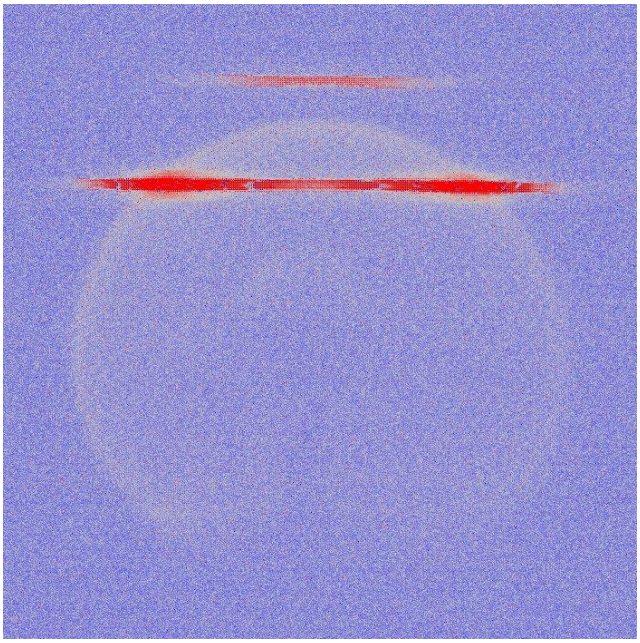


Figure 1. Temperature Measurement of a Flame Design Inverse Flame

In this false-colored image, one 14-micron ceramic fiber was inserted into the flame where a second fiber is visible just above. Both fibers are hot enough to visibly glow, where the temperature can be quantitatively determined from their hue using a camera characterized for that purpose. The flame temperature is determined from where the fiber passes through the flame shell and glows most brightly.

Week 17: Feb. 4-10

On GMT 035, we performed the final test points for the second round of testing for ACME Flame Design. We ignited 15 inverse flames. For these tests, we filled a combustion chamber with ethylene/nitrogen mixtures and then flowed a mixture of oxygen/nitrogen from the burner. Tests were conducted at both elevated and reduced pressures. The spherical symmetry of the flames, resulting from the absence of buoyant flows created by Earth's gravity, simplifies data analysis and can lead to new understanding.

These were our final test points with ACME, which has been successfully performing science in CIR since 2017!

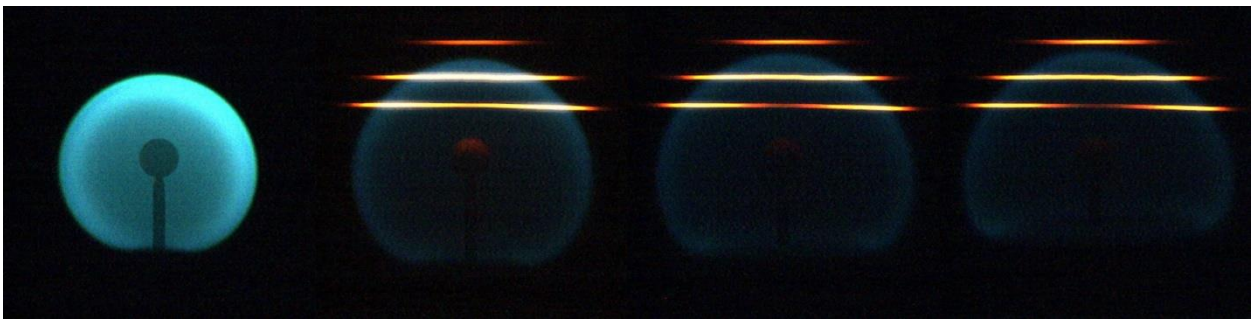
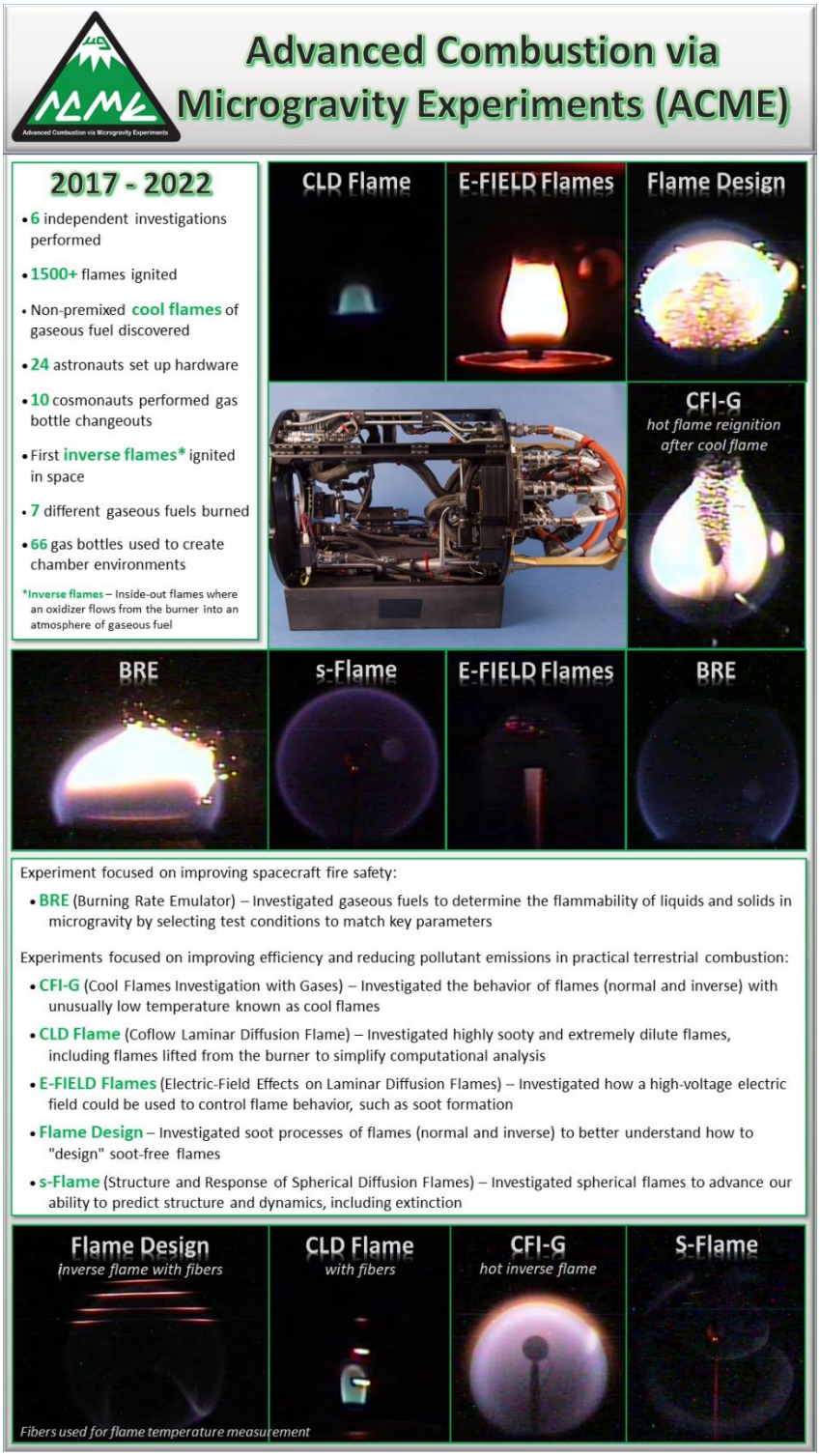


Figure 1. Time Sequence of the Final Flame Design Inverse Flame

Figure 1 shows the inverse spherical flame growing with time to reach enough fuel to consume the oxygen flowing from the porous spherical burner. As the flame grew bigger, the radiative heat loss increased, which caused the flame to weaken and become dimmer. The flame's temperature dropped, which is indicated by the hue of the glowing ceramic fibers, where yellow is hotter than orange, which in turn is hotter than red.



Appendix G. Crew Activity per Space Station Science Highlights

The subsequent text and images are from the weekly highlights prepared by the Lead Increment Scientist and made available via e-mail and online via www.nasa.gov/mission_pages/station/research/.

With the major exception of the Space Station Science Highlights, links were generally not included in the excerpts herein because they typically go to secondary sources of information for the ACME project, for which the primary web site is https://www1.grc.nasa.gov/space/iss-research/iss-fcf/cir/acme/. ***Clarifications have been added in red.***

2019

First round (normal flames)

Week of April 1, 2019

Other investigations on which the crew performed work:

- The Combustion Integrated Rack (CIR) includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion investigations in microgravity.

Week of April 8, 2019

Other investigations on which the crew performed work:

- The Combustion Integrated Rack (CIR) includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion investigations in microgravity.



NASA astronaut Nick Hague replaced an igniter in the Advanced Combustion via Microgravity Experiment (ACME), a set of six studies of gaseous flames conducted in the Combustion Integrated Rack (CIR). The CIR includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion experiments in microgravity.

Credits: NASA

While Hague did support Flame Design operations, this photo is from April 3 work in support of ACME's Burning Rate Emulator (BRE) experiment. That study's final test day was on April 4, while Christina Koch began Flame Design preparations on April 5.

Week of April 15, 2019

Other investigations on which the crew performed work:

- The Combustion Integrated Rack (CIR) includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion investigations in microgravity.

Week of July 15, 2019

Cleaner combustion

The crew completed final runs for the Advanced Combustion via Microgravity Experiments (ACME) Flame Design, which studies the production and control of soot to optimize oxygen-enriched combustion and the design of robust, soot-free flames. Soot can adversely affect efficiency, emissions, and equipment lifetime. Flame Design is part of a series of independent ACME experiments using the space station's Combustion Integrated Rack (CIR).

Week of July 22, 2019

Other investigations on which the crew performed work:

- ACME Flame Design, which studies the production and control of soot to optimize oxygen-enriched combustion and the design of robust, soot-free flames, is part of a series of independent ACME experiments using the space station's Combustion Integrated Rack (CIR).



NASA astronaut Nick Hague services the Combustion Integrated Rack (CIR), which supports safe flame and fuel research with potential benefit for fire safety on Earth and in space as well as applications in the design of advanced combustion systems for spacecraft and Earth-bound vehicles.

Credits: NASA

Flame Design, and the ACME project of which it was a part, is generally focused on practical terrestrial combustion and it is not specifically focused on combustion for vehicles. None of the ACME investigations were directed toward combustion systems for spacecraft, and improved fire safety was the primary goal of only one of ACME's studies, the BRE experiment, where its focus was specific to spacecraft.

Week of July 29, 2019

Other investigations on which the crew performed work:

- ACME Flame Design, which studies the production and control of soot to optimize oxygen-enriched combustion and the design of robust, soot-free flames, is part of a series of independent ACME experiments using the space station's Combustion Integrated Rack (CIR).

While the crew did work on ACME during this week, it was for the s-Flame experiment. First-round operations for Flame Design ended on July 26.

Week of August 5, 2019

Other investigations on which the crew performed work:

- ACME Flame Design, which studies the production and control of soot to optimize oxygen-enriched combustion and the design of robust, soot-free flames, is part of a series of independent ACME experiments using the space station's Combustion Integrated Rack (CIR).

While the crew did work on ACME during this week, it was for the s-Flame experiment. First-round operations for Flame Design ended on July 26.

2021

Second round (inverse flames expect for the first test day)

Week of October 25, 2021

Other investigations involving the crew:

- Flame Design studies the production and control of soot, which can adversely affect combustion efficiency, emissions, and equipment lifetime. Part of the ACME series of studies, this investigation could lead to more efficient and cleaner burner designs.

Week of November 1, 2021

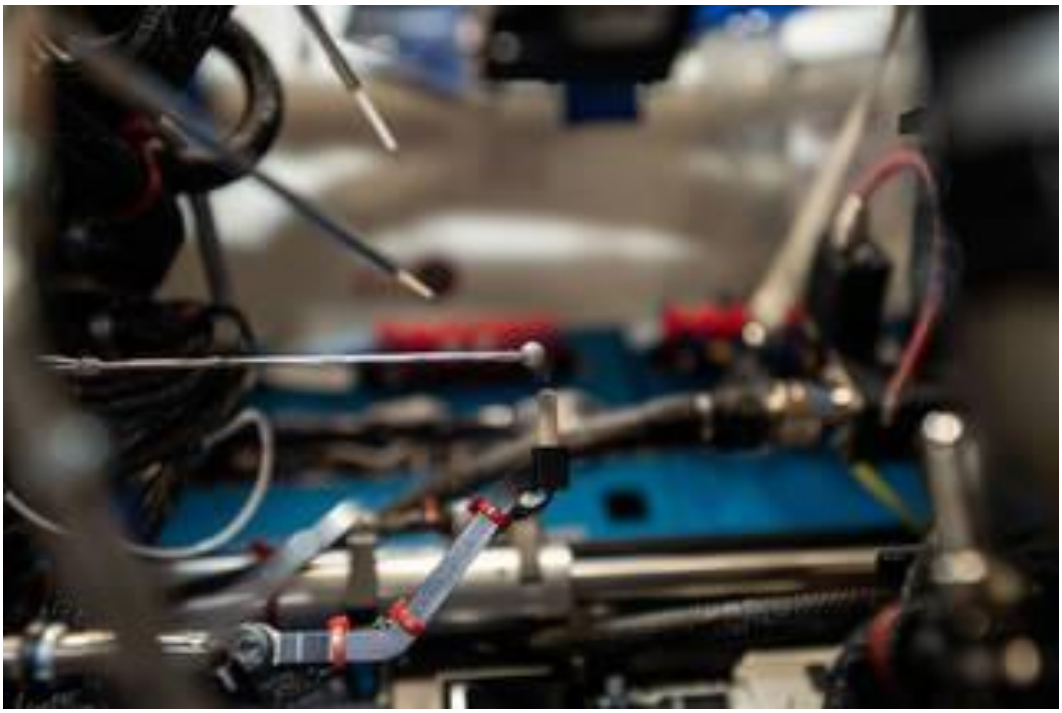
Other investigations involving the crew:

- Flame Design studies the production and control of soot, which can adversely affect combustion efficiency, emissions, and equipment lifetime. Part of the ACME series of studies, this investigation could lead to more efficient and cleaner burner designs.

Week of December 13, 2021

Better burning

The crew replaced fuel bottles in support of the on-going Flame Design investigation, which studies the production and control of soot in oxygen-enriched combustion and the design of soot-free flames. Soot can adversely affect efficiency, emissions, and equipment lifetime, so this research may lead to cleaner and more efficient burner designs for combustion applications on Earth. Findings also could aid the development of future space-based combustion devices for tasks such as solid waste processing or to improve spacecraft fire safety. The experiment is conducted with spherical flames of gaseous fuels in the Combustion Integrated Rack (CIR) as part of the Advanced Combustion via Microgravity Experiments (ACME) project.



Set-up for the ACME Flame Design experiment, which studies the production and control of soot-free flames in an effort to create more efficient and cleaner burner designs for combustion applications on Earth and in space.
Credits: NASA

Week of December 20, 2021

Other investigations involving the crew:

- Flame Design studies the production and control of soot in oxygen-enriched combustion and the design of soot-free flames. This research may lead to cleaner and more efficient burner designs for combustion applications on Earth and aid the development of future space-based combustion devices for tasks such as solid waste processing or to improve spacecraft fire safety.

Week of January 17, 2022

Other investigations involving the crew:

- Flame Design, part of the ACME series of payloads, studies the production and control of soot in oxygen-enriched combustion and the design of soot-free flames. This research may lead to cleaner and more efficient burner designs for combustion applications on Earth and aid the development of future space-based combustion devices for tasks such as solid waste processing or to improve spacecraft fire safety.

Week of January 24, 2022

Other investigations involving the crew:

- Flame Design, part of the ACME series of payloads, studies the production and control of soot in oxygen-enriched combustion and the design of soot-free flames. This research may lead to cleaner and more efficient burner designs for combustion applications on Earth and aid the development of future space-based combustion devices for tasks such as solid waste processing or to improve spacecraft fire safety.

Appendix H. Crew Activity per the ISS Daily Summary Report

*This appendix was compiled from https://blogs.nasa.gov/stationreport. Headings were added identifying the investigation's two rounds of testing, but Flame Design operations listed under the headings such as "*All activities are complete unless otherwise noted*" or "Completed Task List Activities" were not included in this report. Some days of crew activity were only listed in those ways and as a result this section does not comprehensively document crew activity in support of the Flame Design experiment.* ***Clarifications can be found in red.***

First round (normal flames)

2019

2019 April 9: GMT 099

Combustion Integration Rack (CIR)/Advanced Combustion via Microgravity Experiments (ACME) Hardware Replacement: The crew changed the CIR/ACME system from the BRE (Burning Rate Emulator) experiment configuration to the Flame Design experiment configuration. The Flame Design investigation studies the production and control of soot to optimize oxygen-enriched combustion and the design of robust, soot-free flames. Soot can adversely affect efficiency, emissions, and equipment lifetime, so this may lead to more efficient and cleaner burner designs. The experiment is conducted with spherical flames of gaseous fuels in the CIR as part of the Advanced ACME project. Flame Design is one of five experiments hosted by CIR/ACME.

ACME has six experiments when including the Cool Flames Investigation with Gases (CFI-G), which was selected for spaceflight by the Center for Advancement of Science in Space (CASIS) and National Science Foundation (NSF) rather than NASA.

2019 May 10: GMT 130

Combustion Integration Rack (CIR)/Advanced Combustion via Microgravity Experiments (ACME): The crew replaced the ACME Igniter Tip and installed the ACME Short Thermocouple Array, which will allow the continuation of the ACME/Flame Design investigation. The Flame Design investigation studies the production and control of soot to optimize oxygen-enriched combustion and the design of robust, soot-free flames. Soot can adversely affect efficiency, emissions, and equipment lifetime, so this may lead to more efficient and cleaner burner designs. The experiment is conducted with spherical flames of gaseous fuels in the CIR as part of the ACME project. Flame Design is one of five experiments hosted by CIR/ACME.

ACME has six experiments when including the Cool Flames Investigation with Gases (CFI-G).

2019 May 31: GMT 151

CIR (Combustion Integrated Rack): The crew replaced the CIR ACME (Advanced Combustion via Microgravity Experiments) controller and replaced the manifold-2 O2 and N2 bottles. The CIR includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion experiments in microgravity.

ACME has six experiments when including the Cool Flames Investigation with Gases (CFI-G).

2019 June 14: GMT 165

CIR (Combustion Integration Rack)/ACME (Advanced Combustion via Microgravity Experiments)/Flame Design hardware replacement: The crew opened the CIR rack doors, opened the CIR combustion chamber front end cap, and removed the ACME chamber insert to replace the burned out igniter tip. The

CIR team also took this opportunity to replace the burner head, since the current burner head seems to be having difficulty producing gas flows which encourage a spherical flame. The on-going Flame Design investigation studies the production and control of soot in to optimize oxygen-enriched combustion and the design of robust, soot-free flames. Soot can adversely affect efficiency, emissions, and equipment lifetime, so this may lead to more efficient and cleaner burner designs. The experiment is conducted with spherical flames of gaseous fuels in the CIR as part of the ACME project. Flame Design is one of five experiments hosted by CIR/ACME.

ACME has six experiments when including the Cool Flames Investigation with Gases (CFI-G).

2019 July 11: GMT 192

Combustion Integration Rack/Advanced Combustion via Microgravity Experiments (CIR/ACME): The crew exchanged the mass flow controller and manifold gas bottle. The different flow controllers allow the ACME experimenters to study combustion under different gas flow regimes. The currently on-going Flame Design investigation studies the production and control of soot in order to optimize oxygen-enriched combustion and the design of robust, soot-free flames. Soot can adversely affect efficiency, emissions, and equipment lifetime, so this may lead to more efficient and cleaner burner designs. The experiment is conducted with spherical flames of gaseous fuels in the CIR as part of the ACME project. Flame Design is one of five experiments hosted by CIR/ACME.

ACME has six experiments when including the Cool Flames Investigation with Gases (CFI-G).

2019 July 17: GMT 198

Combustion Integration Rack (CIR)/Advanced Combustion via Microgravity Experiments (ACME) /Flame Design: The crew replaced the igniter tip which burned out this week to complete the final Flame Design runs. The investigation studies the production and control of soot to optimize oxygen-enriched combustion and the design of robust, soot-free flames. Soot can adversely affect efficiency, emissions, and equipment lifetime, so this may lead to more efficient and cleaner burner designs. The experiment is conducted with spherical flames of gaseous fuels in the CIR as part of the ACME project.

2019 July 22: GMT 203

Combustion Integration Rack (CIR)/Advanced Combustion via Microgravity Experiments (ACME)/Flame Design: The crew replaced N2 and O2 bottles in the ACME facility. The Flame Design investigation studies the production and control of soot in order to optimize oxygen-enriched combustion and the design of robust, soot-free flames. Soot can adversely affect efficiency, emissions, and equipment lifetime, so this may lead to more efficient and cleaner burner designs. The experiment is conducted with spherical flames of gaseous fuels in the Combustion Integrated Rack (CIR) as part of the Advanced Combustion via Microgravity Experiments (ACME) project.

2019 July 30: GMT 211

Combustion Integration Rack (CIR)/ Advanced Combustion via Microgravity Experiments (ACME) Manifold Bottle Exchange: The crew replaced the 40% O2 gas bottle with a 30% O2 bottle. They also replaced the ethylene fuel bottle with a hydrogen/methane mixture fuel bottle. Telemetry showed a slow leak in the hydrogen/methane bottle and teams are discussing a recovery plan. Flame Design part 1 investigation is complete and preparations are in work for transition to the s-Flame investigation. The purpose of s-Flame is to advance our ability to predict the structure and dynamics, including extinction and instabilities, of both soot-free and sooty flames. The results may contribute to the development of lean-burn engines for improved efficiency and reduced pollutant emissions here on Earth.

This crew activity was in support of the s-Flame investigation, but the entry was included because it references the end of Flame Design's first round operations, for which the final test day was on July 25.

Second round (inverse flames except for the first day)
2021

2021 Oct 26: GMT 299

Combustion Integrated Rack/Advanced Combustion via Microgravity Experiments (CIR/ACME): The crew removed a used bottle with 30% butane/50% N2 and replaced it with a 100% propane gas bottle. This was performed in preparation for a series of calibration burns prior to the Flame Design part 2 experiment. The Flame Design investigation studies the production and control of soot in order to optimize oxygen-enriched combustion and the design of robust, soot-free flames. Soot can adversely affect efficiency, emissions, and equipment lifetime, so this may lead to more efficient and cleaner burner designs.

2021 Nov. 2: GMT 306

Advanced Combustion via Microgravity Experiments (ACME): A crewmember replaced the ACME Fiber Arm 1, Pressure Switch 2 with the ACME Pressure Switch Jumper, and replaced the ACME Igniter Tip 2. This is in preparation of Flame Design part 2 operations. The ACME project is a set of six independent studies of gaseous flames to be conducted in the Combustion Integrated Rack (CIR). ACME's primary and secondary goals are (1) improved fuel efficiency and reduced pollutant production in practical combustion on Earth, and (2) spacecraft fire prevention through innovative research focused on materials flammability.

2021 Nov. 3: GMT 307

ACME (Advanced Combustion via Microgravity Experiments): A crewmember performed the second part of configuration for Flame Design part 2 by replacing two ACME controllers and setting the CIR Valve Timers to limit oxygen flow. The ACME project is a set of six independent studies of gaseous flames to be conducted in the Combustion Integrated Rack (CIR). ACME's primary and secondary goals are (1) improved fuel efficiency and reduced pollutant production in practical combustion on Earth, and (2) spacecraft fire prevention through innovative research focused on materials flammability.

CIR (Combustion Integrated Rack): A crewmember replaced the Manifold #2 CIR High Percentage Oxygen Bottle with a new bottle containing 85% O2 and 15% N2. The CIR Manifold #4 Bottle was also replaced with a new bottle containing 100% C2H4. The CIR includes an optics bench, combustion chamber, fuel, and oxidizer control, and five different cameras for performing combustion investigations in microgravity.

2021 Dec. 3: GMT 337

ACME (Advanced Combustion via Microgravity Experiments): Crewmembers replaced the Manifold 4 bottle with a new bottle containing 100% Ethylene. The ACME project is a set of six independent studies of gaseous flames to be conducted in the Combustion Integrated Rack (CIR). ACME's primary and secondary goals are (1) improved fuel efficiency and reduced pollutant production in practical combustion on Earth, and (2) spacecraft fire prevention through innovative research focused on materials flammability.

2021 Dec. 13: GMT 347

Combustion Integrated Rack/Advanced Combustion via Microgravity Experiments (CIR/ACME): The crew replaced a 100% ethylene (CH2=CH2) fuel bottle with a new bottle of the same composition. This was performed in support of the on-going Flame Design part 2 experiment. The Flame Design investigation studies the production and control of soot in order to optimize oxygen-enriched combustion and the design of robust, soot-free flames. Soot can adversely affect efficiency, emissions, and equipment lifetime, so this may lead to more efficient and cleaner burner designs.

2021 Dec. 21: GMT 355

Combustion Integrated Rack/Advanced Combustion via Microgravity Experiments (CIR/ACME): The crew replaced an 85% oxygen/15% nitrogen bottle with another bottle of the same composition. They also replaced a 100% ethylene (CH_2=CH_2) fuel bottle with a new bottle of the same composition. This was performed in support of the on-going Flame Design part 2 experiment. The Flame Design investigation studies the production and control of soot in order to optimize oxygen-enriched combustion and the design of robust, soot-free flames. Soot can adversely affect efficiency, emissions, and equipment lifetime, so this may lead to more efficient and cleaner burner designs.

2022

2022 Jan. 5: GMT 005

Combustion Integrated Rack (CIR): CIR Manifold #4 bottle was replaced with a new Bottle containing 100% Ethane (C_2H_4). The CIR includes an optics bench, combustion chamber, fuel, and oxidizer control, and five different cameras for performing combustion investigations in microgravity.

2022 Jan. 17: GMT 017

Combustion Integrated Rack/Advanced Combustion via Microgravity Experiments (CIR/ACME): After removing the ACME experiment insert, a pressure switch and igniter tip were replaced, and a mis-aligned fiber array was brought back into alignment. This was performed in support of the on-going Flame Design part 2 experiment. The silicon carbide fibers in the array allow temperature determination of various areas in and around the flame based on the color they glow. The Flame Design investigation studies the production and control of soot in order to optimize oxygen-enriched combustion and the design of robust, soot-free flames. Soot can adversely affect efficiency, emissions, and equipment lifetime, so this may lead to more efficient and cleaner burner designs.

2022 Jan. 24: GMT 024

Combustion Integrated Rack/Advanced Combustion via Microgravity Experiments (CIR/ACME): After removing the ACME experiment insert, a pressure switch and igniter tip were replaced, and a mis-aligned fiber array was brought back into alignment. This was performed in support of the on-going Flame Design part 2 experiment. The silicon carbide fibers in the array allow temperature determination of various areas in and around the flame based on the color they glow. The Flame Design investigation studies the production and control of soot in order to optimize oxygen-enriched combustion and the design of robust, soot-free flames. Soot can adversely affect efficiency, emissions, and equipment lifetime, so this may lead to more efficient and cleaner burner designs.

2022 Jan. 26: GMT 026

Combustion Integrated Rack (CIR): The CIR high percentage oxygen bottle was replaced with a new bottle containing 85% O2 and 15% N2. The fuel bottle in Manifold 4 was replaced with a new bottle containing 100% C2H4. The CIR includes an optics bench, combustion chamber, fuel, and oxidizer control, and five different cameras for performing combustion investigations in microgravity.

2022 Jan. 31: GMT 031

Combustion Integrated Rack (CIR): The CIR Manifold #4 Bottle was swapped out with a bottle containing 100% C2H4. The CIR includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion investigations in microgravity.

2022 Feb. 8: GMT 039

Combustion Integrated Rack (CIR): The crew performed steps in support of the multi-part rack reconfiguration from the Advanced Combustion via Microgravity Experiments (ACME) experiment insert to the Solid Fuel Ignition and Extinction (SoFIE) experiment insert. Specifically, the crew exchanged bottles at manifold 2 and 4, exchanged the adsorber cartridge, set the CIR valve timers, and disconnected the static mixer hose inside the CIR combustion chamber. SoFIE is a hardware insert for CIR that enables a wide range of solid-material combustion and fire suppression studies, providing common infrastructure including sample holders, flow control, test sections, external radiant heaters, igniters, and diagnostics for multiple investigations. While SoFIE initially meets the requirements of five investigations, is intended for use by any researchers who propose related solid combustion studies to NASA in the future.

The transition from ACME to the SoFIE project is not included in this log, except for this initial example, because it is considered here as SoFIE setup.

Appendix I. Crew Activity per the NASA Space Station Blog

This section includes an excerpt from https://blogs.nasa.gov/spacestation/. Hyperlinks were generally not included.

Second round (inverse flames except for the first day)

2021

2021 Oct. 26 - Station Gets Ready for Russian Cargo and SpaceX Crew Missions

Science as usual is continuously ongoing aboard the space station. NASA Flight Engineer Mark Vande Hei swapped fuel bottles inside the Combustion Integrated Rack.

Appendix J. Crew Activity in Publicly Available NASA Photos

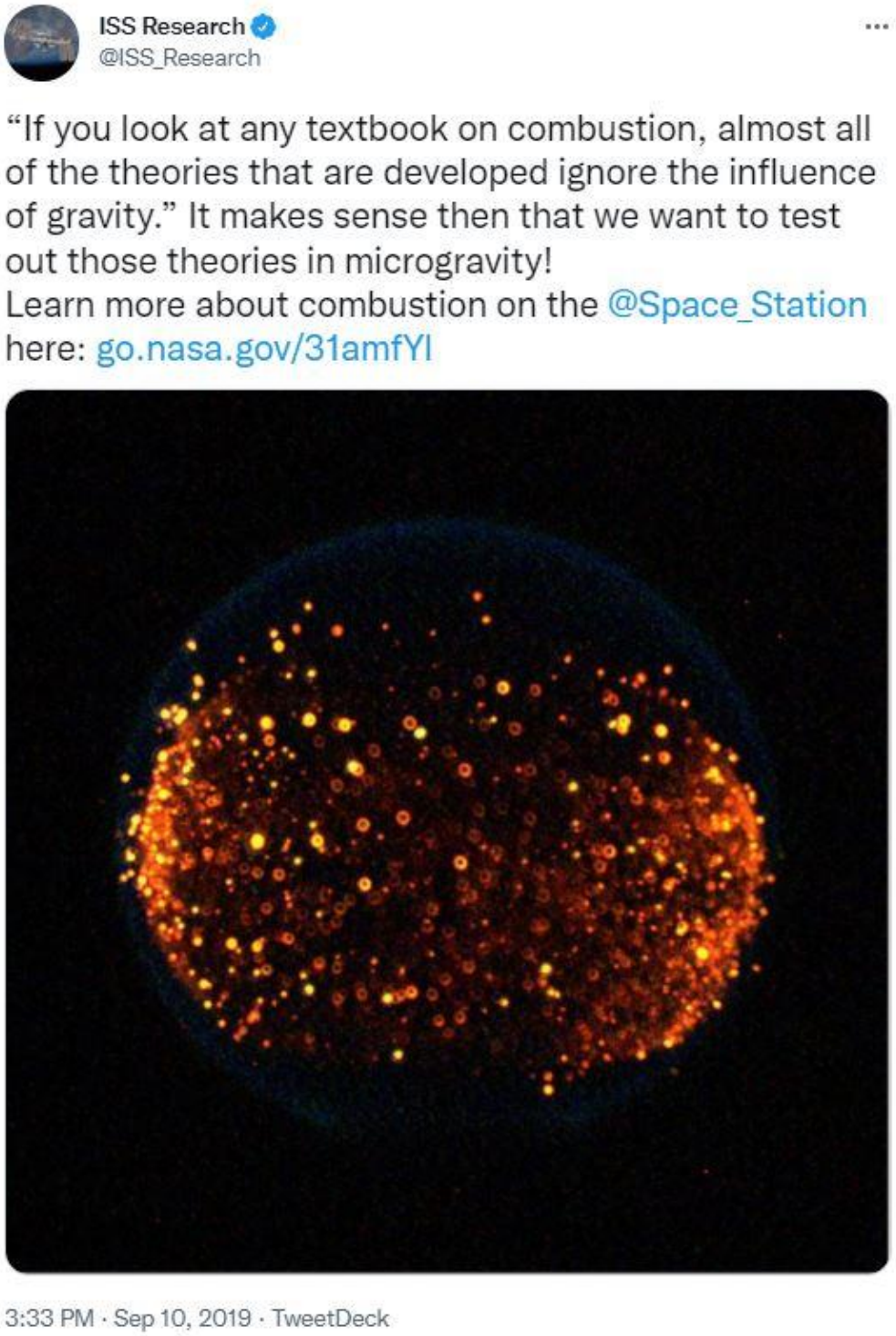
These photos of Flame Design operations are available from https://images.nasa.gov/. Additional photos from the ACME project, CIR facility, etc. can be found by searching that website.

Photo ID	Date taken	Summary description
iss059e017072	2019-04-09	Christina Koch (NASA) with the ACME chamber insert on workbench where a BRE burner can be seen (prior to its replacement with a spherical burner)
iss059e017127	2019-04-09	Christina Koch (NASA) with the ACME insert within open Combustion Integrated Rack (CIR) chamber
iss059e060847	2019-05-10	David Saint-Jacques (CSA) with the ACME insert partially withdrawn from open Combustion Integrated Rack (CIR) chamber
iss059e085880	2019-05-31	Nick Hague (NASA) with the ACME insert within open Combustion Integrated Rack (CIR) chamber
iss059e103188	2019-06-14	spherical burner within ACME chamber insert
iss060e011502	2019-07-17	Nick Hague (NASA) in mask and goggles with the ACME insert within open CIR chamber
iss066e029031	2021-11-02	Shane Kimbrough (NASA) with the Combustion Integrated Rack (CIR) with the chamber closed
iss066e031702	2021-11-02	Shane Kimbrough (NASA) with the ACME insert held outside of open Combustion Integrated Rack (CIR) chamber
iss066e095422	2021-12-21	Raja Chari (NASA) with the top door of the Combustion Integrated Rack (CIR) open for bottle swap(s)
iss066e114301	2022-01-17	Raja Chari (NASA) with the ACME insert partially withdrawn from open CIR chamber
iss066e114415	2022-01-17	Raja Chari (NASA) with the ACME insert partially withdrawn from the CIR, of which only the hatch door is visible

Appendix K. ACME Operations per @ISS_Research on Twitter

These tweets (from https://twitter.com/ISS_Research/) featured Flame Design images. There may be others which are not included here.

2019 Sept. 10: GMT 253

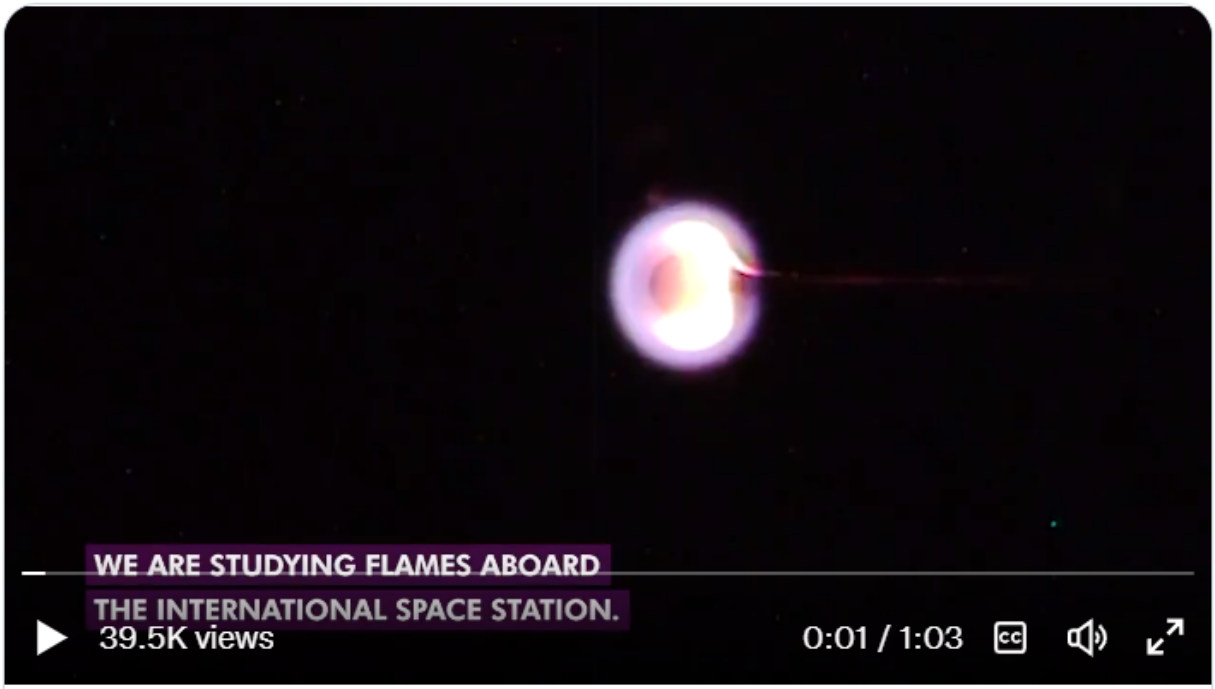


2019 Sept. 12: GMT 255



ISS Research ✓
@ISS_Research

Flames are shaped differently in microgravity, looking like spheres rather than a typical candle flame you might be familiar with.
The forces that change the flame's shape also make the @Space_Station an intriguing place to study combustion. Read: go.nasa.gov/31amfYl



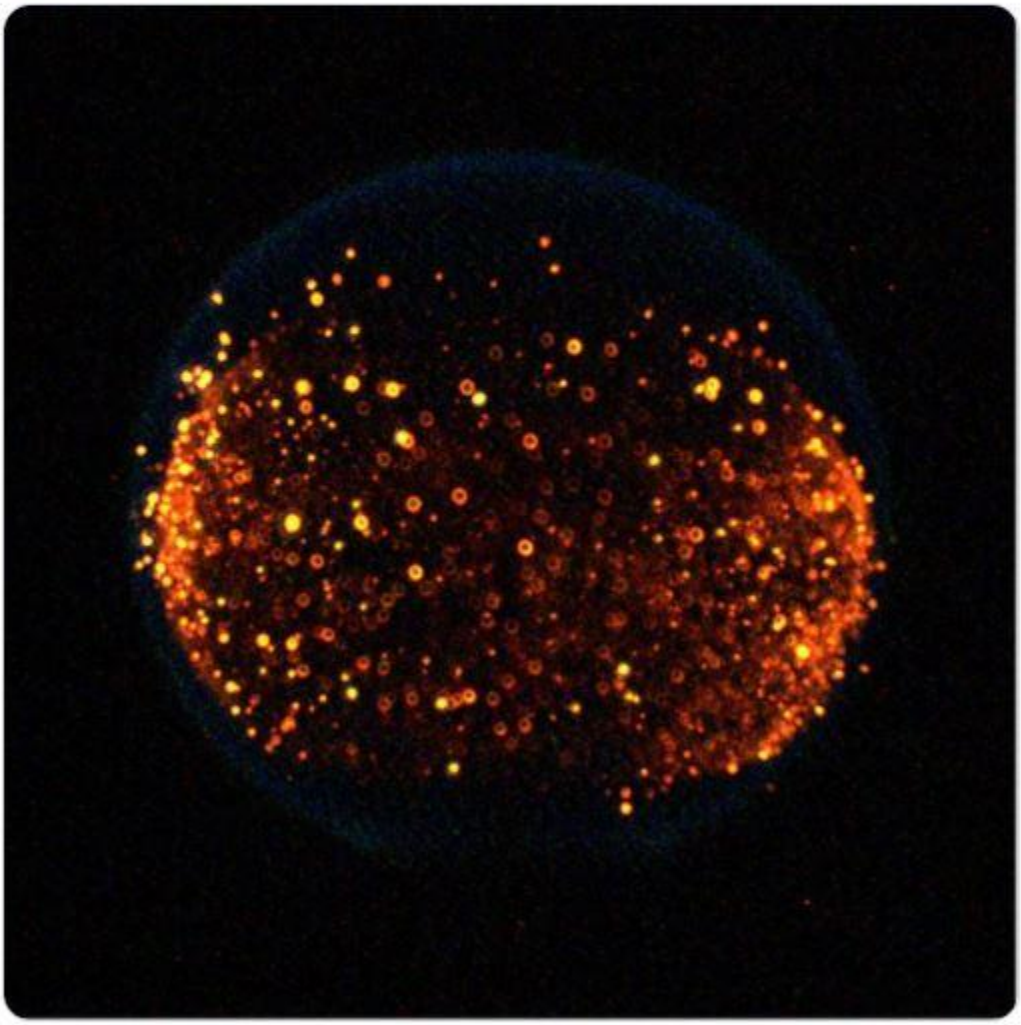
Designing Flames Aboard the International Space Station
The Flame Design investigation is studying the quantity of soot produced under different flame conditions. The results of this experiment could enable the design of flames that are sooty or soot-free.

3:25 PM · Sep 12, 2019

2019 Oct. 18: GMT 291

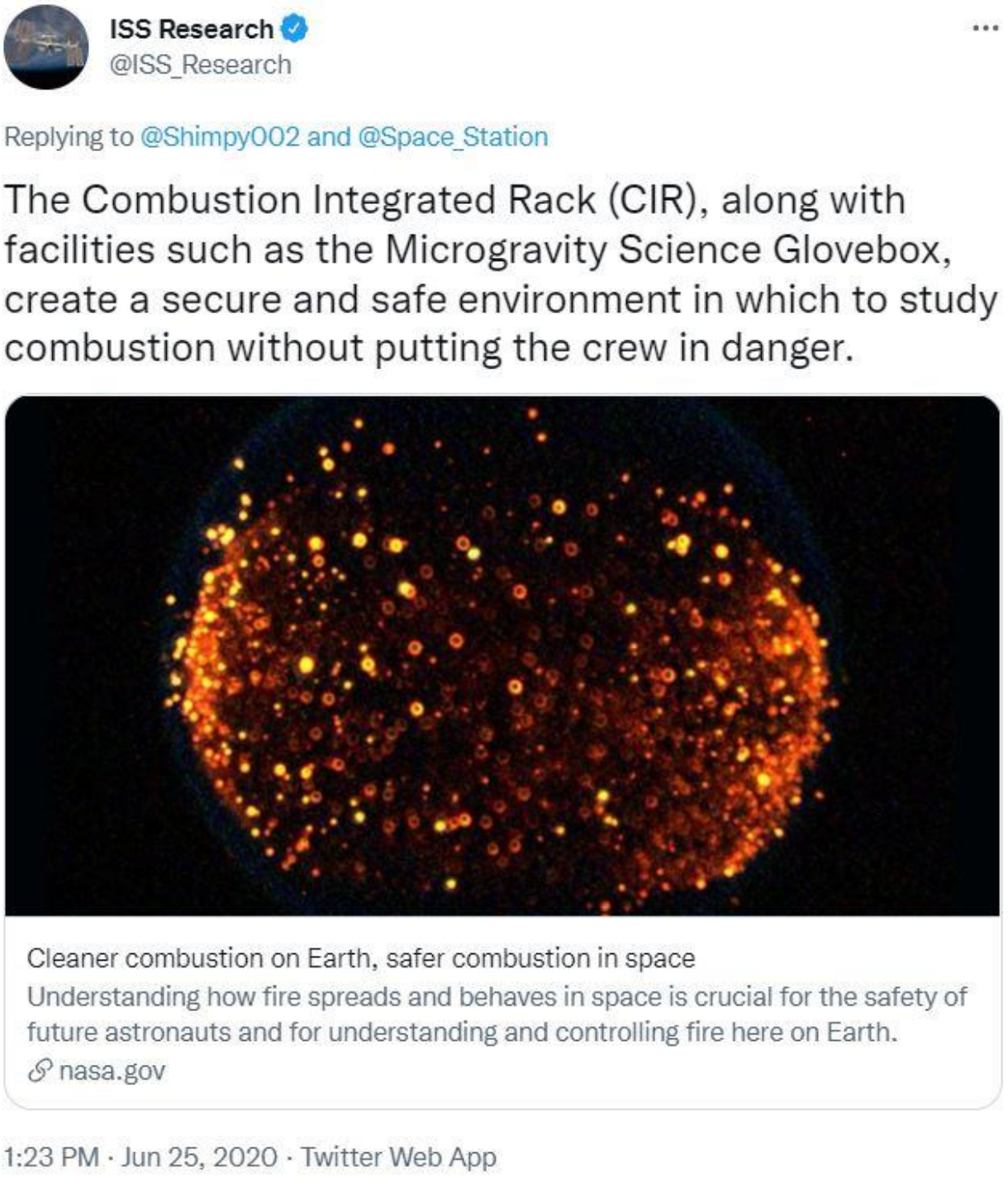
ISS Research
@ISS_Research

The crew has helped with a number of combustion experiments on the @Space_Station which can help give us a better understanding of flames here on Earth. go.nasa.gov/31amfYl



12:20 PM · Oct 18, 2019 · Twitter Web App

2020 June 25: GMT 177



2020 July 23: GMT 205

ISS Research
@ISS_Research

We have now lit more than 1,000 flames as a part of ACME, a series of microgravity combustion experiments aboard the @Space_Station.

What researchers learn from these thousand points of light could improve the use of combustion on Earth. go.nasa.gov/3hsXa2l



Designing Flames Aboard the International Space Station

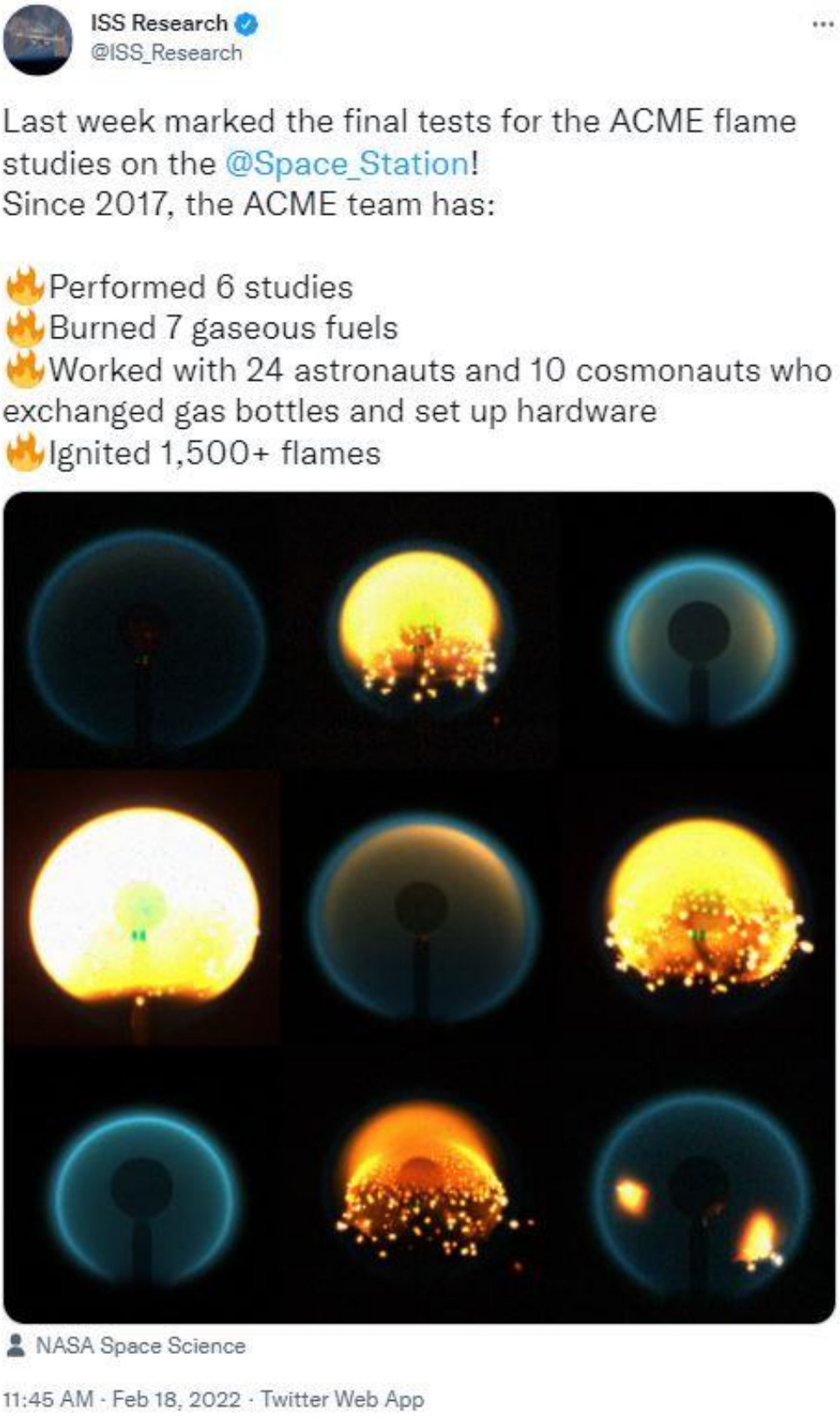
The Flame Design investigation is studying the quantity of soot produced under different flame conditions. The results of this experiment could enable the design of flames that are sooty or soot-free.

1:13 PM · Jul 23, 2020

2021 March 29: GMT 088



2022 Feb. 18: GMT 049



2022 Feb. 18: GMT 049



Appendix L. ACME Operations per Astronauts on Twitter

This tweet is from an ISS crew member who conducted operations in support of the Flame Design experiment. There may be others which are not included here.

Christina H. Koch (@Astro_Christina) on Twitter

https://twitter.com/astro_christina/

2019 April 23: GMT 113



Appendix M. Summary of the ISS data collection

The Flame Design data is all digital and includes both image and non-image data, where that is generally true for the ACME experiments. To be clear, there was no collection of soot or other physical samples that were subsequently transported to the ground.

The Flame Design image data includes images from the following cameras:

1. ACME DataCam, used with a clear fused-silica 'filter' and various zoom and focus settings (but those settings were consistent within each individual test)
2. CFI intensified camera, used at least generally with a CH* filter
3. High-Bitdepth Multispectral (HiBMs) camera, equipped with a red filter corresponding to the opposing CIR illumination package

The raw DataCam images are in a Halcon Iconic Object (hobj) format as well as a converted Tagged Image File (tif) format. NASA's post-processing of the raw images was carried out without each of the following:

a. color balancing (to avoid possible saturation)
b. dark frame subtraction
c. removal of hot spots from dead pixels (e.g., resulting from radiation damage)

Furthermore, the conversion employed a standard bilinear demosaicing process which is detrimental when determining temperature profiles via Thin Filament Pyrometry (TFP) because it will average pixels both on and off the glowing fibers.

The images from the CFI and HiBMs cameras are provided in a tif format. Meanwhile, the analog operations (ops) video was recorded on the ground in a Transport Stream (ts) format.

The key non-imaging data from a test is a comma-delimited file that is described in detail in the *Software Interface Design Description* (ACME-DOC-010 Rev. D).

The CIR's Gas Chromatograph Instrumentation Package (GCIP) was used during a limited number of Flame Design test days and the resulting raw data files are in a text (i.e., ASCII) format.

Consult the forthcoming *ACME/Flame Design Data Interpretation* document to learn more.

The raw Flame Design data is currently archived at NASA and work is underway to make it available publicly through NASA's Physical Science Informatics System (PSI).

Appendix N. Publications, presentations, awards, and outreach

N.1 Publications and Presentations with ISS Results

Refereed Journal Articles

29. P.H. Irace, K. Waddell, D. Constales, P.B. Sunderland, R.L. Axelbaum, *Critical temperature and reactant mass flux for radiative extinction of ethylene microgravity spherical diffusion flames at 1 bar*, Proc. Combust. Inst. (in press) https://doi.org/10.1016/j.proci.2022.07.043
28. P.H. Irace, K Waddell, D. Constales, G. Yablonsky, M. Kim, P.B. Sunderland, R.L. Axelbaum, *On the existence of steady state gaseous microgravity spherical diffusion flames in the presence of radiation heat loss*, Proc. Combust. Inst. (in press) https://doi.org/10.1016/j.proci.2022.07.049
27. P.H. Irace, H. Lee, K. Waddell, L. Tan, D.P. Stocker, P.B. Sunderland, R.L. Axelbaum, *Observations of long duration microgravity spherical diffusion flames aboard the International Space Station*, Combustion and Flame, 229 (2021) 111373. https://doi.org/10.1016/j.combustflame.2021.02.019

Conference Proceedings, Abstracts, Posters, Etc.

64. P.H. Irace, K. Waddell, Z. Xu, D. Constales, P.B. Sunderland, R.L. Axelbaum, *Microgravity Spherical Diffusion Flames: The Critical Point for Radiative Extinction and the Dynamics to Reach It*, U.S. National Combustion Meeting (2021).
63. K. Waddell, P.B. Sunderland, S. Medvedev, S. Frolov, P.H. Irace, R.L. Axelbaum, *Flame and Burner Temperature Measurements and Predictions in Spherical Diffusion Flames*, U.S. National Combustion Meeting, 2021, 9 pp.
62. K. Waddell, P.B. Sunderland, S. Medvedev, S. Frolov, P.H. Irace, R.L. Axelbaum, *Flame and Burner Temperature Measurements and Predictions in Spherical Diffusion Flames*, Abstract, ASGSR, Baltimore, 2021.
61. P. Irace, L. Tan, K. Waddell, H.J. Lee, D. Stocker, D.L. Urban, P.B. Sunderland, R.L. Axelbaum, *Flame Ignition in Space*, Art Competition Poster, ASGSR, 2020.
60. P.H. Irace, H.J. Lee, K. Waddell, L. Tan, D.P. Stocker, P.B. Sunderland, R.L. Axelbaum, *Observations of Long Duration Microgravity Spherical Diffusion Flames Aboard the International Space Station*, Abstract, Annual ASGSR Meeting, 2020.
59. P. Irace, H.J. Lee, K. Waddell, L. Tan, D.P. Stocker, D.L. Urban, P.B. Sunderland, R.L. Axelbaum, *Steady-State and Radiatively Extinguishing Spherical Diffusion Flames in Microgravity*, Abstract, ASGSR, Denver, 2019.
58. P. Irace, L. Tan, K. Waddell, H.J. Lee, D.P. Stocker, D.L. Urban, P.B. Sunderland, R.L. Axelbaum, *A Galaxy of Soot Particles*, Art Competition Entry, ASGSR, Denver, 2019.

Theses and Dissertations

15. Phillip H. Irace, *Particle Formation and Thermal Radiation in Laminar Diffusion Flames with Applications to Energy and Materials*, Ph.D. Dissertation, Washington University in St. Louis, Dept. of Mechanical Engineering and Materials Science, 200+ pp, May 2022.

N.2 Publications and Presentations (comprehensive list)

Papers, etc. in this section which include ISS results are enclosed in borders like this.

Refereed Journal Articles

30. P.H. Irace, A. Gopan, R.L. Axelbaum, *An investigation of thermal radiation from laminar diffusion flames in a tri-coflow burner with central oxygen*, Combust and Flame, 242 (2022) 112158. https://doi.org/10.1016/j.combustflame.2022.112158

29. P.H. Irace, K. Waddell, D. Constales, P.B. Sunderland, R.L. Axelbaum, *Critical temperature and reactant mass flux for radiative extinction of ethylene microgravity spherical diffusion flames at 1 bar*, Proc. Combust. Inst. (in press) https://doi.org/10.1016/j.proci.2022.07.043
28. P.H. Irace, K Waddell, D. Constales, G. Yablonsky, M. Kim, P.B. Sunderland, R.L. Axelbaum, *On the existence of steady state gaseous microgravity spherical diffusion flames in the presence of radiation heat loss*, Proc. Combust. Inst. (in press) https://doi.org/10.1016/j.proci.2022.07.049
27. P.H. Irace, H. Lee, K. Waddell, L. Tan, D.P. Stocker, P.B. Sunderland, R.L. Axelbaum, *Observations of long duration microgravity spherical diffusion flames aboard the International Space Station*, Combustion and Flame, 229 (2021) 111373. https://doi.org/10.1016/j.combustflame.2021.02.019
26. Z. Wang, P.B. Sunderland, R.L. Axelbaum, *Double Blue Zones in Inverse and Normal Laminar Jet Diffusion Flames*, Combustion and Flame 211 (2020) 253-259. https://doi.org/10.1016/j.combustflame.2019.09.014
25. Z. Wang, P.B. Sunderland, R.L. Axelbaum, *Dilution Effects on Laminar Jet Diffusion Flame Lengths*, Proceedings of the Combustion Institute 37 (2019) 1547-1553. https://doi.org/10.1016/j.proci.2018.06.085
24. W. Wu and R.L. Axelbaum, *A new method of flame temperature measurement utilizing the acoustic emissions from laser-induced plasmas,* Proceedings of the Combustion Institute 37 (2019) 1409-1415. https://doi.org/10.1016/j.proci.2018.07.096
23. M.K. Rodenhurst III, B.-H. Chao, P.B. Sunderland, R.L. Axelbaum, *Structure and Extinction of Spherical Burner-Stabilized Diffusion Flames that are Attached to the Burner Surface*, Combustion and Flame 187 (2018) 22-29. https://doi.org/10.1016/j.combustflame.2017.08.024
22. A. Gopan, Z. Yang, B.M. Kumfer, R.L. Axelbaum, *Effects of Inert Placement (Zst) on Soot and Radiative Heat Flux in Turbulent Diffusion Flames*, Energy and Fuels 31 (2017) 7617-7623. https://doi.org/10.1021/acs.energyfuels.7b01092
21. W. Wu, G. Yablonsky, R.L. Axelbaum, *Observation of water–gas shift equilibrium in diffusion flames*, Combustion and Flame 173 (2016) 57-64. https://doi.org/10.1016/j.combustflame.2016.07.025
20. S. Krishnan, B.M. Kumfer, W. Wu, J. Li, A. Nehorai, R.L. Axelbaum, *An Approach to Thermocouple Measurements That Reduces Uncertainties in High-Temperature Environments*, Energy and Fuels 29 (2015) 3446-3455. https://doi.org/10.1021/acs.energyfuels.5b00071
19. V.R. Lecoustre, P.B. Sunderland, B.H. Chao, R.L. Axelbaum, *Modeled Quenching Limits of Spherical Hydrogen Diffusion Flames*, Proceedings of the Combustion Institute 34 (2013) 887-894.

https://doi.org/10.1016/j.proci.2012.07.029

18. F. Xia, R.L. Axelbaum, *Simplifying the Complexity of Diffusion Flames Through Interpretation in C/O Ratio Space,* Computers and Mathematics with Applications, 65 (2013) 1625-1632. https://doi.org/10.1016/j.camwa.2013.01.008
17. F. Xia, G.S. Yablonsky, R.L. Axelbaum, *Numerical Study of Flame Structure and Soot Inception Interpreted in Carbon-to-Oxygen Atom Ratio Space*, Proceedings of the Combustion Institute, 34 (2013) 1085-1091. https://doi.org/10.1016/j.proci.2012.06.043
16. F. Yi, R.L. Axelbaum, *Stability of Spray Combustion for Water/Alcohols Mixtures in Oxygen-Enriched Air*, Proceedings of the Combustion Institute, 34 (2013) 1697-1704. https://doi.org/10.1016/j.proci.2012.05.088
15. V.R. Lecoustre, P.B. Sunderland, B.H. Chao, R.L. Axelbaum, *Numerical Investigation of Spherical Diffusion Flames at their Sooting Limits*, Combustion and Flame 159 (2012) 194-199. https://doi.org/10.1016/j.combustflame.2011.05.022
14. Q. Wang, B.H. Chao, *Kinetic and Radiative Extinctions of Spherical Burner-Stabilized Diffusion Flames*, Combustion and Flame 158 (2011) 1532-1541. https://doi.org/10.1016/j.combustflame.2010.12.007
13. V.R. Lecoustre, P.B. Sunderland, B.H. Chao, R.L. Axelbaum, *Extremely Weak Hydrogen Flames*, Combustion and Flame 157 (2010) 2209-2210. https://doi.org/10.1016/j.combustflame.2010.07.024
12. S.A. Skeen, G. Yablonsky, R.L. Axelbaum *Characteristics of Non-Premixed Oxygen-Enhanced Combustion: II. Flame Structure Effects on Soot Precursor Kinetics Resulting in Soot-Free Flames*, Combustion and Flame 157 (2010) 1745-1752. https://doi.org/10.1016/j.combustflame.2010.04.013
11. S.A. Skeen, G. Yablonsky, R.L. Axelbaum *Characteristics of Non-Premixed Oxygen-Enhanced Combustion: I. The Presence of Appreciable Oxygen at the Location of Maximum Temperature*, Combustion and Flame 156 (2009) 2145–2152. https://doi.org/10.1016/j.combustflame.2009.07.009
10. Kumfer, B.M., Skeen, S.A., Axelbaum, R. L., *Soot Inception Limits in Laminar Diffusion Flames with Application to Oxy-Fuel Combustion*, Combustion and Flame 154 (2008) 546-556. https://doi.org/10.1016/j.combustflame.2008.03.008
9. Santa, K.J., Chao, B.H., Sunderland, P.B., Urban, D.L., Stocker, D.P., Axelbaum, R.L., *Radiative Extinction of Gaseous Spherical Diffusion Flames in Microgravity*, Combustion and Flame 151 (2007) 665-675. https://doi.org/10.1016/j.combustflame.2007.08.009
8. Santa, K.J., Sun, Z., Chao, B.H., Sunderland, P.B., Axelbaum, R.L., Urban, D.L., Stocker, D.P., *Numerical and Experimental Observations of Spherical Diffusion Flames*, Combustion Theory and Modeling 11 (2007) 639-652. https://doi.org/10.1080/13647830601161567
7. Kumfer, B.M., Skeen, S.A., Chen, R., Axelbaum, R.L., *Measurement and Analysis of Soot Inception Limits of Oxygen-Enriched Coflow Flames*, Combustion and Flame 147 (2006) 233-242. https://doi.org/10.1016/j.combustflame.2006.08.004
6. Chen, R., Axelbaum, R.L., *Scalar Dissipation Rate at Extinction and the Effects of Oxygen-Enriched Combustion*, Combustion and Flame 142 (2005) 62-71. https://doi.org/10.1016/j.combustflame.2005.02.008

5. Liu, S., Chao, B.H. and Axelbaum, R.L., *A Theoretical Study on Soot Inception in Spherical Burner-Stabilized Diffusion Flames*, Combustion and Flame 140 1-23 (2005). https://doi.org/10.1016/j.combustflame.2004.08.016
4. Sunderland, P.B., Urban, D.L., Stocker, D.P., Chao, B.-H., Axelbaum, R.L., *Sooting Limits of Microgravity Spherical Diffusion Flames in Oxygen-Enriched Air and Diluted Fuel*, Combustion Science and Technology 176 (2004) 2143-2164. https://doi.org/10.1080/00102200490514994
3. [1]Sunderland, P.B., Axelbaum, R.L., Urban, D.L., Chao, B.H., Liu, S., *Effects of Structure and Hydrodynamics on the Sooting Behavior of Spherical Microgravity Diffusion Flames*, Combustion and Flame 132 (2003) 25-33. https://doi.org/10.1016/S0010-2180(02)00424-8
2. Chao, B.H. and Axelbaum, R.L*., On By-Product Production in Nonpremixed Flames with Implications towards Emission of Organics during Waste Incineration*, Combustion Science and Technology, 160, 191-220 (2000). https://doi.org/10.1080/00102200008935802
1. Chao, B.H., Liu, S., and Axelbaum, R.L., *On Soot Inception in Nonpremixed Flames and the Effects of Flame Structure*, Combustion Science and Technology, Vol. 138, pp. 105-135 (1998). https://doi.org/10.1080/00102209808952065

[1] Recipient of Best-Paper Award from Northern Ohio section of AIAA, 2003.

Conference Proceedings, Abstracts, Posters, Etc.

67. K.A. Waddell, G. Yablonsky, D. Constales, R.L. Axelbaum, P.B. Sunderland, *The kinetics and warm flame chemistry associated with radiative extinction of spherical diffusion flames*, U.S. National Combustion Meeting, College Station, Texas, 8 pp., 2023.
66. P.H. Irace, R.L. Axelbaum, *The effect of stoichiometric mixture fraction on soot formation and thermal radiation in an aromatic compound relative to aliphatic compounds*, 38th International Symposium on Combustion, Poster, Adelaide, Australia, 2021.
65. P.H. Irace, D. Khatri, A. Gopan, R.L. Axelbaum, *Measurements of Thermal Radiation from Laminar Diffusion Flames with Central Oxygen Enrichment in a Triple Port Burner*, 12th U.S. National Combustion Meeting, Virtual, 2021.
64. P.H. Irace, K. Waddell, Z. Xu, D. Constales, P.B. Sunderland, R.L. Axelbaum, *Microgravity Spherical Diffusion Flames: The Critical Point for Radiative Extinction and the Dynamics to Reach It*, U.S. National Combustion Meeting (2021).
63. K. Waddell, P.B. Sunderland, S. Medvedev, S. Frolov, P.H. Irace, R.L. Axelbaum, *Flame and Burner Temperature Measurements and Predictions in Spherical Diffusion Flames*, U.S. National Combustion Meeting, 2021, 9 pp.
62. K. Waddell, P.B. Sunderland, S. Medvedev, S. Frolov, P.H. Irace, R.L. Axelbaum, *Flame and Burner Temperature Measurements and Predictions in Spherical Diffusion Flames*, Abstract, ASGSR, Baltimore, 2021.
61. P. Irace, L. Tan, K. Waddell, H.J. Lee, D. Stocker, D.L. Urban, P.B. Sunderland, R.L. Axelbaum, *Flame Ignition in Space*, Art Competition Poster, ASGSR, 2020.
60. P.H. Irace, H.J. Lee, K. Waddell, L. Tan, D.P. Stocker, P.B. Sunderland, R.L. Axelbaum, *Observations of Long Duration Microgravity Spherical Diffusion Flames Aboard the International Space Station*, Abstract, Annual ASGSR Meeting, 2020.

59. P. Irace, H.J. Lee, K. Waddell, L. Tan, D.P. Stocker, D.L. Urban, P.B. Sunderland, R.L. Axelbaum, *Steady-State and Radiatively Extinguishing Spherical Diffusion Flames in Microgravity*, Abstract, ASGSR, Denver, 2019.
58. P. Irace, L. Tan, K. Waddell, H.J. Lee, D.P. Stocker, D.L. Urban, P.B. Sunderland, R.L. Axelbaum, *A Galaxy of Soot Particles*, Art Competition Entry, ASGSR, Denver, 2019.
57. P.H. Irace, Z. Yang, A. Gopan, R.L. Axelbaum, *Soot Formation and Radiation Heat Transfer in Tri-axial Diffusion Flames.* U.S. National Combustion Meeting, Pasadena, 2019.
56. Z. Wang, P.B. Sunderland, R.L. Axelbaum, *Double Luminous Zones in Laminar Jet Diffusion Flames*, U.S. National Combustion Meeting, Pasadena, 2019, 9 pp.
55. Z. Wang, P.B. Sunderland, R.L. Axelbaum, *Double Blue Zones in Inverse Laminar Jet Diffusion Flames*, Poster, 37th International Symposium on Combustion, Dublin, 2018.
54. Z. Wang, P.B. Sunderland, R.L. Axelbaum, *Double Luminous Zones in Inverse Laminar Jet Diffusion Flames*, Eastern States Section of the Combustion Institute, State College PA, 2018, 6 pp.
53. Z. Wang, P.B. Sunderland, R.L. Axelbaum, *Images of Co-Axial Gas Jet Diffusion Flames*, Art Competition Poster, U.S. National Combustion Meeting, College Park, 2017.
52. Z. Wang, P.B. Sunderland, R.L. Axelbaum, *Observations of Double Flames in Inverse Gas Jet Diffusion Flames*, U.S. National Combustion Meeting, College Park, 2017, 6 pp.
51 Z. Wang, P.B. Sunderland, R.L. Axelbaum, *The Effects of Reactant Dilution on Lengths of Laminar Jet Diffusion Flames*, U.S. National Combustion Meeting, College Park, 2017, 6 pp.
50. Z. Wang, W. Wu, P.B. Sunderland, R.L. Axelbaum, *Ground-Based Research Supporting Flame Design*, ASGSR Abstract, Seattle, 2017.
49. Z. Wang, M. El Makawy, P.B. Sunderland, B. Chao, R.L. Axelbaum, *The Effects of Reactant Dilution on Lengths of Laminar Jet Diffusion Flames*, Abstract, ASGSR, Cleveland OH (2016).
48. Z. Wang, M. El Makawy, P.B. Sunderland, B. Chao, R.L. Axelbaum, *The Effects of Reactant Dilution on Lengths of Laminar Jet Diffusion Flames*, Poster, ASGSR, Cleveland OH, 2016.
47. Z. Wang, P.B. Sunderland, B. Chao, R.L. Axelbaum, *Color Images of Normal and Inverse Diffusion Flames*, Art Competition Poster, ASGSR, Cleveland OH, 2016.
46. W. Wu, R.L. Axelbaum, *Interpreting Flame Structure by Conserved Scalar Measurement Using Laser-Induced Breakdown Spectroscopy*, Combustion Institute Central States Meeting, Knoxville (2016).
45. W. Wu, R.L. Axelbaum, *Interpreting Flame Structure by Mixture Fraction Measurement Using Laser-Induced Breakdown Spectroscopy*, Abstract, ASGSR, Cleveland OH (2016).
44. V.R. Lecoustre, P.B. Sunderland, B.-H. Chao, R.L. Axelbaum, *Numerical Investigations of Temperature Effects on Soot Precursors in Ethylene Premixed Flames*, 9th U.S. National Combustion Meeting, Cincinnati, 10 pp. (2015).
43. W. Wu, R.L. Axelbaum, *Interpreting Diffusion Flame Structure by Measuring C/O Ratio Using Laser-Induced Breakdown Spectroscopy*, 9th U.S. National Combustion Meeting, Cincinnati (2015).
42. W. Wu, R.L. Axelbaum, *Interpreting Flame Structure by Scalar Measurement Using Laser-Induced Breakdown Spectroscopy*, 29th Israeli Combustion Symposium (2015).
41. W. Wu, R.L. Axelbaum, *Simultaneous Measurement of Conserved Scalars in Flames Using LIBS*, SciX, The Great Scientific Exchange, Providence (2015).
40. W. Wu, P.B. Sunderland, B.-H. Chao, D.L. Urban, R.L. Axelbaum, *Recent Advances in Flame Design, A Spaceflight Combustion Experiment*, Abstract, American Society of Space and Gravitational Research, Arlington VA (2015).

39. W. Wu, G. Yablonsky, R.L. Axelbaum, *New Observation of Water Gas Shift Equilibrium in Flames*, 9th International Conference on Chemical Kinetics, Ghent, Belgium (2015).
38. V.R. Lecoustre, P.B. Sunderland, B.-H. Chao, R.L. Axelbaum, *Study of Temperature Effects on Soot Precursor Formation in Nonbuoyant, Laminar Ethylene Premixed Flames*, American Society of Space and Gravitational Research, Abstract, Pasadena (2014).
37. V.R. Lecoustre, P.B. Sunderland, B.-H. Chao, R.L. Axelbaum, *Effects of Temperature and Equivalence Ratio on Soot Precursors in Rich Premixed Flames*, Poster, Eighth U.S. National Combustion Meeting, Salt Lake City (2013).
36. M. Dehnavi, M.K. Rodenhurst, B.H. Chao, V.R. Lecoustre, P.B. Sunderland, R.L. Axelbaum, *A Numerical Study on the Soot Free Conditions of Spherical Diffusion Flames*, Central States Section of the Combustion Institute, Dayton (2012) 8 pp.
35. V.R. Lecoustre, P.B. Sunderland, B.H. Chao, R.L. Axelbaum, *Numerical Investigation of Hydrogen Diffusion Flame Quenching Limits in Microgravity*, Central States Section of the Combustion Institute, Dayton (2012) 9 pp.
34. S. Lieb, S.A. Skeen, B.M. Kumfer, F. Xia, R.L. Axelbaum, *Characterization of the High-Temperature Soot Inception Zone Boundary in Nonpremixed Flames* 7th U.S. National Combustion Meeting, Atlanta (2011) 6 pp.
33. V.R. Lecoustre, P.B. Sunderland, B.H. Chao, R.L. Axelbaum, *Characterization of Microgravity Spherical Diffusion Flame Sooting Limits*, 7th U.S. National Combustion Meeting, Atlanta (2011) 6 pp.
32. V.R. Lecoustre, P.B. Sunderland, B.H. Chao, R.L. Axelbaum, *Numerical Study of Laminar Spherical Hydrogen Diffusion Flames at Low Flow Rates*, 7th U.S. National Combustion Meeting, Atlanta (2011) 6 pp.
31. V.R. Lecoustre, C.W. Moran, P.B. Sunderland, B.H. Chao, R.L. Axelbaum, *Experimental and Numerical Investigation of Extremely Weak Hydrogen Diffusion Flames*, 6th U.S. National Combustion Meeting, Ann Arbor (2009) 12 pp.
30. V.R. Lecoustre, P.B. Sunderland, B.H. Chao, R.L. Axelbaum, *Numerical Simulations of Soot Kinetics in Spherical Diffusion Flames,* Eastern States Meeting of the Combustion Institute, College Park (2009) 8 pp.
29. V.R. Lecoustre, P.B. Sunderland, B.H. Chao, D.L. Urban, D.P. Stocker, R.L. Axelbaum, *Numerical Simulations of Spherical Diffusion Flames at their Sooting Limits,* 6th U.S. National Combustion Meeting, Ann Arbor (2009) 9 pp.
28. V.R. Lecoustre, P.B. Sunderland, B.H. Chao, D.L. Urban, D.P. Stocker, R.L. Axelbaum, Effects of C/O Ratio and Temperature on Sooting Limits of Spherical Diffusion Flames, 46th Aerospace Sciences Meeting, Reno, Paper AIAA-2008-827 (2008) 9 pp. https://arc.aiaa.org/doi/pdf/10.2514/6.2008-827
27. V.R. Lecoustre, P.B. Sunderland, B.H. Chao, D.L. Urban, D.P. Stocker, R.L. Axelbaum, Sooting Limits of Ethylene Spherical Diffusion Flames, Poster, 32nd International Symposium on Combustion, Montreal (2008).
26. P.B. Sunderland, D.L. Urban, D.P. Stocker, B.H. Chao, R.L. Axelbaum, Quasi-Steady Microgravity Spherical Ethylene Diffusion Flame, Fire Science Image, International Association of Fire Safety Science, Karlsruhe (2008).
25. Q. Wang, and B.H. Chao, , Kinetic and Radiative Extinction of Spherical Diffusion Flames, 5th U.S. Combustion Meeting, San Diego (2007) 12 pp.
24. V.R. Lecoustre, B.H. Chao, P.B. Sunderland, D.L. Urban, D.P. Stocker, R.L. Axelbaum, A Computational Investigation of Sooting Limits of Spherical Diffusion Flames, 5th U.S. Combustion Meeting, San Diego (2007) 8 pp.

23. V.R. Lecoustre, B.H. Chao, P.B. Sunderland, D.L. Urban, D.P. Stocker, R.L. Axelbaum, Effects of C/O Ratio and Scalar Dissipation Rate on Sooting Limits of Spherical Nonpremixed Flames, Eastern States Section of the Combustion Institute, Charlottesville (2007) 10 pp.
22. K.J. Santa, B.H. Chao, P.B. Sunderland, D.L. Urban, D.P. Stocker, R.L. Axelbaum, Radiative Extinction of Gaseous Spherical Diffusion Flames in Microgravity, 5th U.S. Combustion Meeting, San Diego (2007) 13 pp.
21. K.J. Santa, Z. Sun, B.H. Chao, P.B. Sunderland, R.L. Axelbaum, D.L. Urban, D.P. Stocker, Effects of Lewis Number on Temperatures of Spherical Diffusion Flames, 45th Aerospace Sciences Meeting, Reno, Paper AIAA-2007-736 (2007) 9 pp. https://arc.aiaa.org/doi/pdf/10.2514/6.2007-736
20. [2]P.B. Sunderland, D.L. Urban, D.P. Stocker, B.H. Chao, R.L. Axelbaum, Spherical Ethylene Diffusion Flame in Microgravity, Art Competition Poster, 5th U.S. Combustion Meeting, San Diego (2007).
19. [3]R.L. Axelbaum, D.L. Urban, P.B. Sunderland, B.H. Chao, Spherical Ethylene Diffusion Flame in Microgravity, Art Entry, Science and Engineering Visualization Contest, National Science Foundation (2006).
18. K.J. Santa, B.H. Chao, P.B. Sunderland, J.L. Taylor, D.L. Urban, D.P. Stocker, R.L. Axelbaum, Radiative Extinction of Gaseous Spherical Diffusion Flames in Microgravity, 44th Aerospace Sciences Meeting, Reno, Paper AIAA-2006-747 (2006) 9 pp. https://arc.aiaa.org/doi/pdf/10.2514/6.2006-747
17. K.J. Santa, Z. Sun, B.H. Chao, P.B. Sunderland, R.L. Axelbaum, D.L. Urban, D.P. Stocker, Numerical and Experimental Observations of Spherical Diffusion Flames, Work-in-Progress Poster, 31st International Symposium on Combustion, Heidelberg (2006).
16. P.B. Sunderland, J.L. Taylor, K.J. Santa, B.-H. Chao, D.L. Urban, D.P. Stocker, R.L. Axelbaum, Extinction of Normal and Inverse Spherical Diffusion Flames in Microgravity, 4th U.S. Combustion Meeting, Philadelphia (2005) 6 pp.
15. K.J. Santa, Z. Sun, R.L. Axelbaum, B.-H. Chao, P.B. Sunderland, D.L. Urban, D.P. Stocker, Numerical and Experimental Observations of Spherical Diffusion Flames, 4th U.S. Combustion Meeting, Philadelphia (2005) 6 pp.
14. Z. Sun, R.L. Axelbaum, B.-H. Chao, P.B. Sunderland, D.L. Urban, D.P. Stocker, Effects of Radiation Heat Loss and Lewis Number on the Temperature of Spherical Microgravity Diffusion Flames, Spring Technical Meeting of the Combustion Institute Central States Section, Austin (2004) 5 pp.
13. [4]J.L. Taylor, P.B. Sunderland, Z. Sun, D.P. Stocker, D.L. Urban, B.-H. Chao, R.L. Axelbaum, Images of Spherical Diffusion Flames, Art Exhibit Poster, 30th International Symposium on Combustion, Chicago (2004).
12. J.L. Taylor, P.B. Sunderland, Z. Sun, D.P. Stocker, D.L. Urban, B.-H. Chao, R.L. Axelbaum, Sooting Limits and Extinction of Spherical Diffusion Flames, Work-in-Progress Poster, 30th International Symposium on Combustion, Chicago (2004).
11. R.L. Axelbaum, P.B. Sunderland, D.L. Urban, D.P. Stocker, B.-H. Chao, Sooting Limits of Diffusion Flames with Oxygen-Enriched Air and Diluted Fuel, abstract, 22nd American Association of Aerosol Research Conference, Anaheim (2003).
10. P.B. Sunderland, D.L. Urban, B.–H. Chao, D.P. Stocker, R.L. Axelbaum, Sooting Limits of Normal and Inverse Microgravity Spherical Diffusion Flames, 3rd U.S. Combustion Meeting, Chicago (2003) 6 pp.

9. P.B. Sunderland, D.L. Urban, D.P. Stocker, B.H. Chao, R.L. Axelbaum, Sooting Limits of Diffusion Flames with Oxygen-Enriched Air and Diluted Fuel, Seventh International Microgravity Combustion Workshop, Cleveland, NASA CP-2003-212376 (2003) pp. 49-52.
8. P.B. Sunderland, D.L. Urban, D.P. Stocker, B.–H. Chao, R.L. Axelbaum, Sooting Limits of Microgravity Spherical Diffusion Flames, Poster, Twenty-Ninth Symposium (International) on Combustion, The Combustion Institute, Sapporo (2002).
7. P.B. Sunderland, D.L. Urban, D.P Stocker, B.–H. Chao, R.L. Axelbaum, Sooting Limits of Microgravity Spherical Diffusion Flames, Spring Technical Meeting of the Combustion Institute Central States Section, Knoxville (2002) 6 pp.
6. R.L. Axelbaum, R. Chen, P.B. Sunderland, D.L. Urban, S. Liu, B.H. Chao, Effects of Flame Structure and Hydrodynamics on Soot Particle Inception and Flame Extinction in Diffusion Flames, Sixth International Microgravity Combustion Workshop, Cleveland, NASA CP-2001-210826 (2001) pp. 153-156.
5. S. Liu and B.-H. Chao, "Effects of Structure and Hydrodynamics on Soot Inception in Spherical Microgravity Diffusion Flames," Poster Paper #313, Second Joint Meeting of the U.S. Sections of the Combustion Institute, Oakland (2001).
4. R.L. Axelbaum, P.B. Sunderland, D.L. Urban, S. Liu, B.–H. Chao, Effects of Structure and Hydrodynamics on the Sooting Behavior of Spherical Microgravity Diffusion Flames, Poster P467, Twenty-Eighth Symposium (International) on Combustion, The Combustion Institute, Edinburgh (2000).
3. P.B. Sunderland, R.L. Axelbaum, D.L. Urban, Effects of Structure and Hydrodynamics on the Sooting Behavior of Spherical Microgravity Diffusion Flames, Fifth International Microgravity Combustion Workshop, Cleveland, NASA CP-1999-208917 (1999) pp. 475-478.
2. P.B. Sunderland, R.L. Axelbaum, D.L. Urban, Understanding Permanently-Blue Flames Through Microgravity Combustion Experiments, 1st U.S. Combustion Meeting, Washington, (1999) pp. 453-456.
1. P.B. Sunderland, R.L. Axelbaum, D.L. Urban, Understanding Permanently-Blue Flames Through Microgravity Experiments, Poster W5A04, Twenty-Seventh Symposium (International) on Combustion, The Combustion Institute, Boulder (1998).

[2]Recipient of 3rd Prize, Art Competition, 5th U.S. Combustion Meeting, San Diego, 2007.
[3]Finalist, Science and Engineering Visualization Challenge, National Science Foundation, 2006.
[4]Recipient of Third Prize, Art Competition, 30th International Symposium on Combustion, 2004.

Theses and Dissertations

15. Phillip H. Irace, *Particle Formation and Thermal Radiation in Laminar Diffusion Flames with Applications to Energy and Materials*, Ph.D. Dissertation, Washington University in St. Louis, Dept. of Mechanical Engineering and Materials Science, 200+ pp, May 2022.
14. Zhengyang Wang, *Shapes and Double Blue Zones in Laminar Co-flow Diffusion Flames*, University of Maryland, Dept. of Mechanical Engineering, 128 pp, August 2019. https://www.proquest.com/openview/28862866b50531bad30944a84dad40a7/
13. Wendong Wu, *Fundamental Studies of Flame Structure Through Laser Plasma Diagnostics,* Ph.D. Dissertation, Washington University in St. Louis, May 2017. https://search.proquest.com/docview/1952045801
12. El Makawy, Mohamed H., *The Effects of Reactant Dilution on Lengths of Laminar Jet Diffusion Flames*, M.S. Thesis, Ghent Univ., Department of Flow, Heat and Combustion, 54 pp., Aug. 2016.

11. Mahmood Dehnavi, *A Numerical Investigation of Microgravity Ethylene/Oxygen/Nitrogen Spherical Diffusion Flames at Their Soot Free Limit*, M.S. thesis, Department of Mechanical Engineering, University of Hawaii (2012).
http://hdl.handle.net/10125/100951
10. Melvin Rodenhurst, *Kinetic Extinction Limit of a Spherical Diffusion Flame Attached to a Burner Surface*, M.S. thesis, Department of Mechanical Engineering, University of Hawaii (2012).
http://hdl.handle.net/10125/101022
9. Vivien R. Lecoustre, *Numerical Investigations of Gaseous Spherical Diffusion Flames*, Ph.D. Dissertation, University of Maryland, Department of Mechanical Engineering, December (2009).
http://hdl.handle.net/1903/9966
8. Scott A. Skeen, *Oxygen-Enhanced Combustion: Theory and Applications,* Ph.D. Dissertation, Department of Energy, Environmental and Chemical Engineering, Washington University, August (2009).
https://doi.org/10.7936/K7MC8X35
7. Ben Kumfer, Ph.D. Dissertation, Department of Energy, Environmental and Chemical Engineering, Washington University (2009).
6. Chad Unrau, Ph.D. Dissertation, Department of Energy, Environmental and Chemical Engineering, Washington University (2009).
5. Wang, Q., *Kinetic and Radiative Extinctions of Spherical Diffusion Flames*, M.Sc. Thesis, University of Hawaii (2007).
http://hdl.handle.net/10125/20698
4. Santa, K.J., *Structure and Extinction of Spherical Diffusion Flames in Microgravity*, M.Sc. Thesis, University of Hawaii (2006).
https://scholarspace.manoa.hawaii.edu/items/8503612a-3fe9-4d76-926e-4b28e4732733
3. Chen, R., *Effects of Flame Structure on Soot Inception and Flame Extinction of Diffusion Flames*, M.Sc. Thesis, Washington University (2003).
2. Sun, Z., *Numerical Investigation of Flame Synthesis Involving Multi-Component Aerosols*, D.Sc. Dissertation, Washington University (2002).
https://www.proquest.com/openview/1eba51c9ed20c4f81d6c5f7c8d621a58/
1. Liu, S., *On Soot Inception in Nonpremixed Flames and the Effects of Flame Structure*, Ph.D. Dissertation, University of Hawaii (2001).
https://www.proquest.com/openview/54e315a455d4201fe315e3296f8743f5/

N.3 General ACME Papers and Presentations

Conference Proceedings, Abstracts, Posters, Etc.

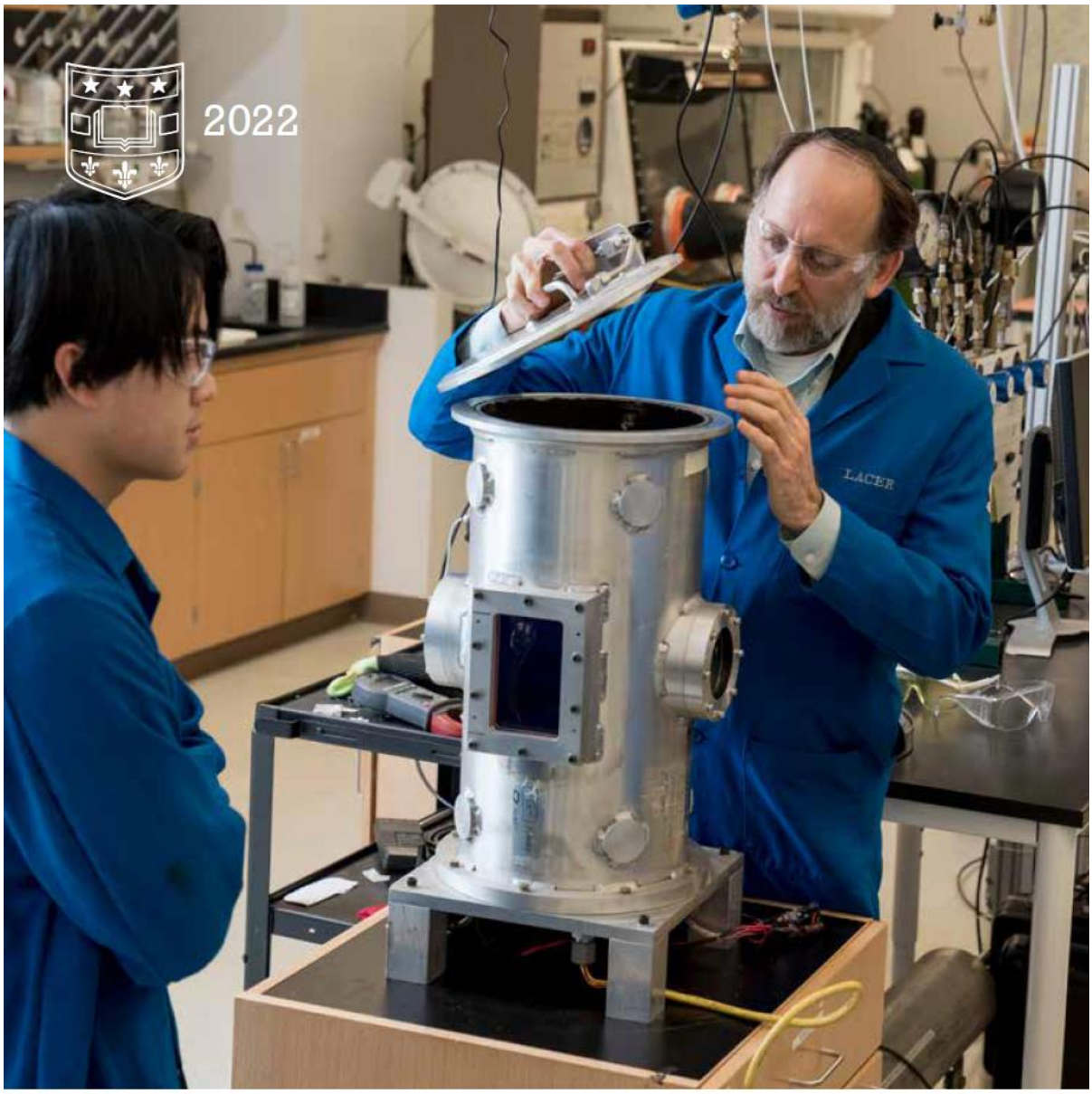
20. Dennis P. Stocker, *ACME: Advanced Combustion via Microgravity Experiments*, Work-in-Progress Poster, 13th U.S. National Combustion Meeting, College Station, TX, 19-22 March 2023.
19. Dennis P. Stocker, *ACME: Advanced Combustion via Microgravity Experiments: ACME*, Poster, 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, 9-12 Nov. 2022.
18. Michael Greshko, *Cosmic Combustion, National Geographic Magazine*, Oct. 2022, p. 22.
www.nationalgeographic.com/science/article/in-space-flames-become-otherworldly
(*online version with different title and formatting*)

17. *International Space Station Benefits for Humanity 2022*, Report No. NP-2022-06-010-JSC, 2022, pp. 117-118. www.nasa.gov/sites/default/files/atoms/files/iss_benefits_for_humanity_2022_book.pdf
16. Dennis P. Stocker, *Laminar Non-Premixed Flames of Gaseous Fuel Aboard the International Space Station*, Spring Technical Meeting of The Central States Section of the Combustion Institute, Detroit, Mi, 15-17 May 2022, 10 pp. https://www1.grc.nasa.gov/wp-content/uploads/ACME-CSSCI-paper-20220402.pdf
15. Doreen B. Zudell, *Final Flames Ignited in Quest for Combustion's Secrets*, Aerospace Frontiers, May 2022, p 3. www.nasa.gov/sites/default/files/atoms/files/ps-03506-05_af_may_2022_508-sp.pdf
14. Peter B. Sunderland, *Transformative Combustion Research in the ACME Suite of ISS Experiments*, 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore MD, 3-6 Nov. 2021.
13. Dennis P. Stocker, *Recent Research on Flames of Gaseous Fuel Aboard the International Space Station*, Conference of the Japanese Society of Microgravity Application (JASMAC-33), virtual, 13-15 Oct. 2021, 11 pp. https://www1.grc.nasa.gov/wp-content/uploads/JASMAC-ACME-paper-20210921.pdf
12. Dennis P. Stocker, *Advanced Combustion via Microgravity Experiments on the International Space Station*, Work-in-Progress Poster, 12th U.S. National Combustion Meeting, virtual, 24-26 May 2021. https://www1.grc.nasa.gov/wp-content/uploads/PS-03403-ACME-on-the-ISS.pdf
11. S. Pichugin, D. Samsonov, S. Frolov, O. Korobeinichev, A. Shmakov, *Flames Investigations at ISS*, Cosmonautics of XXI Century Conference, TSNIIMash, Moscow, Russia, 12-14 May 2021.
10. *On-site Work Underway With ACME*, Aerospace Frontiers, Feb. 2021, p 8. https://www.nasa.gov/sites/default/files/atoms/files/aerospace-frontiers-feb-2021.pdf
9. Sergey Pichugin, Oleg Artemiev, Sergey Prokopiev, Oleg Kononenko, Oleg Novitsky, and Sergey Frolov, *Investigations at ISS on Combustion of Gas Mixtures. Sample Results and Technical Experience*, XIII International Scientific and Practical Conference 'Manned Flights in Space,' Gagarin Cosmonaut Training Centre, Star City, Russia, 13-15 Nov. 2019.
8. S. Jenise Veris, *ACME Update: BRE Experiment Fired Up*, Aerospace Frontiers, April 2019, p. 12. https://www.nasa.gov/sites/default/files/atoms/files/aerospace-frontiers-april-2019.pdf
7. *Suite of Experiments Study Flame Behavior in Microgravity*, Aerospace Frontiers, June 2017, pp. 1, 4-5. https://www.nasa.gov/sites/default/files/atoms/files/af_june2017.pdf
6. S.A. Gokoglu, D.L. Dietrich, D.P. Stocker, P.V. Ferkul, S.L. Olson, M.C. Hicks, *A Researcher's Guide to: International Space Station Combustion Science*, Report No. NP-2015-10-034-JSC, NASA ISS Program Science Office, Johnson Space Center, Houston, TX, USA, 2015, pp. 13-18. https://www.nasa.gov/sites/default/files/atoms/files/np-2015-10-034-jsc_combustion_in_reduced_gravity-iss-mini-book-011116-508.pdf

5. D.P. Stocker, F. Takahashi, J.M. Hickman, and A.C. Suttles, *Advanced Combustion via Microgravity Experiments: Planned International Space Station Research on Gaseous Flames*, Work in Progress Poster, 35th International Symposium on Combustion, San Francisco, CA, 3-8 Aug. 2014. https://www1.grc.nasa.gov/wp-content/uploads/PS-01374-ACME-posterFinal.pdf
4. D.P. Stocker, F. Takahashi, J.M. Hickman, and A.C. Suttles, *Gaseous Non-Premixed Flame Research Planned for the International Space Station*, Spring Technical Meeting of the Central States Section of the Combustion Institute, Tulsa, OK, 16–18 March 2014, 13 pp. https://www1.grc.nasa.gov/wp-content/uploads/ACME_CSSCI_2014_rev.pdf
3. D.P. Stocker, F. Takahashi, and J.M. Hickman, *Burner-Stabilized Gaseous Flames for the Combustion Integrated Rack*, 29th Annual Meeting of the American Society for Gravitational and Space Research and 5th International Symposium for Physical Sciences in Space, Orlando, FL, 3-8 Nov. 2013.
2. D.P. Stocker, F. Takahashi, P.V. Ferkul, and J.M. Hickman, *Gaseous Laminar Diffusion Flames for the Combustion Integrated Rack*, 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, 28 Nov. to 2 Dec. 2012.
1. T.F. O'Malley, W.A. Sheredy, and D.P. Stocker, *Combustion Research on the International Space Station*, 59th International Astronautical Congress 2008, Vol. 2, pp. 756-766 (2008).

N.4 Awards

In 2022, Flame Design Principal Investigator Rich Axelbaum was named a Senior Member of the National Academy of Inventors (NAI). Per the site, “Senior Members are active faculty, scientists, and administrators with success in patents, licensing, and commercialization and have produced technologies that have brought or aspire to bring, real impact on the welfare of society. They are rising stars who foster a spirit of innovation within their communities and institutions while educating and mentoring the next generation of inventors.” As explained in his university’s press release and department’s 2022 brochure about this honor, “Axelbaum holds 10 patents and founded the startup company AP Material Inc., which commercialized a flame-synthesis technology to manufacture high-purity nanopowders.”



Cover of 2022 departmental brochure, where Axelbaum appears on the photo's right with a NASA-loaned combustion chamber.

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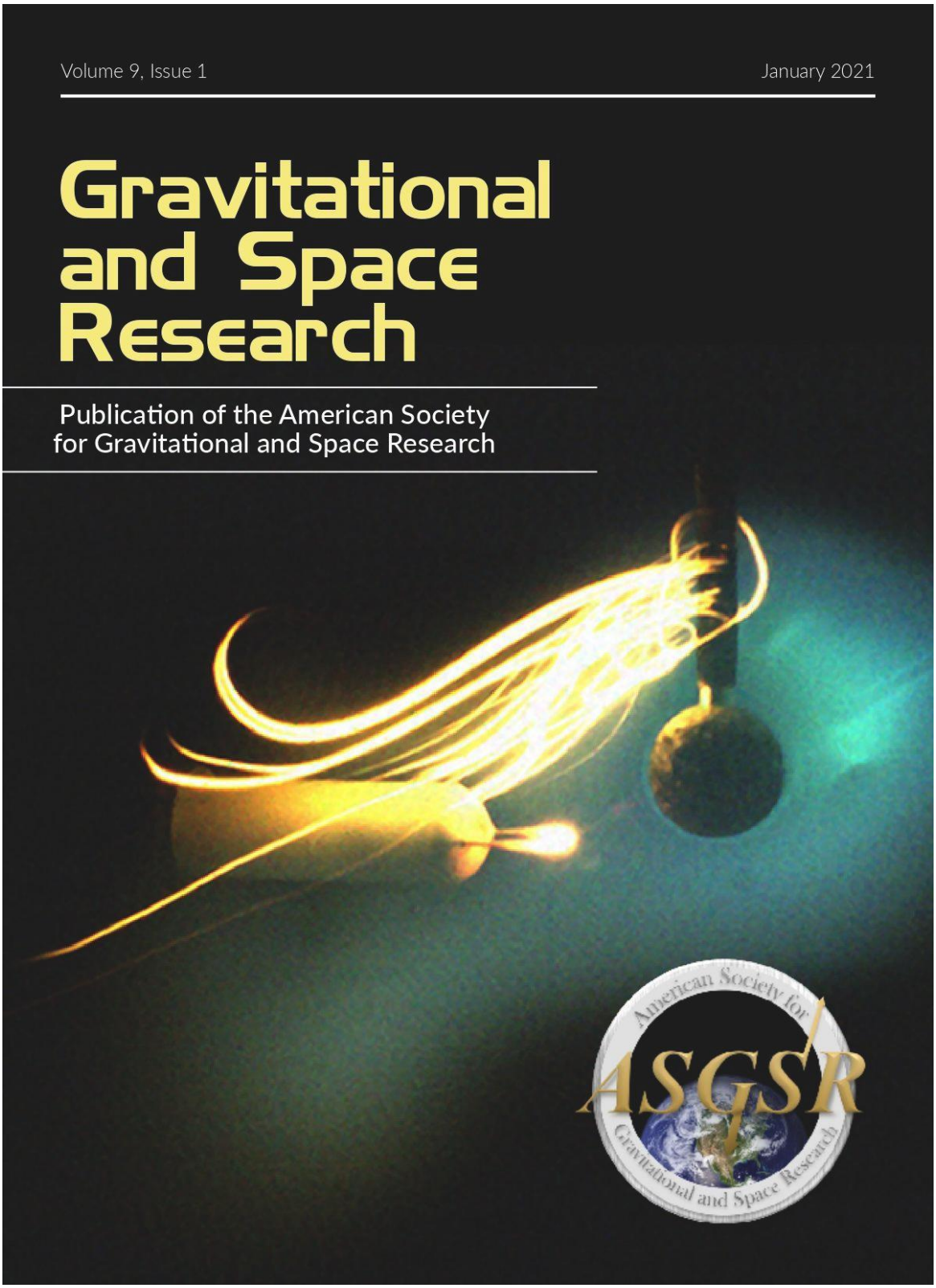
In 2021, Flame Design Co-Investigator Peter Sunderland was inducted as a Fellow of the Combustion Institute for his "significant contributions to understanding laminar flame dynamics and soot formation and oxidation."

Additional awards include the following, where images of the award certificate and journal cover from item #5 appear subsequently:

5. Award for Gravitational and Space Research Journal Cover, Flame Ignition in Space, Jan. 2021.
4. Third Prize, Art Competition, 5th U.S. Combustion Meeting, San Diego, 2007.
3. Finalist, Science and Engineering Visualization Challenge, National Science Foundation, 2006.
2. Third Prize, Art Competition, 30th International Symposium on Combustion, 2004.
1. Best-Paper Award, Northern Ohio section of AIAA, 2003.



Certificate for the winning Flame Design entry, *Flame Ignition in Space*, in the 2020 Art Competition of the American Society for Gravitational and Space Research (ASGSR).



Journal cover featuring *Flame Ignition in Space*, from the Flame Design experiment, which provides a close view of the hot-wire igniter tip and 6.35-mm porous spherical burner.

N.5 Outreach

NASA Project Scientist Dennis Stocker created the Space Flames Facebook page (www.facebook.com/space.flames) and Flickr site (https://www.flickr.com/photos/space-flames), where some entries are specific to Flame Design. He continues to administer these sites, where most content thus far has been about the ACME project of which Flame Design is a part. As an example of educational outreach, a simple board game related to the ACME's spherical flames experiments can be downloaded from https://www1.grc.nasa.gov/space/iss-research/iss-fcf/cir/acme/downloads/. ACME fact sheets, papers, posters, and presentations providing a project overview and status can also be downloaded from that page. Printed copies of fact sheets have been distributed at meetings of The Combustion Institute and the American Society for Gravitational and Space Research (ASGSR).

The investigators, Washington University in St. Louis, and the University of Maryland promoted awareness of the Flame Design investigation, as can be seen by these examples:

- Flame Design
- Fires in Microgravity
- 'Playing' with fire (30 Sept. 2004)
- UMD Experiments Launched into Orbit via SpaceX Capsule (6 June 2017)
- Experiment designed by Washington University engineer launched on SpaceX (8 June 2017)
- Flame design in space may lead to soot-free fire (9 May 2019) – *one of 2019's top-ten stories from the Washington University in St. Louis' McKelvey School of Engineering*
- In the media: International Space Station experiments will search for key to soot-free flames (15 May 2019)
- Engineering Momentum Magazine: Spring 2020 (cover story)
- Unlocking the Mysteries of Fire ... in Space (10 Feb. 2021)
- Fire Protection Experts Available to the Media (22 June 2021)
- Sunderland Gives Invited Talk at ASGSR 2021 Symposium (5 Nov. 2021)

NASA articles, etc. about the study (and ISS combustion research in general) can be seen in these examples:

- Space Life and Physical Sciences Research and Applications SpaceX CRS-11 Payloads and Experiments (1 June 2017)
- The Power of Studying Combustion on the ISS (13 March 2018)
- Studying flames in microgravity is helping make combustion on Earth cleaner, and space safer, where the video can be downloaded from Designing Flames Aboard ISS
- Astronaut Christina Koch's Space Station Science Scrapbook (Flame Design set up seen at *Keeping the Flames Burning*, 3 Feb. 2020)
- A Thousand Points of Light – On The International Space Station (16 July 2020)
- 2021 NASA Science calendar (May image)
- Astronomy Picture of the Day: 2021 August 10
- Studying Flames in Microgravity (23 Feb. 2022)
- Space Station's Quest for the Secrets of Fire (18 April 2022)
- International Space Station Benefits for Humanity 2022 – *ACME is briefly discussed on page 117 of the publication, while page 188 features a Flame Design image*

Other stories about the investigation (and ISS combustion research in general) often resulted from the public engagement by the investigators and Dennis Stocker, where examples can be found here:

- Science in St. Louis Seminar Series: Fire in Space (8 March 2018)
- International Space Station (ISS) Combustion Experiments (15 Oct. 2018)
- International Space Station experiments will search for key to soot-free flames (14 May 2019)
- Lighting A Flame In Space Could Help Fuels Burn More Cleanly On Earth (21 May 2019)
- Don't Light a Fire on the International Space Station…Unless it's for Science (at 1:03:06, 8 Jan. 2020)
- In space, flames become otherworldly (30 Aug. 2022) - *while general to ACME, the photo collage was from 2019 normal-flame testing for Flame Design. Despite its one-paragraph brevity, Sunderland and Stocker were both interviewed for this article which was published under the title Cosmic Combustion on page 22 of the Oct. 2022 issue of the National Geographic Magazine.*

Stocker is involved in a broad range of NASA educational outreach to students and teachers, Scouts of both genders, and the general public. His primary outreach focus is the fundamentals of microgravity and human spaceflight, for example through microgravity design challenges which he helps create and administer. But he periodically addresses other topics, e.g., where he sometimes speaks on 'NASA and the Science of Superman' – for example at the Superman Celebration in Metropolis, IL. In this regard, his NASA outreach is carried out locally, regionally, nationally, and online. For example, he regularly presents on microgravity at Space Center Houston's Space Exploration Educators Conference (SEEC). He has exhibited the last five National Scout Jamborees, five national Boy Scout conferences, the last three national Girl Scout conventions, and the 2019 World Scout Jamboree (which was held in the U.S.). Stocker has been a key lead for the agency in NASA's participation in these Scouting events, where as an example the latter outreach is visually described at www.nasa.gov/feature/stem/world-scout-jamboree.html.